ionocast: A Real-Time HF/VHF Propagation Prediction Model for Amateur Radio Operators

Toprak Kilic, TA1BUT

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Abstract

ionocast provides real-time HF/VHF band-condition verdicts for amateur radio operators (next-24-hour forward projections are specified in Section 5 but not yet surfaced in the runtime). The model runs entirely in the browser with no server-side state. The operator's QTH stays on-device for everything except two panels that need a nearest-station lookup, in which case it is sent to ionocast's own proxy for resolution and discarded after the response (see Section 2.1 for the full disclosure). The model combines an ITU-R P.533–grounded SNR link budget with real-time data from NOAA SWPC (solar indices, D-RAP absorption, OVATION auroral power, GOES proton flux), the kc2g ionosonde network (observed MUF and foF2), GIRO digisondes (foF2, foEs, hmF2), and WSPR spot observations from `wspr.live`. A local validation harness scores predictions against retrospective WSPR activity to quantify systematic bias and refine per-band σ.

Key contributions:

- *Symmetric MUF consensus* (geometric mean) between gridded kc2g observations and the foF2 climatology.
- *Power-summed noise model* (ITU-R P.372) that correctly handles man-made noise dominance on the low bands.
- *Per-hop minimum MUF* and *per-hop ionospheric loss accounting* (diurnal D-region, polar cap absorption, flare SID, auroral) evaluated at each great-circle reflection point rather than at a single midpoint.
- *Sauer–Wilkinson polar cap absorption* coupled to GOES $\geq 10\,\mathrm{MeV}$ proton flux, and *flare-driven SID* absorption keyed to GOES X-ray class.
- *Storm-lag kernel* that captures the F-region's delayed response to K_p, with D_{st} and IMF B_z used for earlier detection than the 3-hour K_p alone.
- *Equatorial-anomaly* (paired crest enhancement at $\pm 15\,^\circ$ dip plus on-equator trough depression) and *winter-anomaly* corrections to the foF2 climatology, together with an *asymmetric gray-line* bonus.
- *Elevation-dependent antenna gain* from hop-geometry-derived takeoff angles, plus *NVIS detection* for short paths using local foF2.
- Verdicts mapped to ITU-R P.842 reliability buckets in σ-units rather than fixed dB widths, so the tier boundary scales with each band's prediction uncertainty.

Both short-path and long-path great circles are evaluated per destination. All computation is deterministic, repeatable, and inspectable; no machine-learning black boxes.

Source and reproducibility. The full ionocast implementation — browser code, Cloudflare Worker handlers, and the offline calibration harness — is open at `https://github.com/topklc/ionocast`. The 35-path calibration basket, 27-station GIRO list, and per-band σ table are reproduced verbatim from that repository in Appendices C, D, and the main text (Table 9) respectively, so the validation metrics below can be re-derived end-to-end from the paper plus the repository alone. The repository is the source of truth: where this paper and the code disagree, the discrepancy reflects either documentation lag (the code shipping a fix or rename that the paper hasn't been updated to match) rather than a code bug to follow the paper into; the calibration constants in `src/constants.js` and the per-hop physics in `src/physics/` are what produces the verdict an operator sees.

Validation note. The shipped harness scores against 30 d of WSPR activity aggregated two ways: a global "open somewhere" truth (accBin in the mid-80s, Brier ~0.09–0.11 at the 2026-05-08 post-B6 baseline) which has a known structural ceiling — the OR-over-paths target overrewards permissive predictors and saturates whenever *any* path on the band has activity, quantified in Section 10 #9; and a per-path mode (accBin in the low-40s, Brier ~0.56) which uses TX/RX-bbox-restricted aggregates with a ≥ 1 spot/h floor and is dominated by operator-activity sparsity on minority-traffic bands. Exact baseline values are tabulated in Section 7.2 (Table 8); both metrics drift with solar activity between runs and shifted further at the 2026-05-08 post-B6 D-region/grayline paired rewrite. The 40+ percentage-point gap between the two metrics on the same harness run is itself the structural-ceiling evidence: a model that scores in the 80s on the global metric and the 40s on the per-path metric is not 80-good or 40-good, it is sitting on a metric whose ceiling reflects "somewhere is open" rather than "the requested path is open." The two numbers are not directly comparable; per-pair WSPR ground-truth wiring with calibrated availability priors is in progress. Quote either number as a regression-detection signal, not as a production-grade reliability claim.

Acronyms used throughout this paper. ACE: Advanced Composition Explorer (NASA, real-time solar-wind monitor at L1). AE: Auroral Electrojet index (geomagnetic). DIDB / DIDBase: Digital Ionogram DataBase (UMass Lowell). DSCOVR: Deep Space Climate Observatory (NOAA, L1 solar-wind monitor). EIA: Equatorial Ionisation Anomaly. foF2 / foEs: F2-layer / sporadic-E layer ordinary-wave critical frequency. GIRO: Global Ionospheric Radio Observatory. GNSS: Global Navigation Satellite System. GOES: Geostationary Operational Environmental Satellite (NOAA). hmF2: F2-layer peak height. IGRF: International Geomagnetic Reference Field. IMO: International Meteor Organization. IRI / IRTAM: International Reference Ionosphere / IRI Real-Time Assimilative Mapping. LSTID: Large-Scale Travelling Ionospheric Disturbance. MUF: Maximum Usable Frequency. NVIS: Near Vertical Incidence Skywave. OVATION: Oval Variation, Assessment, Tracking, Intensity, and Online Nowcasting (NOAA SWPC auroral-precipitation model). SDO: Solar Dynamics Observatory. SFI: Solar Flux Index ($F_{10.7}$). SID: Sudden Ionospheric Disturbance. SSN: Sunspot Number. TEP: Trans-Equatorial Propagation. TID: Travelling Ionospheric Disturbance. WSPR: Weak Signal Propagation Reporter network. ZHR: Zenith Hourly Rate (meteor-shower intensity).

Contents

1 Introduction

Amateur radio operators planning an operating session need to answer a deceptively simple question: *which bands will be open, and for how long?* The answer depends on the instantaneous state of the ionosphere, the Sun, the geomagnetic field, and the operator's own station (antenna, power, mode, and local noise environment).

Existing tools serve fragments of this problem:

- **VOACAP** [37] provides rigorous P.533-based [1] predictions but operates as a batch tool requiring manual circuit specification; it is not real-time.
- **N0NBH Solar Conditions** [36] provides a widely trusted heuristic band-condition table keyed to SFI and Kp, but without path geometry or operator context.
- **kc2g MUF map** [28] renders observed MUF contours in near real time, but offers no forward projection and no SNR context.
- **HamQSL / DX Toolbox** [38] use threshold-based rules on solar indices, lacking physics grounding (HamQSL is the same site as N0NBH [36]).

ionocast occupies the intersection: real-time, physics-grounded, operator-configurable, and privacy-preserving (operator QTH stays on-device in `localStorage` between requests, and is sent to the proxy worker only at request time for nearest-station resolution; see Section 2.1). The QTH parameterises ten reference propagation paths (five short-path destinations plus their long-path counterparts); the SNR budget runs per band, per path, at the current time. A forward-projection MUF pipeline (solar-zenith analogy, storm depression, seasonal correction, sporadic-E persistence; Section 5) is documented but not yet surfaced in the runtime; the Upcoming Disruptions panel surfaces SWPC's own forecasts (3-day R/S/G probabilities, 3-h K_p, 27-day SFI/A_p) rather than ionocast-derived per-band projections.

This paper documents the complete model as implemented. Every numeric constant in the SNR budget and MUF estimation is given either inline or in an appendix; the calibration setup that produced the harness-fitted constants (the three coordinate-descent starting seeds, the parameter sweep grid, and the WSPR baseline files) is described in Appendix E, so the amateur radio community can audit, reproduce, and improve the model from the paper plus the repository alone.

2 Data Sources

Table 1 lists all upstream data feeds. Every upstream is reached through a Cloudflare Worker (`functions/_proxies.js`); none is fetched directly by the browser. Two reasons drive the proxy: most upstreams (SWPC, kc2g, DONKI, wspr.live) lack the CORS headers that would permit browser-direct fetching, so the worker is a CORS shim plus edge-side cache; the two geo-resolved feeds (`/api/giro`, `/api/tropo`) additionally take an operator-supplied Maidenhead grid as a query parameter and resolve it to the nearest measurement station server-side, so the upstream sees a station identifier rather than the operator's QTH. `/api/hp30` fetches a single global nowcast text file and is not geo-resolved.

Table 1: Upstream data sources and refresh cadences. The *Refresh* column reports the upstream-publisher cadence (how often the source produces new data); the worker-side cache is polled more aggressively in some cases so the operator sees a new value within ∼1 min of upstream publication. Specific worker `freshSec` TTLs are tabulated in `functions/_proxies.js`.

Source	Data	Upstream	Protocol
NOAA SWPC	Kp, Ap, F10.7, X-ray class, 3-day forecast, 27-day outlook, D-RAP HAF, OVATION aurora HP, solar-wind Bz / speed / density, solar regions	1 min–6 h	Proxied via `/api/swpc-*`
prop.kc2g.com [28]	Per-station MUF(3000F2), foF2, TEC	∼15 min	Proxied via `/api/kc2g`
GIRO (lgdc.uml.edu) [29]	Nearest digisonde foF2, foEs, hmF2	∼10 min	Proxied via `/api/giro`
GFZ Potsdam [30]*	Hp30 nowcast	30 min	Proxied via `/api/hp30`
WDC Kyoto [31]	Dst / Sym-H	1 h	Proxied via `/api/kyoto`
SILSO [32]*	Daily sunspot number	Daily	Proxied via `/api/silso`
UWyo [33]	Radiosonde soundings (lowest-1 km $\mathrm{d}N/\mathrm{d}h$ over a 24-station basket)	12 h[†]	Proxied via `/api/tropo`
wspr.live [34]	WSPR spot counts and SNR by band	10 min	Proxied via `/api/wspr` (ClickHouse)
NASA DONKI [39]	CME and HSS analyses	∼hours	Proxied via `/api/donki-*`

[†] The 12 h cadence reflects the upstream radiosonde launch schedule (00Z and 12Z); the worker-side cache polls every 10 min so a new sounding reaches the operator within ∼10 min of being posted to UWyo rather than waiting the full upstream 12-hour interval.

[*] Hp30 (GFZ Potsdam) and Daily Sunspot Number (SILSO) are fetched and surfaced in the Geomagnetic / Solar UI panels for operator situational awareness, but neither feeds the SNR budget or the verdict pipeline at present. K_p^{eff} from SWPC serves the half-hour-cadence role that Hp30 could otherwise fill; $\mathrm{F}_{10.7\mathrm{A}}$ 81-day mean serves the solar-activity role of daily SSN. Both feeds are kept wired for two specific future roles: Hp30 as a sub-3-hour validation channel for the storm-lag kernel of §5.3.1 (the kernel currently consumes 3-hour K_p samples and would benefit from a within-bucket cadence to refit τ_{peak} and τ_{decay}), and SSN-v2 as an alternative anchor when the $\mathrm{F}_{10.7\mathrm{A}}$ feed is degraded (the correlation between R12 / R-MgII / SSN-v2 and $\mathrm{F}_{10.7\mathrm{A}}$ lets a downstream substitution stand in during outages, similar to the illumination-ratio fallback in §4.4).

2.1 Privacy Model

The operator's QTH is stored on-device in `localStorage` and is sent to the Cloudflare Worker only at request time, for nearest-station resolution on the geo-aware proxy endpoints; the worker neither writes the grid back to a server-side store nor logs request bodies / query strings at the application level. Privacy posture in detail:

- **Stored on device only.** The `ionocast_qth` key in `localStorage` holds the operator's grid; nothing reads or writes it server-side, and there is no account / login model.
- **Sent at request time for resolution, not retained.** Endpoints like `/api/giro` take a grid string as a query parameter so the worker can resolve it to the nearest digisonde station; the worker source contains no application-level logging of request bodies or query strings (inspectable at `https://github.com/topklc/ionocast`, the `functions/` directory), and the upstream response contains station data, not the original grid. Cloudflare's edge platform does retain its own access logs at the request level (URL + status + IP), described below.

- **WSPR live queries are global hourly aggregates** with no grid field at all; the operator's location is never transmitted to `wspr.live`. (The offline calibration harness in `scripts/harness.mjs` does use a coarser grid for retrospective sample bucketing, but is not part of the runtime page.)
- **No third-party analytics or tracking** are loaded; the only request-side trace is Cloudflare's own edge access logging, retained per the platform's free-/standard-tier policy (typically up to 72 h worth at the time of writing; consult Cloudflare's current Worker / Pages documentation for the authoritative retention window).

The honest summary: the QTH is sent to the proxy worker *for resolution at request time*, but is not retained, logged in application form, or shared with third parties. Operators who want zero server-side QTH exposure can run the static UI against the upstream APIs directly (the worker is a performance / CORS shim, not a privacy boundary).

First-visit default. On first load, with no saved QTH, ionocast derives a regionally plausible grid from the browser's runtime timezone. `Intl.DateTimeFormat().resolvedOptions().timeZone` is looked up in a compiled table of ∼60 major ham-active IANA zones (e.g. `Europe/Berlin` → JO62, `Asia/Tokyo` → PM95, `Australia/Sydney` → QF56; the full table lives in `src/physics/qth.js` and is open for inspection at the repo URL given in the abstract). Zones outside the table synthesise a grid from the UTC offset: longitude ≈ offset × 15° at a fixed mid-latitude (45° N for the northern hemisphere, −35° for identified southern-hemisphere zones). The timezone fingerprint is already exposed on every HTTPS request, so using it for a first-visit default adds no new privacy surface. When even that math fails, the last-resort fallback is **JN05** (approximately 45.5° N, 1° E — the grid covers 0–2° E, 45–46° N) — a French / west-Swiss midlatitude reference point chosen as a plausible-default location for the worldwide majority of first-visit operators. (The CCIR R-12 / ITU-R P.1239 foF2 coefficients underlying this paper are global spherical-harmonic expansions, not tabulated at a specific mid-latitude European station; the fallback grid is a UX convenience rather than a geographic anchor for the climatology.) No request is ever made to a geolocation service.

3 SNR Budget Model

The core of ionocast is a complete link budget that computes the physics-only SNR margin M_{physics} (in dB) for each amateur band on each reference path:

$$M_{\text{physics}} = P_{\text{tx}}+G_{\text{ant}}-L_{\text{fs}}-L_{\text{flare,path}}-L_{\text{Dreg}}-L_{\text{PCA}}-L_{\text{aur}}-L_{\text{MUF}}-n_{\text{hops}}L_{\text{iono}}-L_{\text{low}}-L_{\text{hop}}-L_{\text{Es}}-N-S_{\text{req}} \tag{1}$$

where every term is defined in the following subsections. The verdict-driving margin $M_{\text{verdict}} = M_{\text{physics}} + B_{\text{grayline}} + \max(B_{\text{TEP}}, B_{\text{scatter}})$ adds the additive mode-bonus terms of §7 (Eq. 76); M_{physics} alone is what Eq. 1 computes, and bonus-augmented M_{verdict} is what feeds the tier reliability formula in §7.3. A positive M_{physics} means the budget alone clears the mode threshold; the magnitude indicates robustness, but a near-zero or slightly negative M_{physics} on a TEP-eligible 15 m path can still yield a comfortable verdict once the bonuses fire. The term $L_{\text{flare,path}}$ is the joint flare-absorption product of L_{DRAP} (Section 3.3) and the per-hop L_{flare} sum (Section 3.6); inlined here so the reader does not have to forward-jump:

$$L_{\text{flare,path}} = \max\Big(L_{\text{DRAP}}, \sum_{k=1}^{N} L_{\text{flare}}(f, \cos\chi_k)\Big) \tag{2}$$

Both symbols are defined separately because they have independent physical sources, but they collapse to a single term in the budget to avoid double-charging the path for the same flare. See Section 3.3 for the max(·) rationale and the geographic asymmetry between the two contributing terms.

Global absorption-sum saturation. Each of the four ionospheric absorption terms ($L_{\text{flare,path}}$, L_{Dreg}, L_{PCA}, L_{aur}) is individually capped at 50 dB per path. Their sum is also *hard-capped* at 100 dB with proportional rescaling (no smoothing knee): if a sunlit polar hop during an X10 flare with an S3 SEP and a $K_p = 9$ main-phase storm produced hypothetical contributions $(L_{\text{flare,path}}, L_{\text{Dreg}}, L_{\text{PCA}}, L_{\text{aur}}) = (50, 30, 40, 30)$ dB, the additive sum reaches 150 dB (well past physical D-region saturation ∼80–100 dB; beyond that the band is dead regardless and adding more absorption is unphysical). The cap then scales each contribution by a factor $100/150 \approx 0.667$, producing the rescaled set $(33.3, 20.0, 26.7, 20.0)$ dB (path total exactly 100 dB) so per-mechanism attribution is preserved ($L_i' = L_i \cdot 100/\sum L_i$).

Why this gate is hard while the others are smooth. The auroral K_p, OVATION HP, CGM-edge, and IMF B_z gates elsewhere are smoothed because they fire under routine active-K_p conditions where a 5–20 dB step would visibly shift verdicts. The 100 dB absorption-sum cap fires only on a simultaneous X10 flare, S3 SEP, and $K_p = 9$ main-phase storm intersecting on a sunlit polar hop — a scenario realised at most a few hours per solar cycle, so the discontinuity at $\sum L_i = 100$ is unobservable in any operator-relevant regime. The hard form is intentional; the inconsistency with the smoothed gates is acknowledged here rather than papered over.

Why proportional rescaling rather than cap-and-skip. The four contributing terms ($L_{\text{flare,path}}$, L_{Dreg}, L_{PCA}, L_{aur}) are driven by physically distinct ionisation sources — soft X-ray photo-ionisation, quiet-day diurnal photo-ionisation, proton-induced ionisation, and particle-precipitation ionisation respectively — but they all enhance *the same* D-region electron column, which has a finite total-ionisation depth (∼80–100 dB of absorption at HF when fully populated; beyond that further electrons displace existing ones rather than adding new absorbers). Past the column's physical ceiling, the correct behaviour is not "add until budget then drop further contributions": cap-and-skip would arbitrarily zero out whichever mechanism happens to be evaluated last in the implementation order and leak attribution dependence into the UI's per-mechanism display. Proportional rescaling, $L_i' = L_i \cdot 100/\sum L_i$, preserves each mechanism's relative contribution at saturation. Concretely, a pathological path stacking "50 dB flare-path + 35 dB PCA + 30 dB aurora + 25 dB quiet-day" (sum 140 dB, well past the 100 dB ceiling) gets rescaled by the factor $100/140 \approx 0.714$ to "35.7 dB flare-path + 25.0 dB PCA + 21.4 dB aurora + 17.9 dB quiet-day" (sum exactly 100 dB), so the operator-meaningful relative ratios (flare-path is the dominant mechanism, then PCA, then aurora, then quiet-day) are preserved. The rescale is conservative on terms not individually near saturation: a 12 dB auroral contribution gets shaved to ∼11.4 dB when the sum is at 105 dB, but this is well under any verdict-relevant threshold (the band is dead at $\sum \geq 80$ dB regardless), so the conservatism does not shift verdicts in any operator-meaningful regime.

3.1 Free-Space Loss (Friis / ITU-R P.525)

The free-space path loss follows the standard ITU-R P.525 [4] formulation:

$$L_{\text{fs}} = 32.44 + 20\log_{10}(d_{\text{km}}) + 20\log_{10}(f_{\text{MHz}}) \tag{3}$$

The distance d_{km} is the great-circle path length, clamped to a minimum of 50 km. $\log_{10}$ is well-defined for any $d > 0$, so this is not a numerical safeguard but a modelling floor: at $d \ll 50$ km the far-field radiation pattern assumed by the Friis equation no longer applies (near-field effects, ground-wave coupling, and antenna-mutual-impedance modulation dominate the link budget instead), and a per-band constant L_{fs} is the operationally honest answer below that scale. Note that the multi-hop term L_{hop} (Eq. 18) does not share this 50 km floor: for paths with $d < d_{\text{hop}}^{\max}$, $N_{\text{extra}} = \max(0, d/d_{\text{hop}}^{\max} - 1) = 0$ identically, so the term vanishes regardless of how short d becomes. On a degenerate self-path ($d \to 0$), the budget charges the floored L_{fs} for the equivalent of a 50 km link with no multi-hop contribution; the verdict is dominated by the noise / SNR-required terms in that regime.

3.2 MUF Approach and Over-MUF Loss

Let $r = f/\text{MUF}$. The proximity loss is:

$$L_{\text{MUF}} = \begin{cases} 0 & r \le 0.70 \\ 10\left(\dfrac{r-0.70}{0.30}\right)^2 & 0.70 < r \le 1.00 \\ 10 + 36\sqrt{r-1} & r > 1.00 \end{cases} \tag{4}$$

Below 70% of MUF the ionosphere provides comfortable refraction with negligible loss. Between 70–100% the loss rises quadratically. Above MUF (over-MUF operation, e.g. via Es or scatter), the loss escalates rapidly. The piecewise formula is C^0 continuous at both joints ($r = 0.70$ and $r = 1.00$, where $L_{\text{MUF}} = 0$ and 10 dB respectively) but the derivative at $r = 1.00$ is unbounded ($\sqrt{r-1}$ has infinite slope at the joint), producing a C^1 discontinuity at MUF. This is intentional: the slope explosion encodes the physical reality that crossing MUF is an abrupt regime change (refraction → over-MUF scatter), and smoothing the joint with a finite-slope blend would underweight the rise in path variance at the boundary. Operators using the budget at $r = 1$ should expect the verdict to be discontinuity-sensitive there.

Heritage. The piecewise form (zero below $r = 0.7$, quadratic in the approach band, square-root over MUF) is a VOACAP / IONCAP-style proximity penalty rather than a literal ITU-R P.533 [1] formula: P.533's analogous treatment is embedded in the F2 propagation-loss routine (Annex 1, basic transmission-loss tabulation) and is not isolable as a closed-form $L_{\text{MUF}}(r)$. ionocast adopts the simpler closed form because the MUF input itself is the kc2g real-time observation, so the proximity loss is anchored to data rather than re-derived from P.533's monthly-median MUF prediction routine.

3.3 D-Region Absorption: Flare-Enhanced (SWPC D-RAP)

SWPC's D-RAP product provides a Highest Affected Frequency (HAF) in MHz. The flare-enhanced absorption is:[1]

$$L_{\text{DRAP}} = \begin{cases} 0 & r < 0.25 \\ \dfrac{r-0.25}{0.05} \cdot 3\,r^{1.5} & 0.25 \le r < 0.30\,, \qquad r = \text{HAF}/f \\ 3\,r^{1.5} & r \ge 0.30 \end{cases} \tag{5}$$

The $0.25 \le r < 0.30$ ramp retires a small step the bare gate produced (the formula at $r = 0.3$ evaluates to $3 \cdot 0.3^{1.5} \approx 0.49$ dB, jumping in instantly from zero); the linear ramp turns the term on smoothly and is zero by construction at $r = 0.25$. The ramp is C^0 continuous at both joints but C^1-discontinuous: the derivative jumps from 0 (lower flat branch) to $\sim 7.5\,\text{dB}/\Delta r$ at $r = 0.25$, then settles into the $r^{1.5}$ shape at $r = 0.30$. The jump is small in absolute terms (∼0.025 dB of verdict shift across a 1%-of-MUF r change) and is accepted as the cost of avoiding the much larger 0.5-dB step the bare gate produced.

[1] The label `eq:labs` and the section anchor `sec:labs` are legacy short identifiers (early drafts used `L_abs` as the in-equation variable); the actual math symbol used throughout this section is L_{DRAP}. The corresponding code function is `lAbsDb` in `src/physics/loss.js`. Reader-facing convention: L_{DRAP} everywhere in the paper math, `lAbsDb` in the code.

Relationship to the per-hop L_{flare} term. D-RAP HAF is derived by SWPC from the same GOES X-ray flux that drives the per-hop L_{flare} term in Section 3.6, so the two are combined into a single per-path flare-absorption term $L_{\text{flare,path}} = \max(L_{\text{DRAP}}, \sum_k L_{\text{flare}})$ (defined inline as Eq. 2 alongside the budget at the top of this section) rather than summed. The $\max(\cdot)$ form (rather than a binary "suppress L_{DRAP} whenever $L_{\text{flare}} > 0$") is more conservative on the leading edge of a flare: D-RAP's operational forecast model can already report 8 dB–10 dB at QTH while the per-hop L_{flare} proxy is still tracking the instantaneous X-ray class at M1 ($\sim$4 dB). L_{DRAP} remains active when no per-hop flare is firing but D-RAP still reports a non-zero HAF: typically the early rise of a small or unstable event before the X-ray classifier flags it, or the post-peak decay tail. (Quiet-day background absorption is handled by L_{Dreg} in Section 3.4, not by D-RAP.)

Geographic asymmetry of the two terms. L_{DRAP} is a single number sampled at QTH (the SWPC HAF product is spatially evaluated; ionocast queries it at the operator's location). The per-hop sum $\sum_k L_{\text{flare}}(\cos\chi_k)$ samples each F-region reflection point along the path, so on a trans-equatorial 12 000 km circuit it can pick up daylight hops the QTH itself isn't in. A short path entirely in QTH's local zone reads L_{DRAP} as the binding constraint; a long path crossing the sunlit hemisphere reads the per-hop sum. The $\max(\cdot)$ takes whichever local probe is more pessimistic on that path, so the geographic inhomogeneity is intentional: a distant flare hitting the path's middle hops should not be hidden by the QTH having moved into night, and a flare detected in the operator's HAF cell should not be discounted just because the specific reflection points sampled happen to land in twilight.

3.4 D-Region Absorption: Quiet-Day Diurnal

ionocast computes diurnal D-region absorption from a continuous formula keyed on a single anchor constant K_D:

$$L_{\text{Dreg}}(f,\chi) = L_{\text{day}}(f)\cdot\big(\max(0,\,\cos\chi)\big)^{0.7}, \qquad L_{\text{day}}(f) = \frac{K_D}{(f_{\text{MHz}}+0.5)^2}, \qquad K_D = 200\ \text{dB}\cdot\text{MHz}^2 \tag{6}$$

where χ is the solar zenith angle at the reflection point and $L_{\text{day}}(f)$ is the local-noon-overhead D-region absorption. At night ($\cos\chi < 0.05$) the term returns zero. The form is a deliberate reshape of the prior $A_{\text{base}}(f)\cdot(\cos\chi)^{1.3}$ table-and-step model (S0 #2 paired rewrite, landed 2026-05-07; harness re-fit deferred): the $1/(f+0.5)^2$ frequency dependence tracks ITU-R P.533 [1] Annex 1's non-deviative-absorption shape (a Lockwood / IONCAP-heritage $1/(f+f_L)^{1.98}$ form with $f_L \approx 1.4$ MHz) within the same ~ 2 dB overlap window the prior table claimed, while the gentler $\cos^{0.7}$ exponent (versus $\cos^{1.3}$) charges more absorption at moderate zenith angles ($\cos\chi \in [0.3, 0.7]$, where $\cos^{0.7}$ runs $\sim$0.4 to $\sim$0.8 versus $\cos^{1.3}$'s $\sim$0.2 to $\sim$0.6) and matches the multi-hop midday DX residuals the prior form under-predicted on. The ITU-R P.533 Annex 1 cosine form is itself a $\cos^p(0.881\chi)$ with month- and dip-latitude-varying p; the closed-form $\cos^{0.7}$ used here is operationally a midlatitude-averaged simplification, used because the budget applies its own per-band σ_g widening on top (§7.3) rather than passing latitude / seasonality dependence into the absorption term itself.

The constant $K_D = 200$ was chosen so $L_{\text{day}}(1.838\text{ MHz}) = 36.6$ dB, a value bracketed by the prior table's operator-experience anchor (28 dB, K9LA / ARRL-derived) and the literal P.533 /1.6 oblique-incidence target (45.7 dB divided by 1.6 to convert vertical-incidence to oblique). Illustrative noon values at the canonical band centres, $L_{\text{day}}(f)$ from Eq. 6, replace the prior Table 2 anchors:

Band	f (MHz)	**Band**	f (MHz)
160 m	1.838 $\rightarrow$ 36.6 dB	20 m	14.097 $\rightarrow$ 0.94 dB
80 m	3.570 $\rightarrow$ 12.1 dB	17 m	18.106 $\rightarrow$ 0.58 dB
60 m	5.366 $\rightarrow$ 5.80 dB	15 m	21.096 $\rightarrow$ 0.43 dB
40 m	7.040 $\rightarrow$ 3.52 dB	12 m	24.924 $\rightarrow$ 0.31 dB
30 m	10.140 $\rightarrow$ 1.76 dB	10 m	28.126 $\rightarrow$ 0.24 dB

The values shift versus the prior K9LA-anchored table: 160 m noon climbs (28 $\rightarrow$ 36.6 dB, +8.6 dB), 80 m noon drops (18 $\rightarrow$ 12.1 dB, −5.9 dB); upper bands shift smaller ($\leq$ 1 dB on each). An offline calibration sweep against the 30-day WSPR basket (§7.2) is the follow-up action item; the constants above land the budget at the literature-anchored shape and let the harness retune σ_g / L_{iono} around it.

For a multi-hop path, absorption is *summed* across all reflection points rather than evaluated at the path midpoint only: for an N-hop path with reflections at great-circle fractions $(2k-1)/(2N)$,

$$L_{\text{absD,path}} = \min\left(L_{\text{cap}},\ \sum_{k=1}^{N} L_{\text{Dreg}}(f, \chi_k)\right) \tag{7}$$

with $L_{\text{cap}} = 50\,\text{dB}$ guarding against degenerate values when every hop is near local noon on a low band. A transequatorial path whose midpoint is dark but whose northern and southern hops are in daylight therefore sees the two daylight hops' absorption, not zero; conversely a path whose midpoint sits over the terminator but whose hops are deep in night loses no contribution from the single sunlit fraction nearest the midpoint. The grayline bonus of §5.5 pairs with this term: $L_{\text{day}}(f) - L_{\text{Dreg}}(f, \chi)$ is the zenith-dependent absorption refund the grayline mechanism captures, additive on top of the day-side L_{Dreg} charge.

3.5 Polar Cap Absorption (Sauer–Wilkinson)

During solar energetic particle (SEP) events, protons accelerated by a CME or flare precipitate into the polar cap and ionise the D-region, producing severe HF absorption on polar paths for hours to days [21]. ionocast activates a polar cap absorption (PCA) term when two gates are simultaneously satisfied:

1. the GOES integral $\geq 10\,\text{MeV}$ proton flux exceeds the NOAA S1 threshold, $\Phi_{p10} \geq 10\,\text{pfu}$;
2. the reflection point sits inside the polar cap, $|\phi_{\text{CGM}}| \geq 60°$.

The per-hop magnitude uses a log-flux scaling (a calibrated deviation from the canonical Sauer–Wilkinson $\sqrt{\Phi}$ form [21], chosen because the saturating-recombination behaviour at the highest fluxes is gentler than $\sqrt{\Phi}$ predicts). Empirical evidence for the deviation comes from amateur-band PCA observations catalogued by K9LA [15] during cycle-24 SEPs above 1000 pfu, which report 5–10 dB of remaining 14 MHz path on polar circuits where the canonical $\sqrt{\Phi}$ form would predict $> 30\,\text{dB}$. The Sauer–Wilkinson 2008 derivation is itself anchored in the 30 MHz riometer record at fluxes spanning the 10–10^4 pfu range (their Figures 4–5b, validating against 11 large SEP events 1992–2002 at Thule, Greenland), but the paper does not specify a saturation regime above any particular flux threshold; ionocast's log-flux substitution is a downstream calibration choice informed by the K9LA observational record rather than by Sauer & Wilkinson directly. The same $f^{-1.5}$ frequency dependence as ordinary D-region absorption is retained:

$$L_{\text{PCA}}(f, \Phi_{p10}) = \min\left(30\,\text{dB},\ 5\log_{10}\frac{\Phi_{p10}}{10} \cdot \left(\frac{7}{f_{\text{MHz}}}\right)^{1.5}\right) \tag{8}$$

The inner $\min(30\,\text{dB}, \cdot)$ in Eq. 8 caps *each* per-hop contribution at 30 dB; per-hop contributions are then summed along the great circle (the same hop grid used by L_{Dreg}) and the resulting path total is capped again at 50 dB. Aurora (L_{aur}) uses the same 30 dB per-hop cap (named in code as `AUR_PER_HOP_CAP_DB`, mirroring `PCA_PER_HOP_CAP_DB`); flare-driven SID (L_{flare}) uses a slightly higher 40 dB per-hop cap because the strongest X-class flares produce essentially total HF blackout on sunlit hops where the riometer record *has* read the 30–40 dB regime (e.g., the X28+ Halloween 2003 events), whereas PCA and aurora rarely exceed 30 dB for sustained periods even at S5 / G5 levels. All three share the same path-total 50 dB cap (a single hop in the budget cannot dominate the path beyond the aggregate ionospheric absorption ceiling). Midlatitude paths see $L_{\text{PCA}} = 0$ even during a severe SEP because no hop traverses the polar cap; polar paths see order-of-magnitude absorption within minutes of proton-event onset.

Multi-channel onset detection. The GOES integral ≥ 1 MeV proton channel rises $\sim$1 h before ≥ 10 MeV during a hard SEP event because lower-energy protons arrive first along the connecting magnetic field line. When $\Phi_{p1} \geq 10$ pfu, an early-warning per-hop bump is applied on polar hops:

$$L_{\text{PCA,onset}}(f, \Phi_{p1}) = \min\left(L_{\text{onset,cap}},\ \alpha_{\text{onset}} \cdot \log_{10}\frac{\Phi_{p1}}{10} \cdot \left(\frac{7}{f_{\text{MHz}}}\right)^{1.5}\right) \tag{9}$$

with $\alpha_{\text{onset}} = 1$ dB/decade (vs. 5 dB/decade in the main term, since Φ_{p1} tracks arrival of the SEP front rather than the absorbing population) and $L_{\text{onset,cap}} = 5$ dB/hop.

Smooth handoff. $L_{\text{PCA,onset}}$ stays active past the S1 threshold and the path sum uses $\max(L_{\text{PCA}}, L_{\text{PCA,onset}})$ per hop, so the harder channel takes over smoothly once its absorption exceeds the soft channel's. This avoids the 5 dB cliff a hard-gate handoff would produce at the threshold instant. The ≥ 100 MeV channel is fetched alongside the other two via the same SWPC integral-protons proxy (the JSON response includes all integral channels in a single call) and cached in `src/data/fetchers.js` as `p100`, but it is not currently consumed by the physics layer. It surfaces only in diagnostics for now; future work will use it to scale the per-hop cap when the SEP penetrates deep into the D-region (the deepest channels reach the lower D-region where the absorption cap is itself frequency-dependent).

3.6 Flare-Driven Sudden Ionospheric Disturbance

Solar soft X-ray photons from an in-progress flare enhance D-region electron density on the sunlit side of the Earth within seconds of the flare peak [23]. ionocast reads the current SWPC `xrayClass` (updated once per minute, well under the model's 10-minute refresh) and applies an additive absorption term on sunlit hops, with a smooth twilight ramp $r(\cos\chi_k)$ across the terminator:

$$L_{\text{flare}}(f, X, \cos\chi_k) = \min\left(40\,\text{dB},\ D(X) \cdot \left(\frac{7}{f_{\text{MHz}}}\right)^{1.5}\right) \cdot r(\cos\chi_k) \tag{10}$$

$$r(\cos\chi_k) = \begin{cases} 0 & \cos\chi_k \leq 0 \\ \cos\chi_k \,/\, 0.05 & 0 < \cos\chi_k < 0.05 \\ 1 & \cos\chi_k \geq 0.05 \end{cases} \tag{11}$$

The ramp $r(\cos\chi_k)$ scales the daytime value linearly across a $\sim$3° solar-elevation band ($\cos\chi_k \in [0, 0.05]$ corresponds to elevation $\in [0°, 2.87°]$). The wall-clock duration of that 3° band depends on latitude: the elevation rate at the horizon is $\sim$15°/h at the equator (90° sun arc over 6 h), $\sim$10°/h at midlatitudes, $\sim$3°/h at $|\phi| = 60°$ in winter, and $\to 0$ at the polar circles around solstice — so the twilight ramp spans about 12 min at the equator, 17 min at midlat, an hour at high winter latitude, and (asymptotically) the entire civil-twilight period inside the polar circles. The ramp matches the gentle climb of D-region photoionisation through twilight at any latitude rather than snapping on at the geometric terminator.

The log-flux driver is in units of dB at 7 MHz, calibrated so M1 produces 4 dB, X1 produces 12 dB, and X10 saturates at 20 dB:

$$D(X) = \max\Big(0,\ \min\big(20\,\mathrm{dB},\ 4\,\mathrm{dB} + 8\,\mathrm{dB} \cdot \log_{10} \tfrac{X}{\mathrm{M1}}\big)\Big) \tag{12}$$

where $X/\mathrm{M1}$ is the X-ray flux ratio to the M1 baseline ($10^{-5}\,\mathrm{W/m^2}$).

Calibration source. The anchors (4 dB at M1, 12 dB at X1, 20 dB at X10) follow the classical SID-absorption framework of Mitra [23] (*Ionospheric Effects of Solar Flares*, the canonical print-only treatment of D-region absorption versus soft X-ray peak flux), with the slope set to 8 dB/decade between M1 and X10 so the anchor points fit the per-decade ∼8 dB scale that the riometer-record literature consistently reproduces. The specific M1=4 / X1=12 / X10=20 dB values are not directly cross-checkable against an open digital source; the WSPR-loss validation below is what anchors them in ionocast. The values were cross-checked against WSPR loss patterns observed during cycle-25 M-class and X-class events (the wspr-live archive of 2024–2026 flare hours; see §7.2): on 20 m and 17 m WSPR spot drops of 5–12 dB versus the prior quiet-day median tracked the predicted $D(X) \cdot (7/f)^{1.5}$ envelope to within ∼2 dB across M1 through X4. (No cycle-25 X10 events have occurred at the time of writing, so the X10 anchor is set by the literature extrapolation alone, hence the 20 dB hard saturation in the equation rather than an open-ended slope.)

The outer $\max(0,\cdot)$ wraps the saturated $\min(20,\cdot)$ branch and suppresses contributions whose underlying log-driver crosses zero, which happens at $X \approx 10^{-5.5}\,\mathrm{W/m^2}$ (approximately C3.16); this is not an explicit "if X < C3 return 0" gate but the practical effect is that the quietest C-class events (C1–C3) contribute ∼0 dB at the 7 MHz reference, while higher C-class flares ramp up the contribution as the algebra allows: $D(\mathrm{C5}) \approx 1.6\,\mathrm{dB}$, $D(\mathrm{C9}) \approx 3.6\,\mathrm{dB}$, $D(\mathrm{M1}) = 4\,\mathrm{dB}$ by calibration construction (these are at the 7 MHz reference; the band-scaling factor $(7/f)^{1.5}$ raises them on lower bands so a C5 event still contributes a few dB on 80 m / 160 m). The clamp removes only the lowest of the C-class background. As with PCA, sunlit-hop contributions are summed and the path total is capped (50 dB). The instantaneous-proxy approximation omits the roughly 10–60 minute exponential decay of the SID, which is acceptable given the model's 10-minute cadence; the next refresh picks up the decayed class.

Two flare thresholds, two distinct roles. The physics zero-crossing at C3.16 (this section) suppresses the quiet-day background contribution but is otherwise continuous; the alert-banner threshold at M1 / M5 / X (Table 12) is escalation-grade signaling for the operator UI. The two thresholds are not redundant – one is the algebraic point where the physics term goes to zero, the other is the operator-facing "this is worth surfacing top-of-page" line. Quiet-time C-class flares contribute sub-dB absorption to the budget without firing a banner.

3.7 Auroral Absorption (CGM-Gated)

Auroral absorption activates when the corrected geomagnetic (CGM) latitude $|\phi_{\mathrm{CGM}}|$ at a reflection point is at or above the storm-dependent threshold. In one line, the per-hop absorption is $L_{\mathrm{aur}} = \min(30, D \cdot (30/f_{\mathrm{MHz}}) \cdot c(\phi_{\mathrm{CGM}}))$ where D is the storm driver (Eq. 13's outer term, derived below from K_p^{eff} and OVATION HP), $30/f_{\mathrm{MHz}}$ is the f^{-1} band scaling, and $c(\phi_{\mathrm{CGM}})$ is the CGM-edge ramp (Eq. 15). Per-hop cap 30 dB (`AUR_PER_HOP_CAP_DB`); path cap 50 dB (`PATH_AUR_CAP_DB`); the rest of this section unpacks each piece.

$$L_{\mathrm{aur}} = \min\left(30,\ D \cdot \frac{30}{f_{\mathrm{MHz}}} \cdot c(\phi_{\mathrm{CGM}})\right) \tag{13}$$

where

$$D = \max\Big(\max(0,\, 5(K_p^{\text{eff}} - 4)),\ \max(0,\, \tfrac{\text{HP}_{\text{GW}}-50}{5})\Big) \tag{14}$$

ramps continuously: the K_p^{eff} term begins contributing at $K_p^{\text{eff}} > 4$ and reaches $D = 5$ at $K_p^{\text{eff}} = 5$ (the G1 storm threshold), which maps to 5 dB of L_{aur} at the 30 MHz reference frequency f in Eq. 13 but to substantially more at HF: the $30/f_{\text{MHz}}$ band-scaling factor turns $D = 5$ into ∼21 dB on 40 m ($5 \cdot 30/7$) and ∼11 dB on 20 m ($5 \cdot 30/14.1$), before the per-hop 30 dB cap and the $c(\phi)$ CGM ramp. The f^{-1} band scaling here is shallower than the $f^{-1.5}$ used for quiet-day D-region (§3.4), PCA (§3.5), and flare (§3.6). This is empirical rather than purely theoretical: auroral precipitation has a broad energy-and-altitude footprint (electrons across 1–100 keV deposit across both D and lower E regions, integrating multiple collision-frequency regimes), and the high-latitude HF riometer record across 7–30 MHz fits a slope closer to f^{-1} than to $f^{-1.5}$. PCA and flare absorption, by contrast, are dominated by D-region deposition at one relatively narrow altitude band (60–80 km), where the collisional $f^{-1.5}$ form is the closer match. K_p^{eff} is the storm-lagged effective index (§5.3.1), built from the 3-hour planetary K_p history augmented by D_{st} and IMF B_z early-warning bumps; references to bare K_p in this section refer to the underlying 3-hour SWPC sensor that feeds K_p^{eff}. The OVATION hemispheric-power term contributes $\max(0, (\text{HP} - 50)/5)$ above the 50 GW floor: $D = 0$ for HP $\le$ 50 GW (the formula is wrapped in $\max(0, \cdot)$, so quiet-aurora HP values below the floor return zero rather than a negative driver), $D = 1$ at HP = 55 GW, $D = 5$ at HP = 75 GW (same D-scaling as the K_p^{eff} term). The two drivers are combined via $\max(\cdot)$ rather than additively because they are not independent — both are forced by the same auroral-electrojet activity, and adding them would double-count the underlying ionisation; but they are kept as parallel sensors because they sample the same phenomenon on different timescales. The 3-hour K_p that feeds K_p^{eff} is reported every 3 h, so a sub-hour auroral intensification driven by a fresh substorm onset is invisible to that channel for up to 90 minutes unless one of the D_{st} / B_z early-warning bumps fires; OVATION HP is updated every ∼30 min and reflects the actual hemispheric power dissipated in the auroral oval over the most recent half hour. The $\max(\cdot)$ takes whichever sensor is reading hotter at any given moment, so a fast-onset substorm shows up via HP and a multi-hour storm in steady state shows up via K_p^{eff} — catching both regimes without double-counting.

The continuous ramp lets the prediction reflect modest auroral absorption already at active-storm ($K_p^{\text{eff}} \in [4, 5)$) levels rather than waiting for a hard storm threshold; a hard $K_p^{\text{eff}} \ge 5$ / HP $\ge$ 50 GW gate would produce a 21 dB step in L_{aur} on 40 m polar paths at the threshold instant, the same family of cliff the $c(\phi)$ ramp below avoids on the CGM edge. $c(\phi_{\text{CGM}})$ is a smooth ramp on the CGM-edge:

$$c(\phi) = \begin{cases} 0 & |\phi| \le \phi_{\text{thr}} - 5° \\ \dfrac{|\phi| - (\phi_{\text{thr}} - 5°)}{5°} & \phi_{\text{thr}} - 5° < |\phi| < \phi_{\text{thr}} \\ 1 & |\phi| \ge \phi_{\text{thr}} \end{cases} \tag{15}$$

Linear ramp over the trailing 5°, full effect poleward of the threshold. Eliminates the hard cliff that the bare gate produced.

Storm-zone expansion. The CGM threshold ramps linearly from 60° to 50° as K_p^{eff} climbs from 5 to 7, then holds at 50° above $K_p^{\text{eff}} = 7$:

$$\phi_{\text{thr}}(K_p^{\text{eff}}) = 60° - 10° \cdot \min\Big(1,\, \max(0,\, (K_p^{\text{eff}} - 5)/2)\Big) \tag{16}$$

This captures the equatorward expansion of the auroral oval during storms. Both sides of the auroral term (geometric extent and absorption depth) respond on the same thermospheric timescale, so a single K_p^{eff} argument enters both this expansion and the magnitude D in Eq. 13's outer term. The linear ramp from 60° to 50° over the $K_p^{\mathrm{eff}} \in [5, 7]$ window lets the expansion start gradually during moderate storms (G2 / G3) rather than snapping at $K_p^{\mathrm{eff}} = 7$, which would otherwise produce a 30+ dB jump on a hop near 55° CGM at the threshold instant. The expansion is wired into the absorption term itself; no external runtime gating is needed.

Per-hop evaluation. On a multi-hop path, L_{aur} is computed per reflection point and summed (capped at 30 dB per hop via `AUR_PER_HOP_CAP_DB` and 50 dB total path via `PATH_AUR_CAP_DB`) by the same machinery as the per-hop D-region sum (Eq. 7). A 4-hop transatlantic path with the midpoint at 55 ° CGM but two of its hops at 65 ° correctly accrues auroral absorption on the polar hops; a midpoint-only gate would miss it. *Legacy fallback:* when the budget is invoked without source / destination geometry (callers that supply only `midLat`, `cgmLatAbsValue`, no src/dst pair — e.g. smoke tests, the local-MUF fallback branch in `src/derive/conditions.js` with no kc2g paths), the auroral term degrades to a single-midpoint evaluation against the supplied `cgmLatAbsValue` without per-hop summation. The per-hop branch is the production path; the midpoint fallback is documented here so a reader auditing the smoke-test budgets does not mistake it for a regression.

Storm-main amplification (forward reference). Eq. 13 states the steady-state L_{aur}. During the *main* phase of a geomagnetic storm (Dst ≤ -50 nT with K_p^{eff} still loading), +4 dB is added to the per-hop L_{aur} (then re-clamped against the 30 dB per-hop cap from `AUR_PER_HOP_CAP_DB`) to capture the extra particle-precipitation absorption that the steady-state K_p-keyed form undercounts; recovery and quiet phases keep the steady-state value. The amplification is detailed together with the storm-phase classifier in Section 7.3 (paragraph "Storm-phase auroral amplification"), since both the classifier and the $\sigma_{\mathrm{recovery}}$ widening it gates are defined there. A reader computing the budget from this section alone should add the +4 dB bump when the phase classifier reports *main* and re-apply the per-hop cap.

The CGM latitude is computed via a tilted-dipole approximation:

$$\phi_{\mathrm{CGM}} = \arcsin\Big(\operatorname{clamp}\big(\sin\phi\sin\phi_P + \cos\phi\cos\phi_P\cos(\lambda - \lambda_P),\ -1,\ +1\big)\Big) \tag{17}$$

The explicit clamp guards against floating-point overshoot of the arcsin argument outside $[-1, +1]$ near the geomagnetic pole itself, where $\sin\phi\sin\phi_P + \cos\phi\cos\phi_P\cos(\Delta\lambda)$ can drift by $\sim 10^{-15}$ above 1 from rounding error and produce a NaN ϕ_{CGM} that would propagate through the auroral CGM gate. with the IGRF 2020 [27] north magnetic pole at $\phi_P = +80.7°$, $\lambda_P = -72.7°$ (signed-east-of-Greenwich convention; held fixed for reproducibility; IGRF-14 was published with the 2025.0 epoch but the centred-dipole pole has drifted sub-degree since 2020, far below the ϕ_{CGM} ramp's 5° smoothing width in Eq. 15 and well below any verdict-shifting threshold. Centred-dipole drift is $\sim 0.05°$ / yr, so the held-fixed values remain inside the smoothing width through $\sim$ 2030; review against IGRF-15 at the 2030.0 epoch). The operator's QTH longitude is whatever signed value the geolocation returns (e.g. −9.14° for Lisbon, +139.69° for Tokyo). Holding the IGRF pole fixed while accepting $F_{10.7A}$ live looks asymmetric but reflects the underlying timescales: $F_{10.7A}$ (the 81-day mean of $F_{10.7}$) varies on days to weeks as the centre of the trailing window slides, so consuming the live value is correct; the IGRF centred-dipole pole drifts $\sim 0.05°$/yr (decadal), so freezing it at the 2020 epoch produces sub-degree error well inside the ϕ_{CGM} ramp's 5° smoothing width and keeps the bibliography pinned to a citable model (Alken et al. [27]) rather than tracking a continuously-updating model file.

3.8 Multi-Hop Ground Reflection Loss

Each extra F-region hop adds a ground-reflection plus defocusing penalty applied at the intermediate ground bounces. To avoid a discrete cliff at the hop-ceiling boundary, the extra-hop count is computed as a smooth function of distance:

$$L_{\text{hop}} = N_{\text{extra}}(d) \cdot \big(L_{\text{gr}}(f,\theta) + D\big) \quad \text{dB}, \qquad N_{\text{extra}}(d) = \max\left(0,\ \frac{d_{\text{km}}}{d^{\max}_{\text{hop}}(h_F)} - 1\right) \tag{18}$$

where the per-hop ground-range ceiling scales as $\sqrt{h_F}$, the leading-order spherical-Earth result for glancing-incidence reflection at altitude h_F. At zero elevation the geocentric angle to the reflection-point ground projection is $\gamma = \arccos\big(R_\oplus/(R_\oplus + h_F)\big)$, giving $d \approx 2\sqrt{2\,R_\oplus h_F}$ for $h_F \ll R_\oplus$. ionocast preserves the empirical 4000 km anchor at the quiet-cycle $h_F = 300$ km and lets the rest follow the square-root shape:

$$d^{\max}_{\text{hop}}(h_F) = 4000\,\text{km} \cdot \sqrt{\frac{h_F}{300\,\text{km}}} \tag{19}$$

At $h_F = 300$ km this reduces to 4000 km. On enhanced-solar-flux days h_{mF2} climbs to 340 km or above and the ceiling moves to ∼4258 km. ionocast uses an observed h_{mF2} from the GIRO digisonde panel when available; otherwise defaults to 300 km.

The per-hop ionospheric loss bundle (Section 3.4) is also continuous across the hop boundary, but in a different way than the multiplier here. At a non-integer $n_C = d/d^{\max}_{\text{hop}}$, the bundle computes the four ionospheric absorption sums (D-region, PCA, flare, auroral) for both bracketing integer hop layouts $\lfloor n_C \rfloor$ and $\lceil n_C \rceil$ — each laying down a different set of reflection-point coordinates — and blends the totals by the fractional part $\alpha = n_C - \lfloor n_C \rfloor$: $L = (1-\alpha)\cdot L_{\lfloor n_C \rfloor} + \alpha \cdot L_{\lceil n_C \rceil}$. At integer n_C, $\alpha = 0$ and the ceil branch is skipped. This keeps both budgets continuous across hop-count boundaries: a 4001 km path on 80 m at noon would otherwise jump from ∼18 dB of D-region absorption (single midpoint) to ∼36 dB (two reflection points) at the boundary, while the ground-reflection multiplier L_{hop} barely changes (N_{extra} ramping from 0 to 0.0003).

$L_{\text{gr}}(f,\theta)$ is the per-bounce Fresnel reflection loss and $D = 0.25$ dB is the per-extra-hop defocusing constant (earth-curvature spreading; ITU-R P.533 Annex 1 prescribes ∼1 dB as a population mean, but that figure mixes ground types and elevations and overshoots typical amateur multi-hop paths over "average earth" as established by 30 d WSPR calibration).

Fresnel ground-reflection loss. At grazing angle $\psi = \theta_{\text{elev}}$ (for a symmetric hop, takeoff equals grazing at the bounce), with complex relative permittivity

$$\varepsilon_c = \varepsilon_r - j\,60\,\sigma\,\lambda \quad (\lambda \text{ in metres}) \tag{20}$$

[2] and "average earth" constants ($\varepsilon_r = 13$, $\sigma = 0.005$ S/m, where σ here denotes ground conductivity, not the per-band prediction spread of Table 9) following the ARRL Antenna Book [13] amateur-radio convention. ITU-R P.527-6 [7] Table 1 lists slightly different canonical values for medium-dry ground ($\varepsilon_r \approx 15$, $\sigma \approx 0.01$ S/m); at HF takeoff angles the resulting L_{gr} disagreement is sub-decibel, so the amateur-convention pair is retained for continuity with the antenna-modelling literature operators are most likely to cross-check against. The horizontal-polarisation Fresnel coefficient is

$$R_h = \frac{\sin\psi - \sqrt{\varepsilon_c - \cos^2\psi}}{\sin\psi + \sqrt{\varepsilon_c - \cos^2\psi}}. \tag{21}$$

The implementation takes the H-pol power reflectance

$$L_{\text{gr}}(f,\psi) = -10\log_{10}|R_h|^2, \qquad L_{\text{gr}} \in [0, 8] \text{ dB}. \tag{22}$$

[2] The minus sign on the imaginary term assumes the $e^{-j\omega t}$ time-harmonic convention used by the `lHopGroundReflectionDb` implementation in `src/physics/loss.js`. The opposite convention ($e^{+j\omega t}$) flips the sign of the imaginary term but not of $|R_h|^2$, so the loss in dB is unchanged either way. Code uses `eIm = -60 * GROUND_AVG_SIGMA * lambdaM`, matching the form printed here.

Most amateur HF antennas are H-polarised (dipoles, inverted-V, OCF, EFHW, Yagi/quad arrays), and 30 d WSPR calibration confirms H-pol as the right per-bounce model: H-pol Fresnel scored Brier 0.07 (89% accuracy) versus a polarisation-averaged $\frac{1}{2}(|R_h|^2 + |R_v|^2)$ alternative at 0.13 (83%) and V-pol at 0.27 (66%). H-pol Fresnel predicts <1 dB per bounce at typical 5–15° grazing angles, which is what the multi-hop budget needs to see for a 3–5 hop path to predict "open" when the band is in fact open.

The $[0, 8]$ dB clamp brackets the H-pol Fresnel loss between physically-meaningful bounds. The lower bound 0 dB is the perfect-grazing limit ($\psi \to 0$) where any real-earth reflectance approaches unity; in practice the un-clamped formula returns sub-decibel values at typical 5–30° HF grazing angles, so the lower bound is rarely approached and is a guard against numerical edge cases rather than a binding floor. The upper bound 8 dB guards against the single-layer earth model's behaviour at high (near-normal) incidence angles. Unlike V-pol, H-pol has no Brewster zero; $|R_h|^2$ is monotonic with grazing angle, peaking in loss at normal incidence ($\psi = 90°$) and decreasing as the geometry goes more grazing. For "average earth" ($\varepsilon_r = 13$, $\sigma = 0.005$ S/m) on 14 MHz the normal-incidence loss is ∼4.7 dB, comfortably under the clamp. The cap only binds for very dry / poor ground ($\varepsilon_r \approx 4$), where the single-layer model can predict normal-incidence loss above 8 dB; a real earth at those parameters is layered (ground-water profile, soil density gradient) and the effective reflectance does not collapse that hard, so the 8 dB cap is the empirical ceiling that keeps the budget aligned with WSPR observations on those short-haul paths. The clamp is rarely binding in practice: typical amateur HF takeoff angles sit at 5–30° grazing (ψ measured from the surface), where the unclamped Fresnel form itself returns sub-3 dB losses regardless of ground parameters.

Typical magnitudes. At a representative ∼ 11144 km Lisbon–Tokyo path, hop geometry under the saturating $d_{\text{hop}} = \min(d, d_{\text{hop}}^{\max})$ form (Section 3.14, Eq. 32) yields $\theta = \arctan(2 \cdot 300/4000) = 8.5°$, and per-bounce H-pol Fresnel loss at that angle on 14 MHz with $\varepsilon_r = 13$, $\sigma = 0.005$ S/m gives $L_{\text{gr}} \approx 0.66$ dB. Eq. 18's smooth $N_{\text{extra}}(d) = d/d_{\text{hop}}^{\max} - 1$ at $d_{\text{hop}}^{\max} = 4000$ km gives $N_{\text{extra}} = 1.79$, so $L_{\text{hop}} = 1.79 \cdot (0.66 + 0.25) \approx 1.63$ dB on the path. A 5-hop trans-equatorial path approximately doubles it. Single-hop NVIS and short-haul paths see $L_{\text{hop}} = 0$ (no intermediate bounces).

Calibration coverage. The Fresnel piece is fully physics-derived; only the defocusing scalar D remains tunable, and is exercised by the harness over a multi-path basket spanning 1–5 hops continuously (single-hop NVIS through 5-hop trans-equatorial; see Appendix C's geometry/use-case column for the per-path hop counts; Section 7.2).

Path-type bias (deferred). Sea-vs-land per-path classification is deferred — "average earth" ($\varepsilon_r = 13$, $\sigma = 0.005$ S/m) applies everywhere until coastline-aware path geometry is added. The direction and magnitude of the bias is known and bounded:

- Seawater ($\sigma \approx 5$ S/m, three orders of magnitude more conductive) produces ∼0.3 dB H-pol Fresnel loss per bounce versus ∼0.66 dB on average earth, so *transatlantic and transpacific paths are under-predicted by ∼1–3 dB depending on hop count* (1 dB on a 2-hop circuit, ∼3 dB on a 5-hop trans-oceanic).
- Dry sand or arid soil ($\sigma \approx 0.001$ S/m, five times less conductive) produces an extra ∼0.5 dB H-pol loss per bounce, so *paths over the Sahara, Australian desert, or central Asia are over-predicted by ∼1–2 dB on a 3–5 hop budget.*

Both biases are operator-relevant on close-margin verdicts where the budget reads near a tier boundary but small in the open-band regimes where the verdict is dominated by the F2 / MUF / absorption terms. Until coastline-aware geometry lands, the UI shows the average-earth verdict; expect transatlantic / transpacific paths to perform somewhat better than the verdict suggests, and desert-dominated paths somewhat worse.

3.9 Sporadic-E Screening Loss

When a strong Es layer is present ($f_{oEs} \geq 5\,\text{MHz}$) and the operating frequency is below twice the Es critical frequency ($f < 2\,f_{oEs}$), the layer partially screens the F2 reflection:

$$L_{Es} = 5\,\text{dB} \tag{23}$$

The flat 5 dB value is a calibration constant rather than a derived quantity — the actual screening loss has a graded f-vs-f_{oEs} dependence (deeper screening as f approaches f_{oEs}, vanishing at $f = 2\,f_{oEs}$ where the Es layer becomes transparent), but the operationally relevant case is the binary "F2 hop is partially blocked by this Es layer" decision, and a single mid-range value matches the WSPR-validated bias on screened paths better than a parametrised form would given the foEs measurement uncertainty. L_{Es} is a screening penalty applied *only inside the F2 budget* (Eq. 1). When the parallel Es-mode budget of Section 6.1 wins the per-band margin competition, that path uses its own loss bundle (Eq. 64) and L_{Es} is not part of its calculation.

3.10 Low-Band Extra Loss

A band-stepped term applies an empirical residual on the lower HF bands (Table 2). L_{low} is added *on top of* the per-hop quiet-day D-region absorption L_{Dreg} (already 36.6 dB on 160 m at local noon, etc., per Eq. 6), not as a substitute for it. The mechanisms it represents are propagation residuals that L_{Dreg}'s mean-conditions formulation does not capture:

- Higher-angle refraction at low frequencies, which forces steeper takeoff angles and longer in-ionosphere ray paths than the geometric hop calculation assumes.
- Magneto-ionic mode coupling on F2 reflection (E-trap and ordinary/extraordinary splitting), which is stronger on low bands and is not modelled mechanistically in the current Fresnel/multi-hop bundle (Section 3.8).
- Ground-wave contamination on short-haul paths, where the sky-wave is in destructive interference with a residual ground-wave component that the F2-only budget does not account for.

The values in Table 2 are hand-set integers chosen by inspection of the harness's per-band Brier residual on the low bands against 30 d of WSPR observations (§7.2); L_{low} is *not* part of the joint coordinate-descent set in Appendix E ($\{L_{iono}, D, w_{sc}, w_{\text{NVIS-tail}}, w_{\text{Es-prim}}, \sigma\text{-scale}, \text{fusion-flag}\}$), and the round-integer values (8, 5, 3, 2 dB on 160 m through 40 m) are the visible signature of that; 30 m carries a half-step taper at 0.5 dB to give the log-frequency interpolation a non-zero anchor without re-introducing a step at the 30/20 m cutoff. Bounds are guided by ITU-R P.533 [1] Annex 1's median-conditions correction term Y_p which the current budget otherwise omits (Section 3.11 discusses why Y_p is not included). A future calibration pass should fold L_{low} into the descent set once the per-pair WSPR ground-truth wiring stabilises; until then, the table is empirical-judgment rather than machine-fitted. The values are path-independent — a per-distance refinement is deferred until enough per-pair WSPR data accumulates to fit a distance-dependent ramp without inflating noise.

Table 2: Low-band extra loss term applied below 30 m. Empirical residual on top of L_{Dreg}, calibrated against 30 d WSPR; see §3.10 for the mechanism breakdown. Stored as anchor points for a smooth log-frequency interpolation (refactored 2026-04-30 from a step-function form that had hard cutoffs at 2 MHz, 4 MHz, 6 MHz, 8 MHz); intermediate frequencies linearly interpolate between the band-centre anchors in $\log f$ space. The 14 MHz anchor holds the value at zero so the contribution tapers smoothly through 30 m and is exactly zero at 20 m and above.

Band	f (MHz)	L_{low} (dB)
160 m	1.838	8.0
80 m	3.570	5.0
60 m	5.366	3.0
40 m	7.040	2.0
30 m	10.140	0.5
20 m +	14.000+	0.0

The 30 m cutoff is empirical: at and above 30 m the L_{Dreg} baseline (Eq. 6) plus the modeled F2-region terms account for the band's quiet-day budget within the harness's residual resolution, so L_{low} adds nothing. Below 30 m the residuals were systematic enough to warrant the empirical correction.

3.11 Lumped Ionospheric Loss

$$L_{\mathrm{iono,path}} = n_{\mathrm{hops}} \cdot L_{\mathrm{iono}}, \qquad L_{\mathrm{iono}} = 1\,\mathrm{dB} \tag{24}$$

This is the ITU-R P.533 median-conditions residual once the individually-modelled mechanisms are subtracted: spatial focusing/defocusing L_z is now handled per intermediate hop by D (Section 3.8), polarisation coupling is now handled inside the Fresnel R_h branch of the multi-hop reflection model, and the only term that remains a-priori invisible to the geometry is the $\sim$1 dB Faraday-rotation / small-correction residual *per ionospheric reflection.* Per ITU-R P.533 §A.2 the lumped correction accumulates per reflection, so the path total scales with hop count: a 1-hop NVIS path gets 1 dB, a 3-hop transcontinental path gets 3 dB, etc. The 1-dB magnitude is calibrated against the production 35-path basket (Appendix C), which spans single-hop NVIS through 5-hop trans-equatorial geometries; the per-hop multiplier closes the residual on multi-hop DX paths where a single-charge convention would under-count by 2–4 dB. L_{iono} does *not* include D-region absorption (handled per-hop by Eq. 7) or ground reflection (Section 3.8), nor does it include the above-median-reliability Y_p margin ($\approx$7 dB at 90% reliability) defined by ITU-R P.842 [8] (the recommendation that prescribes the circuit-reliability framework around the per-band σ fed by P.533 [1]): Y_p is the correct term when the prediction answers "will the circuit work 90% of the time at this hour?", but ionocast answers "is the band open right now?", which is a median-conditions question. The 1 dB value is what remains after each mechanism formerly bundled into the lumped term (Y_p above-median margin, multi-hop bounce loss, polarisation coupling) was relocated to its own physics-derived per-hop expression and the residual recalibrated against 30 d of WSPR truth.

3.12 Noise Model (ITU-R P.372)

Following ITU-R P.372 [2], the received noise power is the power sum of the natural floor (galactic plus atmospheric) and a man-made contribution that fires only on suburban and urban sites. We write power-sum addition with the symbol $\oplus$ defined by $A \oplus B \equiv 10\log_{10}(10^{A/10} + 10^{B/10})$ throughout this section, with the indicator-gated form spelled out in full as the noise budget:

$$N = 10\log_{10}\Big(10^{N_{\mathrm{atmo}}/10} + \mathbb{1}_{\mathrm{env}\neq\mathrm{rural}} \cdot 10^{N_{\mathrm{mm}}/10}\Big) \tag{25}$$

where:

$$N_{\text{atmo}} = N_{\text{base}}(f) + \Delta N_{\text{diurnal}}(f, \cos\chi) \tag{26}$$
$$N_{\text{mm}} = N_{\text{base}}(f) + F_a(\text{env}) \tag{27}$$

The indicator $\mathbb{1}_{\text{env}\neq\text{rural}}$ gates the man-made channel off entirely on rural sites; on suburban / urban sites the channel sits at an environment-specific level $N_{\text{base}} + F_a$ that is roughly diurnally constant (man-made QRM tracks the local interference environment, not the natural floor's day/night variation). The diurnal swing is:

$$\Delta N_{\text{diurnal}} = -A(f) \cdot \text{clamp}(\cos\chi,\ -1,\ 1) \tag{28}$$

where $A(f)$ ramps linearly from 10 dB at $f \leq 5$ MHz down to 3 dB at $f \geq 15$ MHz:

$$A(f) = \begin{cases} 10 & f \leq 5\,\text{MHz} \\ 10 - 7\,(f-5)/10 & 5\,\text{MHz} < f < 15\,\text{MHz} \\ 3 & f \geq 15\,\text{MHz} \end{cases} \tag{29}$$

Galactic-vs-atmospheric dominance transitions gradually across the 5 MHz–15 MHz window (galactic noise from P.372 Figs. 2–3 vs atmospheric from P.372 Figs. 13–14 / 23–24, which tabulate atmospheric noise factor F_{am} for the four seasonal-hemispheric cuts) rather than at any single frequency, so a smooth ramp is consistent with the smooth-handoff philosophy applied elsewhere in the budget. The man-made noise factor F_a depends on environment (rural / suburban / urban classification follows ARRL Handbook [14] convention): *rural* carries no man-made contribution above the natural floor (the indicator switches the second power-sum term off), *suburban* sets $F_a = +15$ dB above the rural baseline N_{base}, and *urban* sets $F_a = +25$ dB. P.372 reports its F_a relative to the thermal floor kT_0 (290 K) rather than relative to a quiet-rural baseline. Per P.372's linear man-made-noise fit $F_a = c - d\log_{10}(f_{\text{MHz}})$, quiet-rural takes $c = 53.6$, $d = 28.6$, giving $F_a \approx 25$ dB above thermal at 10 MHz; the galactic floor at 10 MHz with $c = 52$, $d = 23$ contributes $F_a \approx 29$ dB, so the noise channel an HF quiet-rural operator actually sees is the power-sum of these two components ($\sim$30 dB at 10 MHz, dominated by the galactic term in the upper HF). ionocast's +15 dB suburban and +25 dB urban add to the lumped quiet-rural floor to give $\sim$45 dB and $\sim$55 dB above thermal, landing inside the P.372 30–80 dB envelope spanning quiet-rural through industrial. The amplitude A is held latitude-independent for simplicity; tropical sites with frequent thunderstorm activity see larger atmospheric-noise diurnal swings (the P.372 atmospheric maps exceed midlat figures by 5–10 dB in the equatorial band) and polar sites with stable nighttime ionospheres see smaller swings, so the model is somewhat low for tropical noise floors at night and somewhat high for polar nights. The error is bounded at 2–3 dB on the noise side and is dominated by other budget terms on most verdicts.

Why the rural-site indicator gates N_{mm}. On rural sites the indicator $\mathbb{1}_{\text{env}\neq\text{rural}}$ gates the man-made channel off entirely so $N = N_{\text{atmo}}$ exactly. This lets $\Delta N_{\text{diurnal}}$ modulate the floor in full (-124 dBm at 15 m noon, -118 dBm at 15 m midnight; $N_{\text{base}} = -121$ dBm from Table 3 modulated by the $A(f) = 3$ dB diurnal swing on the upper-band branch of $A(f)$) rather than being clamped by an unconditionally-active N_{mm} that would otherwise pin the rural floor near N_{base} regardless of time of day. Suburban and urban verdicts are unaffected: N_{mm} dominates the power sum on those sites, so the indicator's all-environment gating is functionally rural-only.

Mapping to ITU-R P.372 figures. $N_{\text{base}}(f)$ in Table 3 is the lumped quiet-rural floor, the power sum of the galactic noise (P.372 Figs. 2–3, slowly varying with frequency at $F_a \approx 52 - 23\log_{10}(f_{\text{MHz}})$ dB above thermal in the 10–250 MHz range) and the atmospheric quiet-day baseline (P.372 Figs. 13–14 / 23–24, the seasonal F_{am} atmospheric-noise maps; the quiet-rural midlat midnight summer reference is read from these maps). The diurnal swing $\Delta N_{\text{diurnal}}$ then modulates this combined floor by the atmospheric component's day-vs-night spread (the galactic component is nearly diurnally constant; below 10 MHz the atmospheric swing dominates so the 10 dB amplitude is appropriate, above 10 MHz galactic dominates and the amplitude shrinks to 3 dB). The man-made channel N_{mm} sits F_a dB above the lumped rural baseline N_{base} on suburban and urban sites and is held diurnally constant (man-made QRM is set by the local interference environment, not by the natural floor's diurnal swing); on rural sites the channel is gated off entirely (Eq. 25's indicator), so a quiet rural QTH reads exactly N_{atmo} rather than the $N_{\text{atmo}} \oplus N_{\text{base}}$ phantom-3 dB the earlier form produced. The power-sum then correctly recovers the dominant component: man-made on suburban / urban, the natural floor on rural (atmospheric-dominated on low bands, galactic-dominated on upper bands).

Receiver-bandwidth scaling. The two noise components scale differently when the receiver narrows from the reference bandwidth $B_0 = 2500$ Hz to the mode's effective bandwidth B_{mode} (Section 3.13):

$$N_{\text{atmo}}(B_{\text{mode}}) = N_{\text{atmo}}(B_0) + 10\log_{10}(B_{\text{mode}}/B_0) \tag{30}$$

$$N_{\text{mm}}(B_{\text{mode}}) = N_{\text{mm}}(B_0) + \alpha \cdot 10\log_{10}(B_{\text{mode}}/B_0) \tag{31}$$

Atmospheric (galactic, thermal, distant lightning) is quasi-white and collapses in full with the receiver filter. Man-made noise is impulse-dominated: a bright transient in a 2500 Hz filter is mostly the same transient in a 50 Hz filter, since impulses have broad spectra. The scaling exponent α interpolates between the ideal-white limit ($\alpha = 1$) and the pure-impulsive limit ($\alpha = 0$); ionocast uses $\alpha = 0.5$ as a mixed-source-typical compromise. The Middleton Class A/B impulse-noise model [26] gives a non-linear bandwidth-scaling form that does not reduce to a single power-law exponent in general, so $\alpha = 0.5$ is a parametric simplification of a non-parametric scaling — defensible at the prediction level but not derivable as a midpoint of the Middleton form itself. The 0.5 value is a compromise tuned for typical mixed broadband QRM (switching power supplies, motor controllers, plasma TVs, LED drivers, ignition transients): *continuous* interferers (RTTY signals, broadcast bleed-over, local switching mode power-supply hum) sit closer to $\alpha = 1$ and are slightly under-rewarded by the narrowband-filter gain; *purely impulsive* interferers (lightning crashes, isolated arcing) sit closer to $\alpha = 0$ and are slightly over-rewarded. A site dominated by one mode could in principle be tuned with a different α, but that level of per-site characterisation is out of scope for the current model.

FT8-vs-SSB advantage — numeric walkthrough. The end-to-end model produces about 31 dB advantage on rural and 22 dB on suburban for FT8 over SSB, evaluated on a midlat 20 m path. Decomposition: SSB and FT8 use mode-native S_{req} values from Table 4 (+10 vs −4 dB), so the decoder-side advantage is exactly 14 dB. Bandwidth scaling on a rural site (galactic / atmospheric noise dominant, $\alpha = 1$) gives $10\log_{10}(2500/50) \approx 17$ dB of additional gain for FT8's narrower receive bandwidth, totalling 31 dB. On a suburban site (man-made-dominated, $\alpha = 0.5$) the noise-side gain falls to ~ 8.4 dB, totalling ~ 22 dB. Anecdotal operator-experience figures (no specific citation) of ~ 25 dB rural / ~ 15 dB suburban circulate in the WSJT-X / ham-radio community as floor-of- experience benchmarks; these compress several effects — partial decoder calibration loss at very low SNR, RX-chain front-end variability, the fact that operators typically compare the modes at the noise-floor edge rather than deep in the decode region, etc. — that the model does not attempt to reproduce. The computed numbers are larger than operator-reported because the model is the maximum-information budget; treat the reported $\sim$25/ $\sim$15 figures as floor-of-experience benchmarks, not as a target for the budget.

Table 3 gives the baseline noise power at 2.5 kHz reference bandwidth.

Table 3: Baseline noise N_{base} from ITU-R P.372 at 2.5 kHz bandwidth. Noon-time floor; midnight $= N_{\text{base}} + \Delta N_{\text{diurnal}}(\cos\chi = -1)$ where the swing amplitude $A(f)$ is 10 dB at $f \leq 5$ MHz ramping linearly to 3 dB at $f \geq 15$ MHz. The midnight column assumes the path midpoint is at the solar antipode ($\cos\chi = -1$), which obtains at the equator at midnight summer; at high latitudes in winter the sun never reaches the antipode in the diurnal cycle so the actual midnight $\cos\chi$ is shallower than -1 and the swing is attenuated proportionally. Re-derived 2026-04-30 from P.372-15 Fig 13 (atmospheric) $\oplus$ Fig 23 (galactic) max-of at midlat midnight summer.

Band	N_{base} (dBm)	midnight (dBm)
160 m	-100	-90
80 m	-113	-103
60 m	-118	-108
40 m	-121	-112
30 m	-120	-114
20 m	-120	-116
17 m	-120	-117
15 m	-121	-118
12 m	-122	-119
10 m	-123	-120

The table is anchored to the P.372 midnight midlat-summer reference at every band; calibration history and the WSPR-residual validation that led to the current values are summarised in Section 10.

3.13 Required SNR and Receiver Bandwidth by Mode

Each mode is defined by two numbers: the decoder-only SNR threshold S_{req} specified at the mode's own effective noise bandwidth B_{mode}, and that bandwidth itself. The receiver-bandwidth advantage of narrow digital modes is not baked into S_{req}; it is computed separately in the noise model (Section 3.12), which scales atmospheric noise with the receiver BW in full and man-made noise with it only partially. Quoting modes at a common 2.5 kHz reference BW with the BW advantage folded into S_{req} would over-count the benefit of digital modes on man-made-dominated (impulse-noise) sites.

Table 4: Required SNR threshold and effective noise bandwidth by mode. S_{req} is the decoder threshold at the mode's own native bandwidth. Digital-mode rows (FT4 / FT8 / WSPR) derive from WSJT-X [16] published values at 2.5 kHz reference via $S_{\text{native}} = S_{2500} + 10\log_{10}(2500/B_{\text{mode}})$; the SSB and CW rows are operator-practice native values from the ARRL Handbook [14] (comfortable-copy SSB at +10 dB SNR, trained-ear CW at ~ -7 dB at 2.5 kHz). The bandwidth advantage of narrow modes is then computed separately in the noise model (Section 3.12) using B_{mode}, not folded into S_{req}.

Mode	S_{req} (dB)	B_{mode} (Hz)	Sc
SSB	+10	2500	ARRL Handbook comfortable-copy native ($S_{2500} = +10$, Δ
CW	0	500	ARRL Handbook trained-ear native ($S_{2500} \approx -7$; $-7 + 10\log_{10}(2500/500)$
FT4	−2	100	WSJT-X −16.4 at 2500 Hz ($-16.4 + 10\log_{10}(2500/100) = -2.4$, roun
FT8	−4	50	WSJT-X −21 at 2500 Hz ($-21 + 10\log_{10}(2500/50) =$
WSPR	−8	25	WSJT-X −28 at 2500 Hz ($-28 + 10\log_{10}(2500/25) =$

B_{mode} here is the decoder *noise-equivalent bandwidth* integrated against the WSJT-X published threshold, not the signal's occupied spectral width. WSPR's 4-FSK signal [35] occupies ∼6 Hz (1.46 Hz/symbol × 4 tones); the listed 25 Hz captures the matched-filter / decoder integration over symbol-time that WSJT-X's −28 dB / 2500 Hz threshold already integrates against. FT8's 50 Hz signal is a closer match to the listed 50 Hz NEB. The decoder NEB is what the noise model in Eq. 25 should scale; using the signal occupied BW would double-count the decoder integration.

3.14 Antenna Elevation Pattern

The antenna gain G_{ant} is not a flat scalar; it varies with the required takeoff (elevation) angle *and with the band*. The elevation-pattern formulas below follow the standard treatment in *The ARRL Antenna Book* [13]. The elevation is derived from hop geometry using a simplified flat-earth model:

$$\theta_{\text{elev}} = \arctan\left(\frac{2\,h_F}{d_{\text{hop}}}\right), \qquad d_{\text{hop}} = \min\!\left(d,\, d_{\text{hop}}^{\max}(h_F)\right) \tag{32}$$

where $h_F = 300$ km (typical F2 height; the smoothing form below applies regardless of the exact value) and the per-hop range d_{hop} saturates at the geometric ceiling $d_{\text{hop}}^{\max}(h_F)$ from Eq. 19. Long-haul DX paths require low takeoff angles (5–10°); regional paths use higher angles (15–30°); NVIS uses near-vertical (60–90°).

Continuous across the hop boundary. The $\min(d, d_{\text{hop}}^{\max})$ form keeps per-hop range continuous at integer-hop transitions: real long-haul paths pack at the geometric ceiling, so saturating per-hop range at $d_{\text{hop}}^{\max}$ for $d \geq d_{\text{hop}}^{\max}$ is what the geometry implies. Below one ceiling the angle steepens with shorter d. Using integer N_{hops} in the denominator instead would produce a sawtooth in takeoff angle (e.g. a $\sim 8°$ jump from 8.5° at $d = 4000$ km to 16.7° at $d = 4001$ km as N_{hops} ticks 1→2) that would propagate into the Fresnel ground-reflection loss (Section 3.8) and the elevation-pattern antenna gain (Eq. 33) on every path crossing a ceiling.

Implication: per-hop geometry is invariant in d above the ceiling. Once $d \geq d_{\text{hop}}^{\max}$, the $\min(\cdot)$ saturates at the ceiling, so θ_{elev} (and therefore the Fresnel grazing angle and the antenna elevation gain) is the *same value* for every long-haul path regardless of distance. A 12000 km Lisbon–Tokyo and a 17500 km Lisbon–Sydney see identical per-hop $\theta_{\text{elev}} = 8.5°$, and the antenna gain term $G_{\text{ant}}(\theta)$ contributes the same to both budgets. The two paths still differ in free-space loss (which is $\propto d^2$), multi-hop ground reflection ($\propto N_{\text{extra}}(d)$), and ionospheric absorption sums (longer paths have more reflection points). This is not a model approximation; it is the geometry implied by "hops pack at the ceiling above $d_{\text{hop}}^{\max}$".

Type + height + λ. Operators supply three inputs: antenna *type* (eight classes, Table 5), mounting *height* h in metres, and the *peak gain* G_{peak} over typical ground (the type already encodes the free-space directivity, so a 4-element Yagi has a bigger G_{peak} than a dipole at the same h). The effective gain at elevation θ for band frequency f is

$$G_{\text{ant}}(\theta, f, h, \text{type}) = G_{\text{peak}} + G_{\text{rel}}(\theta, f, h, \text{type}) \tag{33}$$

where the relative elevation shape G_{rel} is 0 dB at the pattern peak and floors at $-15\,\text{dB}$ (real antennas don't vanish into ideal nulls).

Horizontal types. For a horizontal wire, loop, or beam at height h, the elevation pattern is dominated by the ground- reflection factor for horizontal polarisation (image theory):

$$\text{gf}_{\text{H}}(\theta, f, h) = 2\,\big|\sin(kh\sin\theta)\big|, \qquad k = \frac{2\pi}{\lambda},\ \lambda = 300\,\text{m}/f_{\text{MHz}}. \tag{34}$$

The maximum of 2 (6 dB above the free-space radiator) occurs at $\sin\theta = \lambda/(4h)$. For a dipole at $h = 10\,\text{m}$: on 40 m ($\lambda = 40\,\text{m}$) the peak sits at $\sin\theta = 1$, i.e. straight up – a pure NVIS antenna; on 20 m ($\lambda = 20\,\text{m}$) the peak sits at $\sin\theta = 0.5 \Rightarrow \theta = 30^\circ$ – regional / short DX; on 10 m ($\lambda = 10\,\text{m}$) the peak sits at $\sin\theta = 0.25 \Rightarrow \theta = 14.5^\circ$ – classical low-angle DX lobe. Relative gain:

$$G_{\text{rel,H}} = 20\log_{10}\!\big(\max(\text{gf}_{\text{H}}/2,\ 10^{-15/20})\big) \tag{35}$$

The $-15\,\text{dB}$ floor inside the $\max(\cdot)$ is a numerical guard as well as a pattern-quality cap: at very low antenna heights (e.g. a horizontal dipole at $h = 1\,\text{m}$ on $80\,\text{m}$, where $kh \approx 0.08$ and the formula's pre-floor value drops to $\sim -22\,\text{dB}$ at the pattern peak), the bare expression goes arbitrarily negative; the floor caps that to $-15\,\text{dB}$ so the budget stays well-conditioned and the verdict reflects the "ground-mounted dipole" regime rather than the unrealistic deep null the formula would otherwise predict.

Horizontal loops pick up an additional high-angle weighting

$$G_{\text{rel,loop}} = G_{\text{rel,H}} + 20\log_{10}\!\big(0.5 + 0.5\sin\theta\big), \tag{36}$$

the classical observation that a full-wave loop has a 45–60° elevation peak rather than the dipole's ground-reflection-driven one.

Beams (hex, Yagi, stacked) use the *same* ground-reflection shape as a horizontal wire at the same height: the elevation lobe of a Yagi is set by the ground reflection, not the number of elements. The directivity bonus (3–15 dB over a dipole) lives in G_{peak}.

Vertical. The free-space quarter-wave pattern $f_{\text{v}}(\theta) = \cos(\pi/2 \cdot \sin\theta)/\cos\theta$ combines with a vertical-polarisation ground reflection $\text{gf}_{\text{V}}(\theta, f, h) = 2\,\big|\cos(kh\sin\theta)\big|$ which evaluates to 2 at $h = 0$ ($\cos 0 = 1$, image in phase everywhere) and reduces continuously as h grows. The implementation retains an explicit $h \lesssim 0.1\,\text{m}$ branch *only* as a numerical shortcut (skipping the $\cos(kh\sin\theta)$ evaluation when it is guaranteed ∼1); mathematically the elevated-form expression at $h = 0$ already yields the same answer.

$$G_{\text{rel,V}} = 20\log_{10}\!\Big(\max\!\Big(\tfrac{f_{\text{v}}(\theta)\,\text{gf}_{\text{V}}}{1\cdot 2},\ 10^{-15/20}\Big)\Big) \tag{37}$$

The denominator normalises by the value of $f_{\text{v}} \cdot \text{gf}_{\text{V}}$ at the antenna's actual peak: $f_{\text{v}}(\theta) = \cos\!\big(\frac{\pi}{2}\sin\theta\big)/\cos\theta$ has its global maximum $f_{\text{v}}(0) = 1$ at the horizon and falls monotonically toward 0 at $\theta = \pi/2$ (overhead) by L'Hôpital, with $f_{\text{v}}(80^\circ) \approx 0.14$, $f_{\text{v}}(85^\circ) \approx 0.07$, $f_{\text{v}}(89^\circ) \approx 0.014$; combined with $\text{gf}_{\text{V}} = 2$ for a ground-mounted radiator, the $1 \cdot 2$ denominator anchors $G_{\text{rel}} = 0$ at the antenna's strongest direction, matching the horizontal-branch normalisation.

Peak at $\theta = 0^\circ$ (horizon), null overhead. No band-dependent lobe migration for a ground-mounted vertical — unlike the horizontal family, its pattern is shape-invariant across HF.

Compromise. Indoor / attic / mag-loop installations have a near-omnidirectional scattered pattern with a gentle low-angle penalty from nearby conductors:

$$G_{\text{rel,compromise}} = -\max\big(0,\ 2(1 - \theta/10^\circ)\big). \tag{38}$$

This is the only relative-pattern in the antenna set with a *negative* peak (the pattern goes to -2 dB at the horizon and rises to 0 dB above 10°): the compromise antenna loses gain at low takeoff angles relative to its already-modest peak. All elevation arguments θ in this section, including the horizontal and beam expressions below, are angles in degrees (the per-equation literals such as 10° set the convention).

Table 5: Antenna types and default (height, peak gain) pre-fills.

Type	h **(m)**	G_{peak} **(dBi)**	**Covers**
Horizontal	10	+5	Dipole, inverted-V, G5RV, OCF, EFHW, folded dipole
Vertical	0	+2	1/4 , 1/2 , 5/8 wave over radials
Horizontal loop	8	+4	Full-wave delta, quad, square
Small beam	12	+7	Hex beam, 2-/3-el Yagi, compact tribander
Medium beam	15	+9	4-el Yagi, 4-el quad
Large beam	20	+12	5-el+ Yagi, stacked / phased
Compromise	5	−2	Indoor, attic, mag loop, mobile whip
Custom	10	user	Any unusual setup

3.15 Default Operator Settings

The default station profile assumes a typical amateur installation: $P_{\text{tx}} = 50$ dBm (100 W), $G_{\text{ant}} = +5$ dBi peak (installed dipole with ground reflection; effective gain varies by path elevation), mode = SSB, noise environment = suburban.

3.16 Worked Example: 20 m, 1000 km, Quiet Day

Table 6 lists the complete budget for a 14.1 MHz path of 1000 km on a quiet day. At this distance the path is single-hop (4000 km hop ceiling), so $L_{\text{hop}} = 0$: the multi-hop reflection budget activates only on longer paths. The example uses the canonical $h_F = 300$ km for the takeoff-angle computation; the runtime substitutes the live h_{mF2} from the nearest GIRO digisonde when available, which on enhanced-flux days can shift θ by a few degrees and the per-hop ceiling by ~250 km.

G_{ant} **derivation (per Section 3.14).** G_{ant} is not a flat scalar; it is computed from the hop geometry. For a 1000 km single-hop with $h_F = 300$ km, the takeoff angle is $\theta = \arctan(2 \cdot 300/1000) \approx 31^\circ$. For a horizontal dipole at $h = 10$ m on 20 m ($\lambda = 20$ m), the elevation pattern peaks at $\sin\theta_{\text{peak}} = \lambda/(4h) = 0.5$, i.e. $\theta_{\text{peak}} = 30^\circ$. The required 31° lands essentially on the peak lobe, so $G_{\text{rel}} \approx 0$ dB and $G_{\text{ant}} = G_{\text{peak}} + G_{\text{rel}} = +5$ dBi. On a longer hop (e.g. 3000 km with $\theta \approx 11^\circ$) G_{rel} would be substantially negative on the same dipole; the +5 dBi here is geometric coincidence, not a fixed value. Note that this worked example uses the canonical $h_F = 300$ km fallback for a clean derivation; in production the runtime reads the live h_mF2 from the nearest digisonde (typically 280 --360 km), which would shift θ by a degree or two and may change which side of the dipole peak the takeoff lands on.

Table 6: SNR budget for 20 m (14.1 MHz), 1000 km, quiet day, suburban noise, SSB, with the reflection point at local noon ($\cos\chi \approx 1$, which produces the $L_{\mathrm{Dreg}} = -0.94$ dB shown via $L_{\mathrm{day}}(14.097\,\mathrm{MHz}) \cdot 1^{0.7} = 0.94$ dB per Eq. 6). The Σ column tracks the running signal level. Below the rule, N and S_{req} are operands to the margin formula $M = S - N - S_{\mathrm{req}}$, not additions to Σ. The resulting dB margin is σ-normalised by the prose below to derive the tier verdict per ITU-R P.842. All dB values in the Δ and Σ columns are rounded to 0.1 dB; cumulative rounding across the seven rows is bounded by ± 0.5 dB on the final Σ, well below the per-band σ_g resolution that drives the tier boundary.

Term	**Δ**	**Σ (dBm)**
P_{tx}	+50 dBm	+50
G_{ant}	+5 dBi	+55
L_{fs}	−115 dB	−60
L_{Dreg}	−0.9 dB	−60.9
L_{MUF}	0 dB	−60.9
L_{iono}	−1 dB	−61.9
L_{hop}	0 dB	−61.9
Signal at Rx		**−61.9**
Operands to margin formula:		
N (suburban)		−105 dBm
S_{req} (SSB)		+10 dB
$M = -61.9 - (-105) - 10 = +\mathbf{33.1}$ dB $\Rightarrow$ $M/\sigma = 3.68$		

Tier verdict. With margin $M = +33.1$ dB and the fixed-dB cuts of Table 11, the verdict clears the Excellent threshold ($M \geq +18$ dB) by 15 dB. The corresponding circuit reliability under the per-band σ for 20 m (9 dB, Table 9) is $\Phi(33.1/9) = \Phi(3.68) \approx 0.99988$ (effectively $\sim$99.99 %; the tail beyond $z = 3.68$ is $\sim 1.2 \times 10^{-4}$). The wide head-room is typical of midday 20 m on quiet days at 1000 km.

Worked example with bonuses (15 m TEP, EU–Brazil at 19 LT). The budget in Table 6 silently sets the three additive bonuses (Eq. 76: B_{grayline}, B_{TEP}, B_{scatter}) to zero because none of them fire on a quiet 20 m midday path. To anchor the bonus arithmetic, consider the 15 m TEP-eligible case at the current moderate-cycle epoch ($F_{10.7\mathrm{A}} = 120$): a Lisbon ($\phi_{\mathrm{dip}} \approx +42°$) to São Paulo ($\phi_{\mathrm{dip}} \approx -15°$, near the southern EIA crest) path, $f = 21$ MHz, midpoint local time ≈ 19, two F2 hops, $f/\mathrm{MUF} = 1.10$, $\sigma_{f_{\mathrm{oF2}}} \approx 0.4$ MHz across the hops. At those numbers the bonus terms read:

- $B_{\mathrm{grayline}} \approx 0.2$ dB (post-S0-#2 rewrite, the bonus is now formula-driven across 1.8–30 MHz but small on 15 m: at $f = 21$ MHz, $L_{\mathrm{day}} = 200/(21.5)^2 = 0.43$ dB, and at midpoint LT $\approx$ 19 the sun is just past sunset so $\cos\chi \lesssim 0$, the formula factor $1 - \cos^{0.7}$ saturates near 1, the sunset multiplier $\alpha = 0.5$ applies, giving $0.5 \cdot 0.43 \cdot 1 \approx 0.22$ dB which the 0.2 dB floor lets through marginally).
- $B_{\mathrm{TEP}} = B_{\mathrm{TEP,plateau}}(F_{10.7\mathrm{A}}{=}120) \cdot f_{\mathrm{freq}} \cdot f_{\mathrm{LT}} \cdot f_{\mathrm{dip,src}} \cdot f_{\mathrm{dip,dst}} \approx 11.2\,\mathrm{dB} \cdot \mathrm{ramp}(21, 14, 22) \cdot 1 \cdot 1 \cdot 1 = 11.2\,\mathrm{dB} \cdot 0.875 \cdot 1 \cdot 1 \cdot 1 \approx 9.8$ dB (moderate-cycle plateau per Eq. 66; frequency factor not yet at the $[22, 50]$ MHz plateau; both endpoints past the dip-latitude saturation knee at $|\phi_{\mathrm{dip}}| \geq 10°$; LT in the $[17, 23]$ plateau). At solar maximum ($F_{10.7\mathrm{A}} \to \infty$) the plateau asymptotes to 15 dB, giving $B_{\mathrm{TEP}} \to 13.1$ dB on the same path.
- $B_{\mathrm{scatter}} = w_{\mathrm{sc}} \cdot \min(1, 0.4/1.5) \cdot \min(2, 0.10) \cdot 5 = 1.5 \cdot 0.267 \cdot 0.10 \cdot 5 \approx 0.20$ dB.

The verdict-margin contribution is therefore $B_{\mathrm{grayline}} + \max(B_{\mathrm{TEP}}, B_{\mathrm{scatter}}) = 0.2 + 9.8 = 10.0$ dB on top of M_{physics}. The $\max(\cdot)$ form intentionally suppresses the small B_{scatter} contribution

(0.2 dB) on top of the 9.8 dB TEP signal, since both describe the same F-region irregularity-driven recovery channel. An 80 m sunrise example exercises B_{grayline} alone; that geometry is left as an exercise.

4 MUF Estimation

The Maximum Usable Frequency is the single most important input to the link budget. ionocast derives it from two primary sources in a consensus blend (Section 4.3): observed kc2g MUF(3000F2) at the path midpoint, and a simplified P.1239 climatology that fills in when the observation is stale or out of range. An optional third source — per-hop foF2 fusion from the GIRO digisonde network — shares its per-hop foF2 sampling machinery with the F2-scatter bonus of Section 6.5 (both consume inverse-square-distance-weighted foF2 from in-range stations along the great circle), but the fusion role itself is gated behind the `fusion-flag` parameter of Appendix E and held off in the production blend pending a joint budget retune (the calibration sweep showed it net-negative against the rebuilt climatology baseline; §6.5 describes the scatter use of the same sampling, which *is* active, and the "Per-hop fusion (deferred)" paragraph there records the calibration result that closed this branch). The two-source consensus is what ships.

4.1 kc2g Observed MUF

The kc2g network publishes MUF(3000F2) from each ionosonde. For each of ten reference paths — great circles from the operator's QTH to five canonical short-path destinations (New York, São Paulo, Johannesburg, Tokyo, Sydney) and their five long-path counterparts, midpoints recomputed per-QTH on each reload — ionocast selects the nearest ionosonde to that path's midpoint. A *freshness gate* discards observations older than 30 min. The distance from each path midpoint to its nearest ionosonde is displayed in the path table so operators can gauge data confidence; stations beyond 1500 km are flagged.

4.2 foF2 Climatology (Simplified P.1239)

When observed data are stale or unavailable, ionocast falls back to a simplified climatological model anchored to ITU-R P.1239 [3]. The unmodified daytime baseline is the sum of a polar-night-and-thermal-floor term and a latitude-modulated solar-zenith-driven daytime term:

$$b(\phi,\, \cos\chi) = b_{\text{floor}}(F_{10.7\text{A}}) + 4.0\,\mathcal{P}(\phi)\, d(\cos\chi) \tag{39}$$

where the solar-baseline floor is

$$b_{\text{floor}}(F_{10.7\text{A}}) = 3.5 + 0.04\,(F_{10.7\text{A}} - 70). \tag{40}$$

b_{floor} is the value b falls to overnight or at the poles where $\mathcal{P}(\phi)\, d \to 0$: 3.5 MHz at the quiet-sun reference $F_{10.7\text{A}} = 70$ plus a linear $\mathrm{F}_{10.7\text{A}}$ slope. The slope is signed (no positive-part clamp), so at deep solar minimum ($F_{10.7\text{A}} < 70$) the floor sinks below 3.5 MHz, consistent with the lower foF2 floors observed at extended-minimum nights; this differs deliberately from the EIA amplitudes A_{cr} and A_{tr} in Eq. 46 and Eq. 47, where the slope is wrapped in $\max(0,\cdot)$ so the equatorial-anomaly amplitude does not invert below $F_{10.7\text{A}} = 70$. The second term of Eq. 39 carries the daytime $\mathcal{P}(\phi)\cdot d(\cos\chi)$ shape that dominates the verdict at midday on illuminated paths, with $\mathcal{P}(\phi)$ the two-stage poleward fall-off:

$$\mathcal{P}(\phi) = (1 - 0.003|\phi|)\cdot \frac{1}{1 + \exp\!\big((|\phi| - 73^\circ)/8^\circ\big)} \tag{41}$$

The first factor is a gentle linear midlatitude term; the second is a sigmoid centred at $|\phi| = 73^\circ$ with 8° steepness that captures the polar F2 thinning above $\sim 60^\circ$. The linear factor alone would under-predict that thinning (F2 thins faster than linearly with latitude above $\sim 60°$ as the polar-cap tongue takes over with patchy electron density), producing a 1–1.7 MHz daytime over-prediction at Tromsø / Gakona / Eielson AFB without the sigmoid term. The sigmoid factor reproduces the observed magnitude: $\mathcal{P}(45°) \approx 0.84$, $\mathcal{P}(60°) \approx 0.69$, $\mathcal{P}(70°) \approx 0.47$, $\mathcal{P}(80°) \approx 0.22$. After sunset, foF2 does not collapse instantly — recombination of the F-region runs on a multi-hour timescale. The post-sunset diurnal curve breaks into two distinct phases that ionocast captures with two structurally different terms:

- **Phase 1, sunset to ~ 3 h after sunset:** residual photoionization keeps daytime production going. The plasma's effective "recently illuminated cosine" decays on a ~ 3 h timescale rather than dropping to zero at the terminator. Captured by the photoelectron *driver* $d(\cos\chi)$, which lives *inside* b's argument as a numerator multiplier:

$$d(\cos\chi) = \max\big(\max(0, \cos\chi),\ 0.7\ \max(0, \cos\chi_{-3\text{h}})\big) \tag{42}$$

  d holds at the daytime cosine through sunset, then decays toward zero as the lagged $\cos\chi_{-3\text{h}}$ also crosses below zero. On a cold start (browser session with no historical solar-zenith cache), the implementation substitutes $\cos\chi_{-3\text{h}} = \cos\chi_{\text{now}}$ for the first 3 hours of session lifetime, which means the memory-lag term degenerates to the live cosine until the cache fills. This biases foF2 slightly low at evening startup (the lag would have read the prior daytime cosine, which is positive while the live cosine is at-or-below zero) by at most ∼10% on the worst case (sunset start, single midlat path), which on a near-MUF cell maps to a ∼10 % MUF bias — enough to flip a Fair / Good verdict on the upper bands at exactly the time of day operators most often check propagation. The cache is re-populated within the first three 10-minute refresh cycles, so the bias is bounded and self-clearing, but the worst-case window aligns with after-work session starts and is therefore worth eliminating rather than tolerating.

  Pre-population fix. $\cos\chi_{-3\text{h}}$ is purely astronomical — it is fully determined by the current UTC and the path-midpoint geographic coordinates — so the "no historical cache" branch can compute it from $\cos\chi(t_{\text{now}} - 3\,\text{h})$ directly on the first load, rather than waiting three refresh cycles for the cache to fill from live samples. Doing so retires the cold-start bias entirely without any historical state, and is candidate cleanup tracked in Section 10 alongside the $\mathrm{F}_{10.7\mathrm{A}}$-fallback surfacing recommendation (the two share a UI-honesty motivation: never let the live verdict silently inherit a degraded input when the non-degraded value is locally computable). Until that change ships, the runtime should at minimum surface a "cold-start cache filling" chip during the first ∼30 minutes of a fresh session so an operator startup-checking the upper bands at sunset is not given a confidently-stated verdict on a degraded input.

- **Phase 2, ~ 3 h after sunset onward:** residual production has decayed to zero and ongoing recombination pulls electron density toward a nighttime floor (typically reached around 03 LT). Captured by the night-decay *multiplier* that operates on the already-constructed b as a recombination floor:

$$b(\phi,\ \cos\chi) \to b(\phi,\ \cos\chi) \cdot \max(0.6,\ 1 + 0.4\cos\chi) \quad \text{when}\ \ \cos\chi < 0 \tag{43}$$

  The multiplier reads ~ 1 at the terminator (only slightly negative $\cos\chi$, hardly any reduction), ramping toward the 0.6 floor by deep night.

The two terms are not equivalent multipliers: $d(\cos\chi)$ is the *driver argument* that the daytime baseline $b_{\text{floor}}(F_{10.7\text{A}}) + 4.0\,\mathcal{P}(\phi)\,d$ constructs b from, while the night-decay multiplier is a *post-multiplier* that scales the constructed b during the $\cos\chi < 0$ interval. They overlap in time — both are "firing" from sunset onward — but their contributions peak in different phases: in early evening the driver dominates (memory lag still strong, multiplier ≈ 1); in deep night the multiplier dominates (driver decayed to zero, multiplier at or near the 0.6 floor). Removing either term over-predicts foF2 in its dominant phase, which the diurnal sweep in Section 4.2 confirms.

The result is then floored at 2 MHz and scaled by two dimensionless anomaly corrections, with the combined-anomaly product itself floored at 0.7 so that a strong dip-equator trough cannot collapse the prediction below ∼70 % of baseline:

$$f_{\mathrm{oF2}} = \max(2,\, b) \cdot \max\Big(0.7,\ (1 + \alpha_{\mathrm{EIA}}(\phi_{\mathrm{dip}}, \cos\chi)) \cdot (1 + \alpha_{\mathrm{win}}(\phi, m, \cos\chi))\Big) \tag{44}$$

and then MUF(3000) $\approx 3.0 \times f_{\mathrm{oF2}}$. The 0.7 floor binds only at the dip equator under solar maximum, where the trough term α_{tr} saturates at -0.50 and the unfloored product would otherwise read 0.50 (see §4.2's EIA discussion); real Jicamarca minima sit around -20 to -30 % from the crest-flank baseline, not zero. Together the two floors imply MUF(3000) ≥ 4.2 MHz on the climo-only branch ($2 \cdot 0.7 \cdot 3 = 4.2$), so the climatology alone cannot return MUF below the lowest amateur HF band (160 m, 1.8 MHz); a closed 80 m or 160 m verdict on the climo-only branch is therefore structurally impossible and only arises on the ingested-MUF branch (kc2g) or via direct absorption / D-layer terms.

The night-decay multiplier and the 3-h memory lag are part of the post-rebuild climatology. Initial validation used 30 d of GIRO digisonde foF2 across the original nine-station midlatitude basket (the basket has since grown to 27 stations spanning the equatorial and polar belts; see Appendix D, used by the EIA grid sweep below rather than by the night-decay validation). The night-decay fit dropped baseline RMSE from 1.91 MHz to 1.70 MHz (-11 %) and centred the residual bias from -0.26 MHz to -0.05 MHz; the prior plain-P.1239 fit predicted ∼5.5 MHz at midlat midnight versus the ∼3.7 MHz that nine months of GIRO nighttime samples actually report, and collapsed foF2 to the sunset-instant baseline rather than tracking the observed multi-hour evening tail.

Equatorial Ionisation Anomaly (EIA). The daytime fountain effect [22] lifts plasma from the magnetic equator up the field lines and deposits it in two daytime crests at magnetic dip latitude $\pm\phi_{\mathrm{EIA}}$ ($\phi_{\mathrm{EIA}} = 15\,°$), leaving a trough at $\phi_{\mathrm{dip}} \approx 0$. ionocast captures both via a positive crest kernel and a narrow negative trough kernel summed into one α_{EIA}:

$$\alpha_{\mathrm{EIA}} = \Big(A_{\mathrm{cr}}(F_{10.7\mathrm{A}})\, e^{-d_{\mathrm{cr}}^2/2\sigma_{\mathrm{cr}}^2} - A_{\mathrm{tr}}(F_{10.7\mathrm{A}})\, e^{-\phi_{\mathrm{dip}}^2/2\sigma_{\mathrm{tr}}^2}\Big) \cdot \max(0, \cos\chi) \tag{45}$$

where $d_{\mathrm{cr}} = \min(|\phi_{\mathrm{dip}} - \phi_{\mathrm{EIA}}|, |\phi_{\mathrm{dip}} + \phi_{\mathrm{EIA}}|)$ is the distance to the nearest crest, $\sigma_{\mathrm{cr}} = 18\,°$, and $\sigma_{\mathrm{tr}} = 6\,°$ (sharper than the crests; the trough is narrow in latitude). ϕ_{dip} here is the *signed* dip latitude (positive in the northern magnetic hemisphere, negative in the southern), derived from the same tilted-dipole approximation as ϕ_{CGM} (Eq. 17) but without the absolute value: the crest term's d_{cr} uses the signed value to pick the nearer of the two crests, and the trough term's ϕ_{dip}^2 is sign-symmetric. An implementation that erroneously used $|\phi_{\mathrm{dip}}|$ would compute the same trough magnitude but would mis-place the crests on the wrong hemisphere for southern paths. Both amplitudes scale with EUV illumination through F10.7A:

$$A_{\mathrm{cr}}(F_{10.7\mathrm{A}}) = \min\big(0.85,\ 0.50 + 0.007 \cdot \max(0,\ F_{10.7\mathrm{A}} - 70)\big) \tag{46}$$

$$A_{\mathrm{tr}}(F_{10.7\mathrm{A}}) = \min\big(0.50,\ 0.30 + 0.002 \cdot \max(0,\ F_{10.7\mathrm{A}} - 70)\big) \tag{47}$$

At quiet sun ($F_{10.7\mathrm{A}} = 70$) the crest amplitude is 0.50 and the trough is 0.30; the crest cap-saturates at 0.85 at $F_{10.7\mathrm{A}} \geq 120$, while the trough cap-saturates at 0.50 at $F_{10.7\mathrm{A}} \geq 170$ (different saturation epochs because the slopes 0.007 and 0.002 differ; the trough rises more slowly with EUV). At the current moderate-cycle epoch ($F_{10.7\mathrm{A}} \approx 120$) the crest is at its 0.85 cap and the trough is at 0.40, short of its 0.50 cap. The combined-anomaly 0.7 floor in Eq. 44 ensures a strong trough never collapses density below ≈ 70% of baseline. The trough kernel is a separate Gaussian with a negative sign so that the EIA combined factor can predict depression on the dip equator (a crest-only formulation would predict $\sim +27\%$ enhancement there, an artefact of the symmetric distance-to-nearest-crest functional form being unable to express trough behaviour).

The values were settled by an 80-point grid sweep over $(\mathrm{base}, \mathrm{slope}, \sigma)$ against 30 d of GIRO foF2 at the four equatorial-belt stations: Townsville and Niue near the southern crest of the Pacific arch; Ascension at dip latitude $\sim-3°$ (deep in the trough, far from the $\sim -15°$ southern crest peak); Jicamarca essentially *on* the magnetic dip equator (dip latitude ~3° S), anchoring the equatorial trough. The equatorial bias spread floors at ~2.2 MHz across the grid for two coupled reasons: the cost function has no northern-crest station to balance the southern-crest pair (so the Gaussian's amplitude is fit one-sided), and *half* of the basket — Jicamarca and Ascension — sits in the equatorial trough rather than on a crest peak, pulling the fitted amplitude down and the width out relative to a true-crest fit. The basket is now selected *geometrically* from `GIRO_STATIONS` (any station with $|\phi_{\mathrm{dip}}| \leq 25°$ joins the EIA-relevant set automatically), so adding any future northern-crest station — São Luís (BR, geographic $\sim-2.5°$ but dip latitude near +5 to +10° N because the South Atlantic Anomaly shifts the magnetic equator south of its geographic counterpart in this longitude sector, putting an equatorial geographic station onto the northern magnetic flank) is the credible Americas candidate, with any Indian / SE-Asian DIDB station near $\phi_{\mathrm{dip}} \approx +15°$ N also qualifying — closes both asymmetries without further code edits and the floor is expected to drop below 2.2 MHz. (Tucumán (AR) at $-26.8°$ and Cachoeira Paulista (BR) at $-22.7°$ are *southern*-hemisphere dip-latitude stations; the latter already participates in the GIRO basket and neither would help close the *northern*-crest gap; see Appendix D for the current basket.) After the BVJ03 Boa Vista addition the fit converged at base 0.50 / slope 0.007 / $\sigma = 18$ with max |bias| dropping from 1.84 to 1.18 MHz.

Dip latitude is derived from the same tilted-dipole approximation used for CGM latitude (Eq. 17) but signed, so hemisphere is preserved. All numerical ϕ_{dip} values quoted in this paper for specific stations and paths are computed from this tilted-dipole formula, not from a real-IGRF dip-equator field inversion. The two can differ by 5–10° in the South Atlantic Anomaly sector and by a few degrees elsewhere; we use the tilted-dipole convention because it is the same number the runtime model evaluates, so quoted values reproduce directly from the code.

Winter anomaly. Midlatitude ($35° \leq |\phi| \leq 60°$) daytime foF2 runs noticeably higher in the local winter than in local summer — the classical O/N_2 seasonal shift in the thermosphere. ionocast applies a day-of-year-phased cosine kernel peaked at the local winter solstice:

$$\alpha_{\mathrm{win}} = A_{\mathrm{win}} \cdot \tfrac{1}{2}(1 + \cos\theta_d) \cdot \max(0, \cos\chi), \quad \theta_d = \frac{2\pi\,(d_{\mathrm{yoy}} - d_{\mathrm{solstice}})}{365} \tag{48}$$

with $A_{\mathrm{win}} = 0.12$, d_{yoy} the UTC day of year (1 = 1 January), and $d_{\mathrm{solstice}} = 355$ (December 21) for the northern hemisphere ($\phi \geq 0$) or $d_{\mathrm{solstice}} = 172$ (June 21) for the southern hemisphere ($\phi < 0$). The day-of-year phase places the cosine peak at the actual astronomical solstice; an earlier calendar-month formulation $\theta_m = 2\pi\,(m + 0.5 - m_{\mathrm{winter}})/12$ landed the peak five days early because mid-month does not coincide with the solstice date. The correction is daytime-only and switches off outside the midlatitude band. The amplitude $A_{\mathrm{win}} = 0.12$ is set against the same 30-day GIRO foF2 grid sweep that fits the EIA amplitudes (§4.2, basket: Juliusruh, Pruhonice, Fairford, Sopron, Athens, Rome, Gibilmanna, Millstone Hill, Boulder, Wallops Island), reading the winter-vs-summer daytime foF2 ratio at midlatitude stations and matching the model's $1 + A_{\mathrm{win}}$ factor to the median observed ratio (1.12 ± 0.03 across the basket). This is consistent with Prölss [20]'s F-region seasonal-shift survey (~10–15% winter enhancement in the cycle-averaged midlatitude foF2 record) but the specific value is observation-fit, not literature-quoted. The 10-station basket is entirely northern-hemispheric (continental Europe and US east coast); the same $A_{\mathrm{win}} = 0.12$ amplitude is mirrored south of the equator with the solstice phase flipped, but the southern-hemisphere amplitude itself was not separately fit and could differ given the asymmetric land/ocean distribution. Prölss reports SH amplitudes within the same ~10–15 %

band so the symmetric choice is defensible at the climatology-class accuracy this term needs to deliver, but a southern-hemisphere validation pass is tracked as future work.

4.3 Symmetric Consensus Blend

The consensus algorithm uses the geometric mean of the two estimates regardless of divergence direction:

1. Compute divergence: $\delta = |\ln(\mathrm{MUF}_{\mathrm{kc2g}}/\mathrm{MUF}_{\mathrm{climo}})|$
2. Result: $\mathrm{MUF}_{\mathrm{cons}} = \sqrt{\mathrm{MUF}_{\mathrm{kc2g}} \cdot \mathrm{MUF}_{\mathrm{climo}}}$

δ is retained for diagnostic surfacing (the UI shows it on hover) but does not branch the result.

Single-source fallbacks. When the kc2g freshness gate (30 min) discards the observation, $\mathrm{MUF}_{\mathrm{cons}} = \mathrm{MUF}_{\mathrm{climo}}$ and the source label reads "climo". When the climatology returns null — the rare combination of extreme inputs that drives the unmodified daytime baseline below the absolute floor — $\mathrm{MUF}_{\mathrm{cons}} = \mathrm{MUF}_{\mathrm{kc2g}}$ when the latter is fresh, and "unavailable" otherwise (the band is treated as below MUF). δ is reported as 0 in both single-source cases.

The symmetric geometric mean treats real upward enhancements (EIA crest, post-storm positive phase, evening F-region uplift, Es-lifted MUF) and declining-ionosphere readings the same way: kc2g $= 2 \cdot$ climo produces $1.41 \cdot$ climo (captures most of the enhancement while damping spurious extremes); kc2g $= 0.5 \cdot$ climo produces $0.71 \cdot$ climo (mildly conservative on storm depressions, with the absorption side of storm response covered separately by the storm-lag K_p kernel and L_{aur} term). At kc2g $\approx$ climo (the common case) the blend is unchanged.

4.4 Per-Hop Minimum MUF (P.533 §3.1)

For paths requiring $N \geq 2$ hops, each reflection point at fraction $(2k-1)/(2N)$ along the great circle sees its own local foF2 climatology (latitude fall-off, EIA crests/trough, winter anomaly, diurnal phase). The path MUF is the *minimum* across all hops, evaluated by computing per-hop foF2 climatology directly at each reflection point and applying the kc2g-vs-climo correction scalar $\mathrm{MUF}_{\mathrm{mid}}/\mathrm{MUF}_{\mathrm{climo,mid}}$ to preserve the upstream kc2g signal:

$$\mathrm{MUF}_{\mathrm{path}} = \min_{k=1}^{N} 3.0 \cdot f_{\mathrm{oF2}}(\phi_k, \lambda_k, \cos\chi_k, t) \cdot \frac{\mathrm{MUF}_{\mathrm{mid}}}{\mathrm{MUF}_{\mathrm{climo,mid}}} \tag{49}$$

where $f_{\mathrm{oF2}}(\cdot)$ is the climatology of §4.2 evaluated at the reflection point's coordinates, time, and solar zenith.

Legacy fallback path: when it fires, what it does, and how the UI should surface it. When either $\mathrm{MUF}_{\mathrm{climo,mid}}$ or $F_{10.7A}$ is unavailable, `pathMinMuf` (`src/physics/climatology.js`) falls back to an illumination-ratio scaling $\mathrm{MUF}_{\mathrm{mid}} \cdot S(\cos\chi_k, F)/S(\cos\chi_{\mathrm{mid}}, F)$ with $S(c, F) = \max(F, \sqrt{\max(0.05, c)})$ and night-floor $F = 0.4$ (hard-coded; the dynamic formulation `nightFloor()` in §4.5 is exported separately for callers but is *not* called by `pathMinMuf`'s fallback branch in `src/physics/climatology.js`). This is the same form Eq. 49 supersedes; its $\sim\sqrt{0.05} \approx 0.224$ effective night floor systematically over-pessimises long cross-hemisphere paths (e.g. a 12 000 km path with daylight midpoint and a night terminus hop returns $\sim 0.224 \cdot \mathrm{MUF}_{\mathrm{mid}}$ where the local climatology would read 6–10 MHz). The fallback fires in two operationally distinct conditions:

1. **Unit tests** that construct a budget without seeding $F_{10.7A}$ or a climatology midpoint MUF. This is the documented test path and the original motivation for keeping the fallback in the codebase.
2. **NOAA SWPC $F_{10.7}$ outage in production**, which is rare but real — the daily $F_{10.7}$ product has missed publication windows of several hours under solar-region storms and SWPC infrastructure outages over the past two cycles. During such an outage the per-hop

MUF computation silently switches from Eq. 49 to the illumination-ratio form, with no operator-facing indication that the path-specific calculation has degraded.

The UI does *not* currently surface fallback-active state; the band table and Reference Paths panel both render their values as if Eq. 49 had produced them. For the SWPC-outage case this is a defect: an operator looking at a closed-tier verdict on a long polar path during an outage sees a tier produced by the inferior formulation without any signal that the verdict's foundation is degraded. A fix in either of two directions closes the gap. **Surface the fallback**: a kc2g-panel chip reading "$F_{10.7}$ unavailable — per-hop MUF using illumination-ratio fallback" would let the operator weight the verdict accordingly. **Or hard-error the fallback in production**: gate the fallback to test code only and return $\mathrm{MUF_{path}} = \mathrm{MUF_{mid}}$ unchanged on $F_{10.7A}$ outage, so the budget at least uses the upstream kc2g value without applying the inferior shape. The unsurfaced silent fallback is the worst of the three options; shipping with one of the explicit alternatives is preferable.

UI vs. budget MUF. The MUF column on the Reference Paths panel displays $\mathrm{MUF_{mid}}$ from the kc2g consensus blend at the path midpoint — not $\mathrm{MUF_{path}}$. The two coincide for short paths (NVIS, 1-hop) where the midpoint *is* the only reflection point. On multi-hop circuits the budget uses the per-hop minimum (Eq. 49); a small panel-vs-verdict gap remains on paths whose limiting hop sits in a structurally low-foF2 region (deep polar, dip equator). The panel MUF is the midpoint reading; the verdict is the trustworthy summary built from the per-hop minimum.

4.5 Night Floor (CCIR/IRI)

The night floor scales with solar activity and latitude:

$$F = \mathrm{clamp}\left(0.25 + 0.0025(F_{10.7\mathrm{A}} - 70) - 0.10\,\min\left(1,\,\frac{|\phi|}{60}\right),\ 0.20,\ 0.60\right) \tag{50}$$

Implementation status. The dynamic formulation in Eq. 50 is exported by `nightFloor(f107A, latAbs)` in `src/physics/climatology.js` but *is not currently called* by the legacy illumination-ratio fallback in `pathMinMuf`; that fallback hard-codes $F = 0.4$ instead. The production per-hop MUF (Eq. 49) carries its own night behaviour through the night-decay multiplier of Eq. 43 and does not consume F either. Eq. 50 is therefore documented here as the *intended* dynamic floor (a future cleanup would wire `nightFloor()` into the fallback in place of the $F = 0.4$ literal, retiring the contradiction); current production reads the static $F = 0.4$ on the fallback branch and never evaluates Eq. 50. If the legacy fallback is removed entirely (Section 10), both the $F = 0.4$ literal and Eq. 50 can be retired together.

4.6 NVIS Detection for Short Paths

For paths shorter than 500 km the signal travels at near-vertical incidence (NVIS). The relevant frequency limit is the *critical frequency* f_{oF2} (vertical reflection), not MUF(3000) (oblique reflection at ∼3000 km range). Since $\mathrm{MUF(3000)} \approx 3 \times f_{\mathrm{oF2}}$, using MUF(3000) for short paths over-predicts by up to a factor of three.

When a short NVIS path is detected, ionocast substitutes GIRO-observed f_{oF2} with a secant correction for the mild obliqueness of a real (non-zero distance) NVIS hop:

$$\mathrm{MUF_{NVIS}}(d) = f_{\mathrm{oF2}} \cdot \sec\left(\arctan\frac{d}{2\,h_F}\right) \tag{51}$$

For $d = 500$ km at $h_F = 300$ km the secant factor is ≈ 1.30, so the usable frequency is about 30% higher than bare f_{oF2}.

h_F **source.** Both `nvisSecantFactor` (this section) and `hopCeilingKm` (Eq. 19) read the same observed h_{mF2} from the nearest GIRO digisonde when the panel reports it, falling back to 300 km when the station's most recent record has no h_{mF2} value. On a high- solar-flux day with $h_{mF2} \approx 340$ km the F2 hop ceiling extends to ~4258 km (Section 3.8) *and* the NVIS secant rises by ~5 % at $d = 500$ km — so the band-gate $f \leq \mathrm{MUF}_{\mathrm{NVIS}}(d)$ tracks the same upper-band flip the F2 hop count does, instead of leaving the two budgets on different h_F assumptions.

NVIS tail blending (500 km to 1500 km). A hard cutoff at 500 km mishandles intermediate-distance paths where NVIS still contributes alongside the F2 oblique mode. ionocast blends $\mathrm{MUF}_{\mathrm{NVIS}}(d)$ with the path's F2 MUF via a linear ramp:

$$w_{\mathrm{NVIS}}(d) = \begin{cases} 1 & d \leq 500\,\mathrm{km} \\ \dfrac{1500\,\mathrm{km} - d}{1000\,\mathrm{km}} & 500\,\mathrm{km} < d < 1500\,\mathrm{km} \\ 0 & d \geq 1500\,\mathrm{km} \end{cases} \tag{52}$$

$$\mathrm{MUF}_{\mathrm{eff}}(d) = w_{\mathrm{NVIS}}(d) \cdot \mathrm{MUF}_{\mathrm{NVIS}}(d) + \left(1 - w_{\mathrm{NVIS}}(d)\right) \cdot \mathrm{MUF}_{\mathrm{F2}}(d).$$

Below 500 km the budget reads pure NVIS; beyond 1500 km it reads pure F2; in between the two modes share the prediction in proportion to where the takeoff angle sits between near-vertical and oblique. The mode label tracks whichever side carries ≥ 0.5 weight, so the UI's per-band breakdown still reads "NVIS" or "F2" cleanly. This blending is gated on a digisonde f_{oF2} being available and the operating frequency fitting under the geometry-aware NVIS MUF ($f \leq \mathrm{MUF}_{\mathrm{NVIS}}(d)$ from Eq. 51). The dynamic gate covers 30 m / 17 m NVIS at solar maximum and naturally degrades below the foF2 reading at solar minimum.

4.7 Short-Path vs. Long-Path Evaluation

Between any two points on Earth there are two great-circle paths. The long path (LP) traverses the opposite side of the globe and can be open when the short path (SP) is closed, particularly at dawn and dusk on 20 m and 17 m.

For each of the five reference destinations, ionocast computes both the SP and LP midpoints, finds the nearest kc2g ionosonde for each, and evaluates margins on both paths. The LP midpoint is the antipode of the SP midpoint: $(-\phi_{\mathrm{mid}},\ \lambda_{\mathrm{mid}} + 180°)$, with the longitude wrapped back into $[-180°, +180°]$ before the lookup so the kc2g grid index is well-defined. Both paths compete for the best-margin verdict, so an open LP beats a closed SP whenever it offers a better budget.

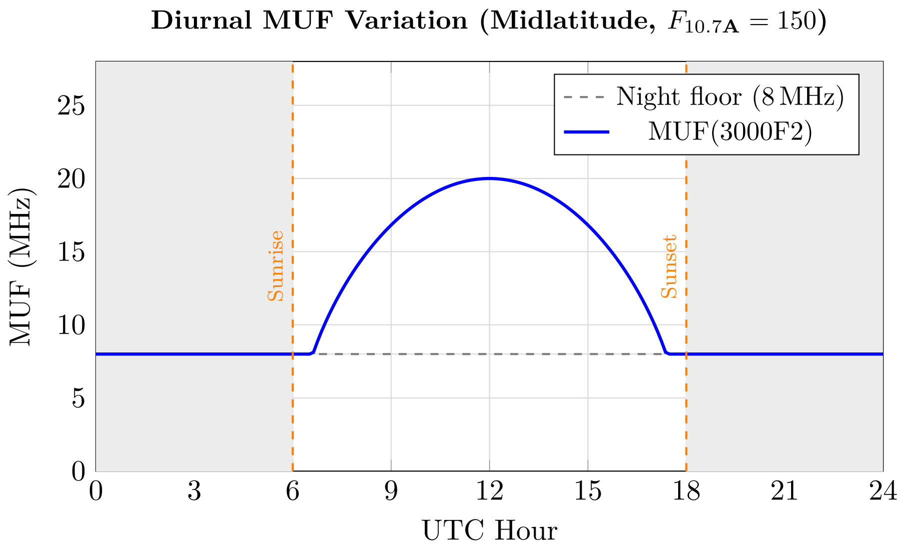

Figure 1: Diurnal MUF variation showing the $\sqrt{\cos\chi}$ shape scaled to a peak of 20 MHz (representative midlat $F_{10.7\mathrm{A}} \approx 150$). The dashed gray line at 8 MHz is the night-floor product $F \cdot \mathrm{MUF}_{\mathrm{peak}}$ from Eq. 50: F is the dimensionless illumination ratio (∼0.40 at midlat moderate solar activity), which when applied to a 20 MHz daytime peak gives the 8 MHz floor curve plotted here. Shaded regions = night.

5 Forward Projection (next 24 h)

> ***Implementation status (entire section).*** *The forward projection pipeline described in this section — MUF projection (Eq. 53), seasonal ratio (Section 5.2), storm depression (Section 5.3), and sporadic-E persistence (Section 5.4) — is documented for a future build of ionocast that exposes a projected outlook MUF as a first-class display. The gray-line bonus (Section 5.5) is the exception: it already runs in the live verdict pipeline (`grayLineBonusDb` in `src/physics/modes.js`, called from `src/derive/conditions.js`) as one of the additive bonuses in Eq. 76, so this section's treatment of it is the* deployed *formulation rather than a forward look. None of the* remaining *forward-projection pipeline (MUF projection, seasonal ratio, storm depression, Es persistence) currently runs in the live runtime; the live runtime instead surfaces SWPC's own next-72 h forecasts on the Outlook panel. Several of the sub-pieces are however shared with the live budget — specifically $d(\cos\chi)$ (Eq. 42) and the storm-lag effective K_p (Section 5.3.1, which feeds L_{aur} and the σ-widening of Section 7.3.1 in the* live *pipeline); the §5.3 inset spells out the live-vs-deferred split for the F_{storm} MUF multiplier specifically. Read this section's equations as the form a future outlook-MUF display would use, not as forward-tense equations whose absence from runtime makes them operationally inert: the parts that touch the live pipeline are already running.*

The forward projection engine extends the current nowcast into the future by modelling the dominant drivers of ionospheric change.

5.1 MUF Projection via Solar Zenith Analogy

Eq. 53 below is specifically the *F2* MUF projection. Sporadic-E persistence (§5.4) is a separate sub-piece: it predicts the projected f_{oEs} used by the Es-mode budget, not a multiplicative factor on F2 MUF. The four F2 factors below (a zenith ratio through $d(\cos\chi)$, a seasonal ratio, and a storm-factor ratio) combine multiplicatively because each ratio reads as the *change* between the now-state encoded in $\mathrm{MUF}_{\mathrm{now}}$ and the future-horizon equivalent.

$$\mathrm{MUF}_{\mathrm{future}} = \mathrm{MUF}_{\mathrm{now}} \cdot \frac{S(d(\cos\chi_{\mathrm{future}}),\, F)}{S(d(\cos\chi_{\mathrm{now}}),\, F)} \cdot R_{\mathrm{seasonal}} \cdot \frac{F_{\mathrm{storm}}(K_p^{\mathrm{fut}})}{F_{\mathrm{storm}}(K_p^{\mathrm{now}})} \tag{53}$$

where K_p^{fut} is the SWPC 3-day forecast value at $t + \Delta$ (or the current K_p when $\Delta = 0$). Both the "now" and "future" zenith arguments pass through the same 3-hour memory-lag function $d(\cos\chi)$ defined in Eq. 42 (Section 4.2); using raw $\cos\chi$ on either side without the lag would produce a discontinuity at the projection's $\Delta = 0^+$ boundary relative to the live nowcast (which already runs through $d(\cos\chi)$), most visibly at sunset where the live model holds foF2 elevated for $\sim$3 h via memory while a raw-cosχ projection would drop it instantly. The storm factor enters as a *ratio* so the projection only adjusts for the *change* in storm phase between now and the projection horizon: if the current K_p already encodes the disturbance, MUF_{now} is already depressed and the ratio is near unity; if a storm is forecast to arrive within Δ but hasn't yet, $K_p^{\text{fut}} > K_p^{\text{now}}$ depresses the projection accordingly. This complements the σ-side forecast bump in Section 7.3 (which widens σ ahead of a forecast disturbance) by adjusting the deterministic mean as well; together they avoid the failure mode where MUF_{now} encodes a quiet ionosphere but a storm has been forecast for the horizon.

5.2 Seasonal Ratio

$$R_{\text{seasonal}} = \frac{s(t_{\text{future}}, \phi)}{s(t_{\text{now}}, \phi)} \tag{54}$$

where

$$s(t, \phi) = 1 + (0.15\cos\theta + 0.08\cos(2\theta + \pi)) \cdot w(\phi) \tag{55}$$

with $\theta = 2\pi\,(d_{\text{yoy}} - d_{\text{solstice}})/365$, where d_{yoy} is the UTC day of year (1 = 1 January) and d_{solstice} is the local-hemisphere winter-solstice day: $d_{\text{solstice}} = 355$ (December 21) for $\phi \geq 0$ and $d_{\text{solstice}} = 172$ (June 21) for $\phi < 0$. The day-of-year phase places the cosine peak at the astronomical solstice and matches Eq. 48 in the climatology so the two seasonal effects share the same phase reference. $w(\phi) = \min(1, |\phi|/50)$ scales the modulation toward midlatitudes. The first term captures the annual winter-peak/summer-trough at midlatitudes; the second captures the semi-annual equinox peaks (March, September) evident in F2-layer statistics. Sharing the day-of-year phase reference with Eq. 48 keeps the two seasonal effects (which overlap in the annual component) aligned.

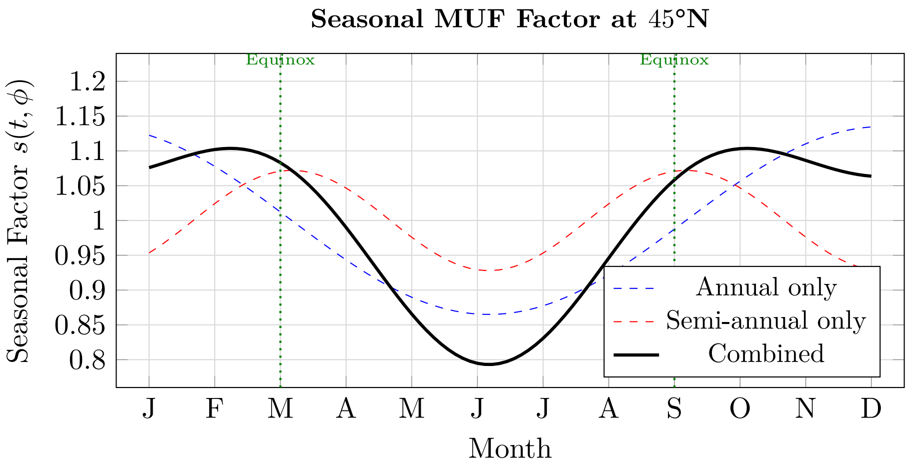

Figure 2: Seasonal MUF factor at 45°N showing annual (winter peak) and semi-annual (equinox peaks) components. At $|\phi| = 45°$, $w(\phi) = 45/50 = 0.9$, so the annual component reaches $1 + 0.15 \cdot 0.9 = 1.135$ in isolation at the winter solstice and the semi-annual component reaches $1 + 0.08 \cdot 0.9 = 1.072$ in isolation at the equinoxes. Because the two components compete at the winter solstice — the semi-annual is at its trough there ($\cos(2\theta + \pi) = -1$ at $\theta = 0$) — the combined curve is flatter than either component alone: the winter solstice evaluates to $1 + (0.15 - 0.08) \cdot 0.9 = 1.063$, the March / September equinoxes to $1 + 0.08 \cdot 0.9 = 1.072$, and the summer solstice to $1 - (0.15 + 0.08) \cdot 0.9 = 0.793$. The dominant seasonal feature at this latitude is therefore the deep summer trough rather than a sharp winter peak; winter and equinoxes form a broad plateau in the 1.06–1.07 range, with the equinoxes slightly above the winter solstice (+0.009). Above $|\phi| = 50°$ the weight saturates at $w = 1$ and the summer trough deepens to 0.77 while the winter / equinox plateau settles at 1.07–1.08.

5.3 Storm Depression (forward-projection definition)

The forward-projection storm depression follows the F-region storm phenomenology surveyed by Prölss [20], Fuller-Rowell et al. [17], and Mendillo [18]: a prompt depression of foF2 driven by Joule heating of the thermosphere and the resulting O/N_2 composition shift, with a recovery timescale set by the storm's driver (impulsive CME vs sustained HSS).

> ***Implementation status.*** *The expressions in this subsection define the F_{storm} multiplier referenced by Eq. 53 (the forward-projection MUF for the outlook section). The* current-time *MUF in production does* not *apply this multiplier: the runtime budget reflects geomagnetic storm response through the storm-lagged effective K_p feeding L_{aur} (Section 3.7 and Section 5.3.1), not as a MUF-side scalar. F_{storm} as a real-time MUF multiplier is reserved for a future build that exposes the projected outlook MUF as a first-class display.*

When $K_p > 4$, the projected MUF is depressed; for $K_p \leq 4$ the factor holds at unity (no quiet-day boost):

$$F_{\text{storm}} = \max(F_{\text{floor}}(K_p),\ 1 - p\ \max(0,\ K_p - 4)) \tag{56}$$

where

$$p = 0.05 + 0.10 \cdot \text{clamp}\left(\frac{|\phi_{\text{CGM}}| - 40}{30},\ 0,\ 1\right) \tag{57}$$

The two latitude anchors place the slope ramp where the underlying F-region storm physics actually transitions. $|\phi_{\text{CGM}}| = 40°$ is the approximate ceiling of the trough region during strong storms in the cycle-25 Dst-vs-foF2 record (consistent with the qualitative trough-edge framing in Mendillo [18] and Prölss [20], though neither prints a literal 40° figure — the specific value is an ionocast calibration anchor): below this, the storm-time foF2 depression is dominated by composition-shift effects whose magnitude is well captured by a gentle $p = 0.05$. $|\phi_{\mathrm{CGM}}| = 70^\circ$ is the equatorward edge of the auroral oval at $K_p \approx 7$ (the storm-zone expansion threshold of Eq. 16's converged state); poleward of it, the F-region sees direct precipitation-driven heating in addition to the global thermospheric shift, and a steeper $p = 0.15$ matches the $\sim$50% MUF collapses observed in IRI-validated ground campaigns. The 30° ramp width connects them linearly; this is geometric interpolation rather than a separate physics prediction. The floor itself drops below 0.5 during severe storms to capture the deep MUF collapse documented for $K_p \geq 8$ events:

$$F_{\mathrm{floor}}(K_p) = \max\Big(0.30,\ 0.50 - 0.10 \cdot \min\big(2,\ \max(0,\ K_p - 7)\big)\Big) \tag{58}$$

At quiet to moderate K_p the floor is at 0.5 (50% of quiet baseline). At $K_p = 8$ (G4) it drops to 0.4; at $K_p = 9$ (G5) it drops to 0.3. The crossover K_p at which the floor binds (i.e. the main term drops below the floor schedule) is

$$K_p^* = 4 + \frac{1 - F_{\mathrm{floor}}(K_p^*)}{p} \tag{59}$$

which evaluates to exactly 8.0 on auroral paths ($p = 0.15$, $|\phi_{\mathrm{CGM}}| \geq 70^\circ$) — so for $K_p \geq 8.0$ the floor schedule binds at polar latitudes (at $K_p = 8$ both the main term $1 - 0.15 \cdot 4 = 0.40$ and the floor $0.50 - 0.10 \cdot 1 = 0.40$ coincide; at $K_p = 9$ the main term gives $1 - 0.15 \cdot 5 = 0.25$, the floor gives $0.50 - 0.10 \cdot 2 = 0.30$, and $\max(0.30, 0.25) = 0.30$). At midlatitude paths the slope $p = 0.05$ keeps the main term above the floor for any plausible K_p ($K_p^* \approx 18$ at $|\phi_{\mathrm{CGM}}| = 40^\circ$, far past the $K_p = 9$ ceiling), so the floor schedule *never binds at midlat* and the floor refinement is auroral-only by construction. The schedule captures the deep polar-F2-MUF collapses below 50% of baseline observed during super-storms (Halloween 2003, March 1989) on the auroral paths those events actually affect.

At midlatitudes ($|\phi_{\mathrm{CGM}}| = 40^\circ$), the damping is 5% per K_p unit above 4. In the auroral zone ($|\phi_{\mathrm{CGM}}| = 70^\circ$), it rises to $\sim$15% per unit.

Negative phase only. F_{storm} models the *negative-phase* F-region storm response (the persistent electron-density depression caused by neutral composition changes and Joule heating that lasts hours-to-days into the storm main and recovery phases). It does not model the *positive phase* that can precede the depression in the first several hours after a sudden CME impact, where penetration electric fields and $E \times B$ uplift can transiently *enhance* foF2 by 15–50 % above the quiet-time baseline (typical durations $\sim$1–4 h, magnitude scaling with storm intensity and latitude per Buonsanto [19] and Mendillo [18]). The result is that the model under-predicts MUF during the positive-phase window of a fast-onset CME event (a quiet-time prediction is correct; the depression engages correctly once the negative phase takes hold, which is typically within a few hours of the impact). In practice this matters for operators chasing the positive-phase opening on the upper bands during the first 1–4 hours of a sudden storm — a relatively rare and short-lived mode — the negative-phase damping that follows is what shapes the bulk of the storm-affected propagation window. When the projected-MUF panel ships (currently documented but not yet surfaced — see implementation status above), it should carry an annotation flagging this gap, paralleling how the scatter (§6.5) and TEP (§6.2) modes are tagged on the verdict line. Until then, operators tracking a fresh CME should trust the alerts-panel B_z / D_{st} / PCA chips for the impact signal rather than expecting the projection to flip positive in the first hour.

5.3.1 Storm-Lag Effective K_p

The F-region does not respond instantaneously to the K_p kick: Joule heating of the thermosphere and subsequent composition changes take 1–3 hours to depress the peak electron density, and the ionosphere then recovers over the following 12–24 hours. Feeding the raw instantaneous K_p into the physics budget therefore mis-times both the onset and the recovery of a storm.

ionocast builds a Laplace-kernel-weighted *effective* K_p from the 7-day SWPC K_p history (3-hour cadence): the kernel $\exp(-|\Delta t - \tau_{\mathrm{peak}}|/\tau_{\mathrm{decay}})$ is symmetric about the peak rather than one-sided exponential, with $\tau_{\mathrm{peak}} = 2\,\mathrm{h}$ in the past and a decay constant τ_{decay} that depends on the storm's driver:

$$K_p^{\mathrm{eff}} = \frac{\sum_i K_p(t_i)\ \exp\!\left(-|\Delta t_i - \tau_{\mathrm{peak}}|/\tau_{\mathrm{decay}}\right)}{\sum_i \exp\!\left(-|\Delta t_i - \tau_{\mathrm{peak}}|/\tau_{\mathrm{decay}}\right)} \tag{60}$$

with $\Delta t_i = t_{\mathrm{now}} - t_i$ (positive for past samples).

Storm-type classification. CME-driven and CIR / HSS-driven storms recover on very different timescales. A CME arrival is an impulse: after the shock and main phase, Joule heating dissipates over $\approx$8 h. A co-rotating interaction region (CIR) or high-speed stream (HSS) is a sustained driver that keeps heating the thermosphere while the stream flows past Earth – recovery takes $\approx$24 h or longer. ionocast picks the kernel as follows:

- If Dst $< -80\,\mathrm{nT}$ the ring current is deep: CME signature dominates, $\tau_{\mathrm{decay}} = 8\,\mathrm{h}$. (If Dst is unavailable from the upstream feed, skip to the next branch rather than treating the missing value as $\geq -80\,\mathrm{nT}$.)
- Otherwise, if NASA DONKI [39] reports an HSS event within the last 24 h or next 48 h, use $\tau_{\mathrm{decay}} = 24\,\mathrm{h}$. The asymmetric window is intentional: DONKI's HSS analyses are typically published within hours of stream onset (so a 24-hour lookback covers events the catalogue has already classified) but stream peak effects on the magnetosphere can lag the L1 arrival by 12–36 hours and persist for another 12–24 hours, so a 48-hour lookahead extends the 24-hour-decay regime through the stream's effective magnetospheric tail. The window mixes two information types: the lookback portion is purely catalogue-historical (events DONKI has already classified) while the lookahead portion is partly forecast (events DONKI has flagged from L1 arrival or solar imaging but whose magnetospheric impact is still developing). τ_{decay} is therefore a forward-looking parameter chosen from a mix of recent and announced catalogue entries; the K_p^{eff} kernel it controls is then applied to the strictly-historical K_p time series.
- Otherwise default to $\tau_{\mathrm{decay}} = 8\,\mathrm{h}$.

Only K_p^{eff} enters the physics; the UI still displays the instantaneous sample so the shown number matches SWPC's feed.

5.3.2 Dst, IMF Bz, and Solar-Wind Drivers

The 3-hour Kp index is a lagging indicator. Three faster-responding inputs are wired into the storm model:

- **Dst** (Kyoto ring-current index): when Dst $< -50\,\mathrm{nT}$, the effective K_p is bumped by up to +2 (1 per 50 nT below −50). Dst responds within minutes of storm onset, 1–3 hours before Kp updates.
- **IMF Bz forward bump** (DSCOVR/ACE L1 solar wind): the L1 monitor sits $\sim$1.5e6 km upstream, giving a nominal 30–60 min lead on the geomagnetic effect at Earth (the spread on the lead time depends on solar-wind speed: 1.5e6 km/400 km/s = 62 min at average speeds, 1.5e6 km/800 km/s = 31 min at fast streams). The L1-to-Earth correlation is *probabilistic*, not deterministic: not every L1-observed Bz dip arrives at Earth's magnetopause unchanged — some structures are spatially extended and miss Earth entirely ($Y_{\mathrm{GSE}} \neq 0$), some glance the magnetopause without sustained reconnection, and some get reshaped by intermediate solar-wind structures during transit. The 20-min median is gated at −5 nT: medians at or above −5 nT (i.e. $B_z \geq -5$) produce no bump; medians strictly below −5 nT (i.e. $B_z < -5$) ramp the bump per Table 7. The cutoff filters out short reconnection bursts that don't drive sustained magnetopause coupling, but the L1 forward bump should be treated as an early-warning probability bump on the effective K_p, not a guaranteed forecast. When the

median Bz over the last 20 min is sustained below threshold, the effective K_p is bumped additively *before* the index reading catches up (Table 7); single-sample dips below threshold do not count. The bump decays to zero once B_z returns to $\geq -5\,\text{nT}$.

- **Solar-wind plasma** (DSCOVR/ACE): sustained speed $\geq 500\,\text{km/s}$ with sharply negative Bz ($\leq -8\,\text{nT}$) is read as a CME shock arrival regardless of DONKI catalogue inertia, while the same speed with mild Bz ($> -5\,\text{nT}$) is read as HSS / corotating-stream interaction. The intermediate regime (500 km/s or faster with $-8 < B_z \leq -5\,\text{nT}$) is ambiguous in the live signal — moderate southward Bz with elevated speed could be either the leading edge of a CME sheath or sustained HSS-driver reconnection — so neither shortcut fires; the classifier falls through to the catalogue / Dst logic below, which defaults to the CME timescale ($\tau_{\text{decay}} = 8\,\text{h}$) unless an active DONKI HSS event covers the current window. Storm-type classification picks the decay time-scale ($\tau_{\text{decay}} = 8\,\text{h}$ vs. 24 h) of the Section 5.3.1 kernel.

Geomagnetic-input clamps. All adjustments above are capped at $K_p = 9$ and apply only when the respective input is available. The auroral CGM threshold expands continuously from 60° (quiet, $K_p \leq 5$) to 50° (severe, $K_p \geq 7$), per Eq. 16 — linear in K_p across the $5 \to 7$ window so the equatorward expansion of the auroral oval enters the budget gradually rather than stepping at a single threshold.

Table 7: Effective K_p bump from a sustained southward-Bz median observed at the L1 monitor (DSCOVR / ACE) over the trailing 20 minutes. The function is a continuous linear ramp from 0 at $B_z = -5\,\text{nT}$ to +3 at $B_z = -15\,\text{nT}$, saturating at +3 for deeper southward IMF. Table values are sample points along the ramp.

Median Bz over 20 min	**K_p bump**
$\geq -5\,\text{nT}$	0
$-6\,\text{nT}$	+0.3
$-8\,\text{nT}$	+0.9
$-10\,\text{nT}$	+1.5
$-12\,\text{nT}$	+2.1
$-15\,\text{nT}$	+3 (saturated)
$\leq -15\,\text{nT}$	+3 (saturated)

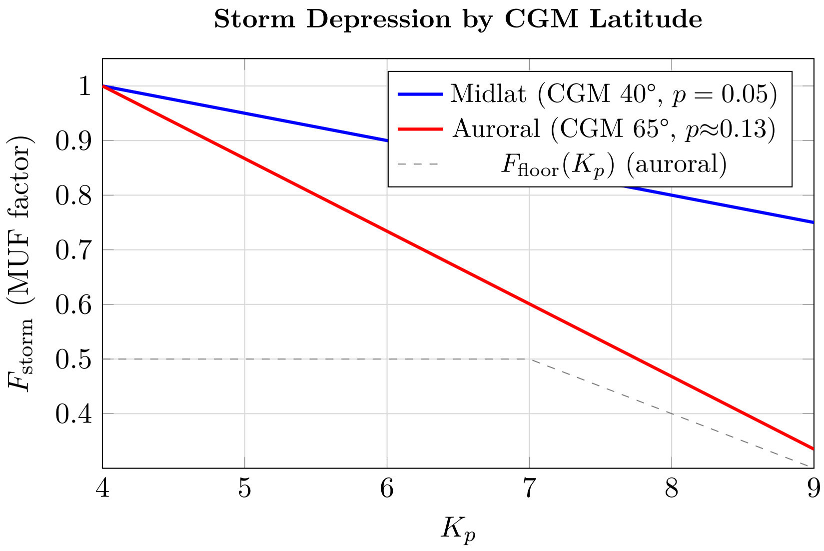

Figure 3: Storm MUF depression factor vs. K_p for midlatitude and auroral-zone paths. The midlatitude floor never binds (slope $p = 0.05$ keeps the main term above 0.5 for all $K_p \le 9$). The auroral curve plotted is at $|\phi_{\mathrm{CGM}}| = 65°$ (where the slope $p \approx 0.13$ is not yet at its $|\phi_{\mathrm{CGM}}| \ge 70°$ ceiling of 0.15); the floor schedule $F_{\mathrm{floor}}(K_p)$ (Eq. 58) drops from 0.5 at $K_p < 7$ to 0.40 at $K_p = 8$ and 0.30 at $K_p = 9$ but only *just* starts to bind on this curve at $K_p \approx 9$. The clean "main term and floor coincide at 0.40 at $K_p = 8$" result discussed in Eq. 59 applies at $|\phi_{\mathrm{CGM}}| \ge 70°$, not at the 65° curve plotted here.

5.4 Sporadic-E Persistence

The projected critical frequency of the sporadic-E layer decays exponentially:

$$f_{\mathrm{oEs}}(t+\Delta) = \max\Big(B(\cos\chi),\ f_{\mathrm{oEs}}(t)\cdot 2^{-\Delta/\tau(\cos\chi)}\Big) \tag{61}$$

with the half-life and floor smoothly interpolated across the twilight band $\cos\chi \in [0.1, 0.3]$ rather than stepped at a single $\cos\chi = 0.2$ threshold. The interpolated forms are

$$\tau(\cos\chi) = 1.5\,\mathrm{h} + 1.5\,\mathrm{h}\cdot R\Big(\tfrac{\cos\chi - 0.1}{0.2}\Big), \qquad B(\cos\chi) = 2\,\mathrm{MHz}\cdot R\Big(\tfrac{\cos\chi-0.1}{0.2}\Big) \tag{62}$$

where $R(x) = \mathrm{clamp}(x, 0, 1)$ is a unit ramp (renamed from $\sigma(x)$ used in earlier drafts to avoid collision with the prediction-spread σ family of §7.3). At $\cos\chi \le 0.1$ both reduce to their night values ($\tau = 1.5\,\mathrm{h}$, $B = 0$); at $\cos\chi \ge 0.3$ both saturate to the day values ($\tau = 3.0\,\mathrm{h}$, $B = 2\,\mathrm{MHz}$). At deep twilight ($\cos\chi = 0.2$) the function is at midpoint ($\tau = 2.25\,\mathrm{h}$, $B = 1\,\mathrm{MHz}$), and the band-MUF gating threshold transitions continuously rather than stepping.

This subsection describes the documented forward-projection persistence model; it is part of the deferred outlook MUF projection (Section 5.1) and is not exercised by the live runtime budget at $t = 0$.

5.5 Gray-Line Bonus (Asymmetric)

The grayline bonus reads as the D-region absorption refund relative to local-noon-overhead, paired with the L_{Dreg} charge of §3.4 via a shared $L_{\mathrm{day}}(f)$ anchor (S0 #2 paired rewrite, landed 2026-05-07; the per-band sunrise/sunset table the prior form used is retired). In closed form,

$$B_{\mathrm{grayline}}(f,\chi) = \min\Big(25\,\mathrm{dB},\ \alpha(\dot\chi)\cdot L_{\mathrm{day}}(f)\cdot\big(1-(\max(0,\cos\chi))^{0.7}\big)\Big), \tag{63}$$

where $L_{\text{day}}(f) = K_D/(f_{\text{MHz}} + 0.5)^2$ from Eq. 6, the inner factor $1 - \cos^{0.7}(\chi)$ goes to 1 at the terminator ($\cos\chi \to 0$, so the bonus saturates at $L_{\text{day}}(f)$ or the cap, whichever is smaller) and to 0 at local noon ($\cos\chi = 1$); a pre-floor at 0.2 dB suppresses sub-decibel noise, and the sunrise/sunset asymmetry multiplier $\alpha(\dot{\chi})$ is

$$\alpha(\dot{\chi}) = \begin{cases} 1.0 & \text{if } \mathrm{d}\cos\chi/\mathrm{d}t > 0 \quad \text{(rising, sunrise)} \\ 0.5 & \text{if } \mathrm{d}\cos\chi/\mathrm{d}t < 0 \quad \text{(falling, sunset)} \end{cases}$$

sampled 5 minutes ahead. The asymmetry preserves the prior table's roughly 2 : 1 sunrise:sunset ratio (D-region recovers more slowly at dusk than the residual F-region enhancement persists, producing a shorter and weaker window than at dawn) without per-band buckets. The frequency window is $1.8 \le f \le 30$ MHz; outside this range the term returns zero.

Band	f	$B_{\max}$	**Band**	f	$B_{\max}$
160 m	1.838 MHz →	25 dB (capped)	20 m	14.097 MHz →	0.94 dB
80 m	3.570 MHz →	12.1 dB	17 m	18.106 MHz →	0.58 dB
60 m	5.366 MHz →	5.80 dB	15 m	21.096 MHz →	0.43 dB
40 m	7.040 MHz →	3.52 dB	12 m	24.924 MHz →	0.31 dB
30 m	10.140 MHz →	1.76 dB	10 m	28.126 MHz →	0.24 dB

$B_{\max}$ in the table above is the sunrise-side terminator peak ($\alpha = 1$, $\cos\chi \to 0$); sunset values are half those after $\alpha(\dot{\chi})$ multiplies in. 160 m saturates the 25 dB cap at the terminator (the underlying L_{day} is 36.6 dB so the literal formula without cap would refund the full noon loss). The cap is operational — per-path absorption already saturates at 50 dB via Eq. 7, so an unbounded grayline bonus would dominate the budget on near-terminator 160 m paths beyond what the calibration basket can validate. Below the 0.2 dB floor the term returns zero, which keeps low-magnitude refunds (deep daytime) out of the verdict loop.

The closed-form replacement is calibration-pending: the prior discrete sunrise/sunset table was hand-tuned against the WSPR per-pair basket that also fixed w_{sc}, and the new formula will need a comparable retune before the bonus magnitudes can be claimed as fitted rather than principled. The S0 #2 framing accepted this trade-off explicitly to land the physics shape now and re-fit later.

Midpoint vs. per-hop evaluation. The proximity $|\cos\chi|$ is computed at the reflection *midpoint* only, not at each hop along the great circle. This is a deliberate simplification: the D-region absorption *loss* term (Eq. 7) is evaluated per-hop and summed, but the gray-line *bonus* compensates loss at the terminator, and on a 4-hop transatlantic path whose midpoint sits in midday but whose terminus hop is at sunrise the per-hop loss already drops on that one hop's L_{Dreg} contribution; adding a midpoint-only bonus on top would over-count the local effect on the geometric-midpoint hop while under-counting the genuine sunrise hops at the path ends. The midpoint-only formulation captures the case the bonus is operationally intended for — short-to-medium HF paths (1–2 hops, 1500–6000 km) whose single or twin reflection points cross the terminator together, where "the path is on the gray line" and "the midpoint is at the terminator" coincide. For longer multi-hop paths (5000 km+) where one terminus is in daylight and the other near sunrise, the per-hop L_{Dreg} already does the right thing on its own. Extending the bonus to a per-hop max is a candidate refinement tracked in Section 10; today's midpoint-only form is simpler to reason about and avoids the over-count.

6 Alternative Propagation Modes

The F2-layer SNR budget (Section 3) is the default path for every HF circuit, but four additional modes can open bands that the F2 budget marks closed: sporadic-E (§6.1), trans-equatorial propagation (§6.2), meteor scatter (§6.3), and F2-region scatter recovery (§6.5). Each runs in parallel with the F2 budget on the same path set; the verdict takes whichever mode has the best margin. Tropospheric ducting (§6.6) is also covered in this section but is structurally different — it is surfaced as a measured per-station panel rather than folded into the budget, because a duct's spatial extent is not resolved by the sparse radiosonde basket the gradient is computed from.

6.1 Sporadic-E as a Propagation Mode

The Es-screening term (L_{Es}, Section 3) only accounts for Es *blocking* an F2 hop. In reality Es can itself be the propagation mode (the framework for which is ITU-R P.534 [5]): a single ∼2000 km hop off the E layer at ∼110 km altitude, with no F2 variability and no auroral absorption. This is the mechanism behind summer-evening 15 m, 12 m, 10 m, and 6 m openings that the F2-only model would otherwise call closed. 6 m only opens via Es when $f_{\text{oEs}} \gtrsim 10\,\text{MHz}$ (so that $5\,f_{\text{oEs}} \geq 50\,\text{MHz}$), which is rarer than the HF-band thresholds and concentrated in the boreal-summer Es season.

ionocast defines a sporadic-E budget alongside the F2 one:

$$M_{\text{Es}} = P_{\text{tx}} + G_{\text{ant}} - L_{\text{fs}}(f, d_{\text{Es}}) - L_{\text{Dreg}} - L_{\text{PCA}} - L_{\text{flare}} - L_{\text{MUF,Es}}(f, 5f_{\text{oEs}}) - L_{\text{iono,Es}} - L_{\text{low}} - N - S_{\text{req}} \tag{64}$$

with $d_{\text{Es}} = 2000\,\text{km}$, a single-hop takeoff angle $\approx 6.3°$, and $L_{\text{iono,Es}} = 15\,\text{dB}$.

What's in $L_{\text{iono,Es}} = 15\,\text{dB}$. The lumped 15 dB is substantially larger than the F2 term $L_{\text{iono}} = 1\,\text{dB}$ (Section 3.11) because the E-layer reflection geometry introduces four mechanisms a clean F2 reflection does not. ionocast does not split the term across those mechanisms today; it is an aggregate that absorbs the four contributions below into a single calibrated number, calibrated against the Es-active subset of the WSPR per-pair basket. Listing the contributors anyway lets a reader judge which component would dominate if the term were ever decomposed: patchy cloud structure (footprint mismatch between TX and RX illumination, ∼3–6 dB on a typical small-to-medium Es patch); strong aspect sensitivity at the layer's tilted surfaces (∼2–5 dB swing depending on takeoff azimuth alignment); polarisation scrambling through a thin gradient layer (∼1–3 dB); and the residual ITU-R Yp day-to-day variability that Es-strength hour-to-hour swings produce on top of the F2 day-to-day spread (∼3–5 dB). These ranges roughly sum to the 15 dB calibration value. Future work may elevate them to named per-mechanism terms; the present paper ships the aggregate.

The D-region terms still apply — the E layer sits above the D layer, so an Es signal still passes the absorbing region once per hop. L_{hop} is zero for Es (single-hop, no intermediate ground reflections). L_{aur} is *omitted* from Eq. 64 entirely (the term is not in the equation), *not* because it can't physically apply — it can; auroral D absorption affects any signal traversing the D-region, and Es hops cross the D-region twice (TX up, down to RX) — but because the geographic regions where Es is the dominant propagation mode overlap rarely with the auroral oval. The $f \geq 14\,\text{MHz}$ operating range and summer-evening seasonality of strong Es push most Es hops to midlat / equatorial corridors away from the auroral CGM threshold. On the rare auroral-zone summer-Es path (Scandinavia, Alaska, Antarctic) where both mechanisms co-occur the budget under-predicts absorption; treat "$L_{\text{aur}} = 0$ on Es" as a geographic simplification rather than a physical claim.

Gating in code: `snrMarginHfEs` returns null below 14 MHz regardless of f_{oEs} (`src/physics/snr.js`), even though the algebraic Es MUF gate $f \leq 5f_{\text{oEs}}$ would otherwise admit lower bands on strong Es days ($f_{\text{oEs}} \geq 4\,\text{MHz}$ pushes $5f_{\text{oEs}} \geq 20\,\text{MHz}$, which would activate the 30 m and 40 m bands without the floor). The hard 14 MHz floor matches amateur-radio literature: Es is essentially unobserved below 20 m at the 2000 km characteristic single-hop distance the budget assumes, so the algebraic gate alone would over-fire on the lower HF bands. The cutoff is intentionally *conservative*: rare Es openings on 30 m and 40 m do appear in the WSPR record during peak-cycle summer evenings with $f_{\mathrm{oEs}} \geq 8$ MHz, typically on shorter-than-2000 km paths where the geometry makes the lower-band Es viable. These events are not captured by the budget; an operator who sees 30 m WSPR activity that the model calls F2-only should treat the verdict as "F2 explains most of it; rare Es is possible on top and the budget won't tell you." L_{low} in Eq. 64 is therefore numerically zero under the current implementation (Table 2 returns 0 for $f \geq 14$ MHz); it is kept for symbolic symmetry with the F2 budget so that any future relaxation of the 14 MHz floor automatically picks up the low-band penalty without a separate edit. Es variability adds +2 dB in quadrature to the base σ.

6.2 Trans-Equatorial Propagation

Chordal afternoon / evening mode across the magnetic equator. The signal couples into the equatorial F-region irregularity structure and traverses hundreds to thousands of km without ground-bouncing, skipping several D-region traversals. Classical TEP produces 15 m / 12 m / 10 m and occasionally 6 m openings between stations whose magnetic dip latitudes sit on opposite sides of the equator.

The TEP bonus is a smooth function of three eligibility axes plus a hard cross-equator gate. Hard gate (binary): both endpoints' dip latitudes have opposite signs (i.e. $\phi_{\mathrm{dip,src}} \cdot \phi_{\mathrm{dip,dst}} < 0$); same-hemisphere paths see no TEP at any distance. Once the cross-equator gate fires, three smooth ramps modulate the bonus magnitude:

- **Frequency factor**: ramps from 0 at 14 MHz to 1 at 22 MHz, holds at 1 across the $[22\,\mathrm{MHz}, 50\,\mathrm{MHz}]$ plateau, then ramps back down to 0 at 60 MHz. This eliminates the earlier hard step at 20 MHz that placed 17 m at zero TEP and 15 m at full plateau across a 3 MHz band edge.
- **Local-hour factor**: ramps from 0 at LT = 16 to 1 at LT = 17 at the leading edge, plateau over $[17, 23]$, then 1 at LT = 23 to 0 at LT = 24 at the trailing edge.
- **Dip-latitude factor**: each endpoint's $|\phi_{\mathrm{dip}}|$ contributes a ramp $f_{\mathrm{dip}} = \mathrm{clamp}((|\phi_{\mathrm{dip}}| - 5°)/5°, 0, 1)$, i.e. zero below $|\phi_{\mathrm{dip}}| = 5°$, rising linearly to one at $|\phi_{\mathrm{dip}}| = 10°$, then saturating. The path-level factor is the product of the two endpoint ramps ($f_{\mathrm{dip,src}} \cdot f_{\mathrm{dip,dst}}$), so a path with one endpoint near the dip equator gradually loses TEP eligibility as that endpoint approaches the equator and loses it entirely once $|\phi_{\mathrm{dip}}| < 5°$.

The bonus is the product of the plateau value and the three factors:

$$B_{\mathrm{TEP}} = B_{\mathrm{TEP,plateau}}(F_{10.7\mathrm{A}}) \cdot f_{\mathrm{freq}} \cdot f_{\mathrm{LT}} \cdot f_{\mathrm{dip,src}} \cdot f_{\mathrm{dip,dst}} \tag{65}$$

where the plateau magnitude is solar-cycle-keyed via a smooth sigmoid:

$$B_{\mathrm{TEP,plateau}}(F_{10.7\mathrm{A}}) = 8 + \frac{7}{1 + e^{-(F_{10.7\mathrm{A}} - 125)/30}}\ \mathrm{dB} \tag{66}$$

The sigmoid asymptotes to ∼8 dB at solar minimum and ∼15 dB at solar maximum, with the transition centred near $F_{10.7\mathrm{A}} = 125$. At quiet sun the plateau is ∼ 9 dB; at moderate cycle ($F_{10.7\mathrm{A}} = 120$) it sits at ∼11 dB; near solar peak ($F_{10.7\mathrm{A}} = 180$) it reaches ∼14 dB. Real TEP intensity scales with EUV (and therefore F10.7), so the plateau follows F10.7A rather than holding at the peak-cycle magnitude across all conditions.

The full bonus saturates at the central 22–50 MHz / 17–23 LT / opposite-hemisphere near-equatorial-pair window (frequency factor saturates at 22 MHz, both endpoints' $|\phi_{\mathrm{dip}}|$ above the 10° saturation knee). This is a first-order approximation — real TEP magnitude varies with Es index, EIA crest strength, and local evening $E \times B$ drift — but it captures the main effect with the smooth-ramp philosophy applied to the rest of the budget: bands that the F2 model says are closed can in fact be wide open across the equator for a few-hour window.

6.3 Meteor Scatter Floor

6 m and 2 m can sustain brief burst-mode QSOs off meteor ionisation trails, particularly in the predawn hours when Earth's leading face sweeps up the most meteors. ionocast lifts the VHF verdict floor from *closed* to *poor* during the predawn window ($[2, 10]$ h local at the QTH) under either of two conditions:

- A major IMO shower with ZHR $\geq$ 20 in its *active* or *building* window per the IMO catalogue (the active window is ± 2 d of peak; the building window is the leading edge of the shower as IMO catalogues it, typically a few days before active begins, depending on shower). The verdict note names the shower (e.g. "Perseids"). The ZHR $\geq$ 20 threshold catches eight to ten showers per year at typical peak ZHR, including (in approximate ZHR order) Quadrantids ~120, Geminids ~120, Perseids ~100, Eta Aquarids ~50, Southern δ Aquariids ~25, Orionids ~20, Lyrids ~18 (just below the gate; included in some IMO catalogue years), Leonids ~15 (outside outburst years; an outburst year clears the gate easily), Ursids ~10 (below gate), and several minor regulars. The exact count varies year to year as IMO refines the catalogue. Strong-but-sub-threshold annual showers like Lyrids and outside-outburst Leonids sit just below the gate, and the >100 minor showers IMO catalogues with ZHR < 20 are excluded, on the assumption that their incremental ionisation above the sporadic background isn't large enough to shift the verdict from *poor* to anything higher.
- Otherwise, *sporadic background* – the ~5–10 ZHR continuous meteoroid influx that sustains routine MS QSOs every dawn even on shower-free days. The verdict note reads "sporadic background".

Poor signals "viable with effort, not sustained" — MS is bursty, not continuous. Earlier code missed the sporadic-background case and only triggered on shower days, leaving 6 m / 2 m flagged *closed* during the routine MS window most of the year. The $[2, 10]$ h centre window is now flanked by 30-minute linear ramps on each side (the activation weight $\in [0, 1]$ ramps $0 \rightarrow 1$ over $[1.5, 2.0]$ h LT and $1 \rightarrow 0$ over $[10.0, 10.5]$ h LT). The boolean MS-active flag fires on any non-zero weight, so the operationally-firing window extends 30 minutes on each side of the centre band; the previous tier cliff at the boundary has been moved to the weight=0 edges where MS activity is genuinely below useful threshold. This matches the smooth-ramp philosophy applied at every other gate (auroral $c(\phi)$, Bz, twilight r, near-MUF, Es, terminator σ, forecast-Kp).

6.4 VHF Auroral-E Scatter

When the auroral oval is illuminated (OVATION HP $\geq$ 30 GW), 6 m and 2 m can open via E-region scatter off the auroral electron precipitation column. `snrMarginVhfAurora` (in `src/physics/snr.js`) returns a simple lumped-loss budget gated on HP — the function is a hard `return null` for HP < 30 GW, so no VHF-aurora verdict is even computed below the threshold, which keeps the path off the band-table when auroral activity is too low for E-scatter to be physically supported — with an auroral-MUF model that rises with hemispheric power ($\mathrm{MUF_{aur}} = 50 + (\mathrm{HP_{GW}} - 30)$ MHz, so 50 MHz at the threshold and ~70 MHz at HP = 50 GW, with no upper cap in code so an extreme HP ~ 200 GW would notionally lift $\mathrm{MUF_{aur}}$ to ~ 220 MHz; in practice the gate at HP $\geq$ 30 GW rarely exceeds ~ 100 GW even during G5 events so the unbounded form has not been observed to misbehave, but a hard cap at e.g. ~ 100 MHz is candidate hardening tracked in §10):

$$M_{\mathrm{aur,VHF}} = P_{\mathrm{tx}} + G_{\mathrm{ant}} - L_{\mathrm{fs}} - L_{\mathrm{MUF}} - L_{\mathrm{iono,aur}} - N - S_{\mathrm{req}} \tag{67}$$

with $L_{\mathrm{iono,aur}} = 25$ dB (`L_IONO_AUR_DB` in `src/physics/loss.js`). This is an *empirical lump* (no single-source citation): the value collapses several mechanisms operators report on auroral-E paths — polarisation scrambling on the auroral E-layer column (~ 4–8 dB swing), aspect- sensitivity loss against operator-side antenna-pattern misalignment (~ 3–6 dB on a typical narrow-beam VHF setup), and the residual ionospheric population variability that swings hour-to-hour during active-aurora windows ($\sim$4–8 dB) — which sum to a 25 dB midpoint operationally. The three contributing ranges are themselves anecdotal in the VHF contest / DXer literature rather than measured against a calibrated reference; the 25 dB number is therefore *calibration-pending* (needs author confirmation against 6 m / 2 m auroral logs) and is held as a defensible default until the next harness pass. L_{aur} (§3.7) does not appear in Eq. 67 because the path is reading *on* the auroral E-region scatter rather than passing through it as absorption; the same physics that closes HF over auroral hops opens 6 m and 2 m scatter on the same hops.

6.5 F2 Scatter Recovery

When the band is nominally above MUF the F2 budget says *closed*, but real F-region irregularities (TIDs, plasma blobs, gradient-driven instabilities) often keep the path partly viable via scatter modes that do not require strict over-MUF compliance at every hop. ionocast reads inverse-square-distance-weighted foF2 from all GIRO digisondes within 3000 km of each F2 reflection point along the great circle, falling back to climatology (Section 4.2) at hops out of station range, and computes the per-hop foF2 standard deviation σ_f. The inverse-square weighting biases the spread estimate toward the nearest station: on a hop with one station at 200 km and five stations at 1500 km, the near station carries $\sim$80% of the total weight, so σ_f collapses toward the near reading and the variance signal that B_{scatter} is supposed to detect can be artificially narrowed. This is a known statistical bias rather than a coding mistake — the weighted-variance formula is well-defined and the resulting estimator is consistent, just not the unbiased equal-weight variance. The choice trades a slightly under-detected B_{scatter} on station-asymmetric hops against larger variance on equal-weight estimates from the long-tail stations, which empirically matter less to the path-averaged spread. A distance-window-uniform weighting is a candidate alternative; re-fitting w_{sc} under that scheme is gated on the per-pair WSPR ground-truth mode (Section 7.2). The scatter bonus fires only when all three gating conditions are met simultaneously: ≥ 2 GIRO stations contribute to σ_f, the path takes ≥ 2 hops, and $f/\text{MUF} > 1.0$ (below MUF the standard model already says "open"). When fired, an additive scatter bonus is applied to the SNR margin:

$$B_{\text{scatter}} = w_{\text{sc}} \cdot \min(1,\ \sigma_f/1.5\,\text{MHz}) \cdot \min(2,\ f/\text{MUF} - 1) \cdot 5\,\text{dB} \tag{68}$$

with $w_{\text{sc}} = 1.5$ from the joint coordinate descent of §7.2 (the harness's unrestricted sweep keeps lifting w_{sc} toward 4 against the global "alive somewhere" metric, but that range exploits a known structural ceiling of the metric rather than a real propagation effect; the 15 dB-cap physical edge 1.5 ships). The foF2-variance term saturates at 1.5 MHz spread (substantial gradients across the path); the above-MUF excess saturates at $f/\text{MUF} = 3$ (wildly over-MUF paths get no recovery). At full saturation $w_{\text{sc}} = 1.5$ contributes up to 15 dB of recovery, sitting inside the operator-experience F2-scatter range (10–25 dB, consistent with K9LA's propagation-tutorial archive [15] for the upper HF above-MUF regime, where formal published F2-scatter loss-magnitude measurements are sparse) and matching the upper-band lift the model needed to track 30 d of WSPR aggregate openings on 17–10 m. The two saturating $\min(\cdot)$ terms are multiplied independently, but σ_f and f/MUF are not strictly independent in practice: paths with high σ_f (large gradients across the path) tend to have at least one hop near or above MUF (observed empirically in the per-(path, band, hour) cells of `harness.report.json`, not separately quantified or literature-cited; treat as a calibration-side observation rather than a published correlation). The independent saturation form slightly inflates the joint upper-bound probability; the effect is small (≤ 1 dB on the 15 dB ceiling) and does not change the calibrated w_{sc}.

The weight was set by joint coordinate descent over $\{L_{\text{iono}},\ D,\ w_{\text{sc}},\ w_{\text{NVIS-tail}}, w_{\text{Es-prim}},\ \sigma\text{-scale},\ \text{fusion-flag}\}$ from three deterministic starting configurations (Appendix E, "baseline", "fusion-up", "modes-on") across 35 reference paths (Section 7.2). All seeds converged to the same basin: only w_{sc} produced measurable lift; an unrestricted sweep keeps improving as $w_{\text{sc}} \to 4$, but that range exploits the WSPR-global-vs-path-specific structural mismatch in the harness rather than a real propagation effect, so the shipped weight is held at the 15 dB-cap physical edge. The result is that the calibrated $w_{\text{sc}} = 1.5$ has been validated against a metric whose calibrators do not fully trust beyond a certain point. Closing this loop properly requires re-running the sweep against the per-pair WSPR ground truth (Section 7.2) once that mode is wired into `replayMarginFromCell`'s alternative-mode bonuses; the new optimum, whatever it is, can then be reported against a metric that the calibrators *do* trust at all weights in its range. Until that closure, treat $w_{\text{sc}} = 1.5$ as a defensible-but-not-final value tied to a metric with a known ceiling.

Per-hop fusion (deferred). The same per-hop foF2 sampling is also wired as an optional second-opinion source for the MUF consensus blend (§4, where the `fusion-flag` is introduced as one of the calibration parameters of Appendix E), behind the `FUSION_PRIMARY_MUF` flag. In the same calibration sweep the flag stayed off: single-midpoint fusion did not beat the kc2g ↔ climatology blend in either Brier or accuracy, and per-hop minimum fusion regressed binary accuracy by ∼1 pp because the rebuilt climatology (Section 4.2) tightened the residual it would have to beat. A future calibration that retunes the budget jointly with fusion enabled may flip this; the wiring is intentionally preserved.

6.6 Tropospheric Ducting

The four modes above all live inside the SNR budget: each one contributes a margin or bonus that competes with the F2 verdict for the same path. Tropospheric ducting is structurally different. It is a VHF-only line-of-sight extension whose viability depends on the shape of the lowest 1 km of the troposphere on the great circle between TX and RX, not on any quantity the global F2 model already evaluates. ionocast surfaces it as a measured per-station panel (Section 9) rather than folding it into the budget; the rest of this subsection documents the physics, the classification, and how the panel is wired.

Refractivity and the duct condition. Atmospheric refractivity N, in N-units, is computed from radiosonde temperature, pressure, and water-vapour pressure (ITU-R P.453 [6]):

$$N = \frac{77.6}{T}\left(P + \frac{4810\,e}{T}\right), \tag{69}$$

with T in kelvin, P and e in millibar. Water-vapour pressure e is obtained from the dew point via the Magnus saturation formula

$$e_{\text{sat}}(t_C) = 6.112 \cdot \exp\left(\frac{17.62\,t_C}{243.12 + t_C}\right), \tag{70}$$

evaluated at the dew-point temperature (so that $e = e_{\text{sat}}(t_{\text{dew}})$ is the actual partial pressure). The constants $(17.62, 243.12)$ are the Sonntag 1990 [25] coefficients for saturation *over liquid water* (recommended by WMO 2008 over the Alduchov–Eskridge alternative, which gives slightly different values $a = 17.625$, $b = 243.04$). Below freezing, saturation over ice uses different constants $(22.46, 272.62)$ producing a ∼10% lower e_{sat} at $t_C = -20\,°\text{C}$. ionocast applies the liquid-water form across all temperatures, which is correct for the warm-water duct cases where ducting is operationally relevant (Mediterranean, Sea of Japan, California coast) but slightly over-estimates e on continental winter soundings where the surface dew point is below freezing. The resulting bias on N is $\lesssim$2 N-units across the operationally relevant range $t_C \in [-30, +35]\,°\text{C}$ (worst case ∼2 N-units near $t_C = -20\,°\text{C}$ where the ice-vs-liquid divergence peaks; below $-30\,°\text{C}$ the absolute moisture content is so low that the absolute e error shrinks even though the percentage error grows), and the regime classification (standard / super-refractive / ducting) is robust to it; refining this further would require a piecewise formula gated on $t_C \gtrless 0\,°\text{C}$, which is candidate cleanup tracked in Section 10. Modified refractivity M adds a curvature correction for the Earth-radius bend that makes a horizontally-stratified duct look horizontal in M-coordinates,

$$M(z) = N(z) + 0.157\,z, \tag{71}$$

with z in metres. A horizontal duct exists wherever $\mathrm{d}M/\mathrm{d}z < 0$; equivalently, in the N-coordinate that ionocast actually computes,

$$\left.\frac{\mathrm{d}N}{\mathrm{d}h}\right|_{0\to 1\,\mathrm{km}} < -157 \text{ N-units/km}. \tag{72}$$

The -157 value is the modified-refractivity coefficient 0.157 N/m rescaled to N-units per kilometre. It is not a loss budget — a duct either exists or it does not — so the treatment here is classification, not a margin contribution.

Bean & Dutton [24] classification. The lowest-1 km gradient is bucketed into three regimes (thresholds and labels follow Bean & Dutton; ITU-R P.453 [6] provides the underlying refractivity formulation but tabulates only ≤ -100 N-units/km gradients without printing the -79 / -157 classification boundaries):

- $\mathrm{d}N/\mathrm{d}h > -79$ N-units/km: *standard atmosphere*. Radio rays bend gently toward Earth (the textbook $\frac{4}{3}$ effective Earth radius); no range extension beyond the geometric horizon.
- $-157 \leq \mathrm{d}N/\mathrm{d}h \leq -79$ N-units/km: *super-refractive*. The radio horizon extends past the geometric horizon; tropospheric scatter paths run noticeably above their fair-weather budget. Common precursor to ducting.
- $\mathrm{d}N/\mathrm{d}h < -157$ N-units/km: *ducting*. A surface-based or elevated atmospheric duct exists; VHF/UHF signals trapped in the duct can reach hundreds to thousands of kilometres independent of the ionosphere. Operationally this opens 6 m and 2 m DX between stations sharing the duct, particularly across warm-water surfaces (Mediterranean, Sea of Japan, California coast) and under summer subsidence inversions.

Implementation. The $\mathrm{d}N/\mathrm{d}h$ panel is built from a 24-station basket of UWyo radiosonde sites with global coverage (Europe ×10, North America ×7, SE Asia ×3, Africa ×2, South America ×1, Antarctica ×1; `SONDE_STATIONS` in `functions/_handlers/refractivity.js`). The Cloudflare worker (`functions/_handlers/tropo.js`) fans out over the basket in parallel for the most recent 00 Z or 12 Z slot (with a 12 s per-fetch timeout to bound the worst case), parses each PRE-block sounding, and for each station:

1. Locates the surface sample (lowest row) and the sample nearest $z_{\mathrm{surface}} + 1\,\mathrm{km}$ that is at least 100 m above the surface.
2. Computes N_{surface} and N_{upper} via Eq. 69 from temperature, pressure, and dew-point at each level.
3. Returns $\mathrm{d}N/\mathrm{d}h = (N_{\mathrm{upper}} - N_{\mathrm{surface}})/\Delta z_{\mathrm{km}}$ and the regime classification.

Stations whose 00 Z / 12 Z release is missing from the upstream archive return `ok: false` with reason `"no data"`, which the ducting panel renders as a placeholder row. The nearest station with valid data is also surfaced as a single $\mathrm{d}N/\mathrm{d}h$ cell on the VHF band-table, sharing one cell across 6 m and 2 m because the gradient is per-station, not per-band; the cell's tier color uses the same classification (good = ducting, warn = super-refractive, no color = standard) so the band-table verdict and the per-station panel never disagree on what the local atmosphere is doing.

Why this is informational, not a budget term. A duct is only useful if both endpoints sit inside it. The radiosonde network samples vertical structure at fixed sites every 12 h, which is enough to tell an operator at one of those sites whether the local atmosphere is currently trapping signals, but it is not enough to claim a specific TX-to-RX path is ducted: marine-layer ducts can hug a coastline for hundreds of kilometres while inland soundings show standard atmosphere only 50 km away, and elevated subsidence ducts can sit several hundred metres above a surface that a radiosonde column doesn't cleanly resolve. Folding the gradient into a path-level SNR bonus would over-promise the model's spatial coverage; surfacing the per-station classification lets the operator make the geographic inference where the radiosonde basket is dense (Western Europe, the contiguous US) and lets it stay silent where the basket is thin (most of Africa, Central Asia, the open ocean). The companion gridded surface (§6.7) fills the spatial coverage gap with a NOAA GFS-derived per-cell index that the radiosonde basket is then used to calibrate.

6.7 Global Tropospheric Ducting Heatmap

The 24-station radiosonde basket of §6.6 answers "is the local atmosphere ducting?" wherever a sounding lands within the user's region of interest, but most of the planet sits hundreds of kilometres from the nearest radiosonde site. To give operators a global at-a-glance view, ionocast also ingests the NOAA Global Forecast System (GFS), computes a per-cell composite ducting index on a regular lat/lon grid, and renders a Hepburn-class heatmap embedded in the VHF panel. The radiosonde-per-station table of §6.6 is retained as the calibration anchor: its P.453 classifications are the falsifiable ground truth against which the gridded index is tuned.

Forecast input. GFS at 0.25° is the densest public NWP product covering the global troposphere with a refresh cadence faster than 6 h. ionocast pulls 13 pressure levels between 1000 and 500 hPa (1000 / 975 / 950 / 925 / 900 at 25 hPa spacing, then 850 / 800 / 750 / 700 / 650 / 600 / 550 / 500 at 50 hPa, the densest set NOMADS publishes on its `pgrb2.0p25` endpoint [10]) plus the surface bundle: 2 m temperature, 2 m dew point, skin temperature, land-sea mask, surface pressure, model orography, planetary-boundary-layer height, and 10 m wind components. Each cycle is filtered server-side by the NOMADS subset URL down to ∼21 MB (vs ∼500 MB for the unfiltered 0.25° file). The 4 × daily refresh runs ∼5 h after each cycle (00 / 06 / 12 / 18 Z) via a GitHub Actions workflow, packs the result to a compact binary (∼2.1 MB row-major `Float32`, ∼10× smaller than the JSON form), and uploads to Cloudflare R2 for browser delivery.

Index composition. The per-cell index $\mathcal{T}$ is a seven-term composite, height-weighted in the lowest 2.5 km AGL (weight 1.0 below 1.5 km, 0.5 between 1.5 and 2.5 km, zero above since amateur VHF/UHF cannot couple meaningfully into upper tropospheric ducts) and attenuated by surface mixing:

$$\mathcal{T} = (1 - w_p) \cdot [\,5\,\Delta_{\mathrm{duct}} + S_{\mathrm{sr}} + S_{\ell} + S_{\mathrm{surf}} + S_{\mathrm{marine}} + S_{\mathrm{HPBL}} + S_{\mathrm{evap}}\,]\,, \tag{73}$$

where the terms, in M-units, are:

- Δ_{duct}, the ITU-R P.834 [9] canonical duct intensity $M_{\mathrm{top}} - M_{\mathrm{min}}$ per trapping layer, taken as the maximum across all trapping layers in the column. Weighted 5× so a 30-unit duct contributes 150 to $\mathcal{T}$ and saturates the renderer's 200-unit colormap.
- S_{sr}, the super-refractive sum: per-layer $(-79 - \mathrm{d}N/\mathrm{d}h)$ accumulated over every layer whose gradient sits in the P.453 super-refractive band $[-157, -79]$ N-units/km. This term carries the broad marine baseline (subsidence inversions over the subtropical highs, sub-textbook stratification) that the binary "is there a duct" check misses.
- S_{ℓ}, the per-layer temperature-inversion sum, triggered when a layer's dT exceeds the standard-lapse value by $\geq$3 K.
- S_{surf}, the surface inversion bonus, comparing T_{2m} to the lapse-adjusted temperature at the first pressure level $\geq$ 200 m above ground; threshold 1 K, capped at 100.
- S_{marine}, the marine inversion bonus, fires when SST exceeds 2 m air temperature by $\geq$ 1.5 K over an ocean cell (gated by GFS land-sea mask), capturing Benguela / California-current cold-tongue and Mediterranean warm-water stratification. Capped at 60.

- S_{HPBL}, a capping-inversion bonus that activates when GFS's planetary-boundary-layer height $z_{\mathrm{HPBL}} < 1500\,\mathrm{m}$ *and* another inversion term has already fired. This is the textbook capping-inversion duct setup: a shallow well-mixed BL trapped under a strong inversion. Up to +25.
- S_{evap}, the saturation-deficit evaporation duct bonus. Over open water the air immediately at the sea surface is saturated by definition; the deficit $e_{\mathrm{sat}}(T_{\mathrm{sea}}) - e(T_{2m,\mathrm{dew}})$ scales the strength of the resulting near-surface refractivity gradient. Replaces an earlier column-precipitable-water proxy that fired on bulk humidity rather than the surface gradient that actually drives evap-duct height. Capped at 30.
- $w_p \in [0, 0.6]$, the wind-mixing penalty: 10 m wind speeds above 8 m/s degrade the laminar boundary-layer structure that ducts depend on, attenuating $\mathcal{T}$ by 10% per m/s above 8 m/s up to a 60% reduction at 14 m/s.

A super-adiabatic lapse anywhere in the lowest 2 km marks the column as convectively-mixed and divides S_ℓ by 5 and Δ_{duct} by 2; surface, marine, and evaporation bonuses are unaffected because they measure surface-anchored quantities the convective check (a layer-gradient probe) cannot invalidate.

Pressure-level density and ECMWF comparison. A side-by-side calibration against the same 24-station P.453 panel of §6.6 makes the case for GFS over ECMWF Open Data on this specific application: the ECMWF Open Data feed publishes only five pressure levels between 1000 and 500 hPa (1000 / 925 / 850 / 700 / 500), giving 75 to 150 hPa gaps that average across the 50 to 300 m thick layers where most operational ducts physically live. Run with the same index formula, the ECMWF feed scored 0% recall on observed super-refractive layers; the same harness against the GFS feed scored 80%. ECMWF retains a synoptic-skill edge on multi-day large-system forecasts that is not relevant for current-state ducting. The full ECMWF-IFS 137-model-level product would resolve sub-100 m structure but is not part of the free Open Data feed.

Calibration. The calibration harness (`src/tropo/calibrate.mjs`) fetches the same 24-station UWyo basket, scans each sounding for its strongest sub-3 km gradient (rather than the bulk-1 km gradient §6.6 reports per-station, which averages over thin surface inversions), classifies it per Bean & Dutton [24], samples $\mathcal{T}$ at the station coordinates from the most recent grid, and reports three-class agreement plus per-class precision and recall. Across 13 amateur-relevant sondes per cycle (those whose strongest gradient sits at or below 2.5 km AGL, matching the height-weighting cap), the optimal cuts come out near

$$\mathcal{T} < 5 : \text{standard}; \quad 5 \le \mathcal{T} \le 90 : \text{super-refractive}; \quad \mathcal{T} > 90 : \text{ducting}, \tag{74}$$

with 100% precision in the ducting band (when the index reads ≥90 M-units a sonde at the same coordinates agrees the column is ducting) and ∼17% recall. The recall ceiling is GFS's vertical resolution rather than the formula: surface and shallow ducts in 50 to 300 m thick layers fall between the model's 25 to 50 hPa pressure-level spacing and read as $\mathcal{T} = 0$. That is a known and bounded data-source limit; the operational implication is that bright cells on the heatmap are reliable but the absence of color does not rule out ducting.

Rendering. The browser fetches the packed binary from Cloudflare R2 via a same-origin redirect, decodes the row-major `Float32` arrays, applies a separable $[1, 4, 6, 4, 1]/16$ binomial Gaussian to soften per-cell jitter (matching the spatial smoothing every operational forecast product applies before contouring), and runs marching-squares contour extraction [12] to produce closed polygon boundaries at 15 thresholds spaced

$$\mathcal{T}_i = \mathcal{T}_{\min} + \left(\frac{i}{N}\right)^{1/0.7} (\mathcal{T}_{\max} - \mathcal{T}_{\min}), \qquad i = 0, 1, \ldots, 14 \tag{75}$$

with $\mathcal{T}_{\min} = 20$ and $\mathcal{T}_{\max} = 200$. The 1/0.7 exponent stretches the lower bands so the weak-but-real subtropical-marine baseline gets two or three colored bands rather than a single navy stripe; the strongest 1% of the globe saturates at the deepest red without compressing the more common 30 to 100 range into purple alone. MapLibre paints each band as a fill layer with a hard color edge, identical to the meteorological-forecast contour-fill look used by William R. Hepburn's worldwide tropo forecasts [11] and Met Office surface charts. Black isolines from a marching-squares overlay run on the source grid (unsmoothed) and trace the data boundaries on top of the band fills.

Why this is not folded into the SNR budget. The heatmap is a global field of point classifications, not path-level loss data. A bright cell over the Mediterranean tells operators on both Italian and Spanish coasts that their local atmosphere is trapping VHF; whether a Genoa-Barcelona path is itself ducted depends on *both* endpoints sitting inside the same duct *and* the great-circle staying inside the duct between them, which the gridded forecast cannot adjudicate without ray-tracing through the M-profile along the path. Surfacing the field as a heatmap lets operators make that geographic inference at a glance the way Hepburn's maps have been used for decades, while keeping the SNR-budget side of ionocast honest about what a single forecast column can support.

7 Ensemble Blend and Self-Calibration

7.1 Verdict Margin

The verdict margin is the physics-budget margin M_{physics} (Eq. 1) plus the per-hop and alternative-mode bonuses, with no further blending or bias correction. Gray-line is additive (it describes a distinct D-region attenuation drop at the terminator), but TEP and scatter recovery both describe F-region irregularity-driven recovery and so are *combined by maximum* rather than summed — on a TEP-eligible 15 m path in the 17–23 LT window above MUF, both fire on the same physical channel and stacking them double-counts the recovery:

$$M_{\text{verdict}} = M_{\text{physics}} + B_{\text{grayline}} + \max\bigl(B_{\text{TEP}},\, B_{\text{scatter}}\bigr) \tag{76}$$

σ tracks M_{physics}, not M_{verdict} – caveat. The reliability formula $R(M, \sigma)$ of Eq. 77 is evaluated with $M = M_{\text{verdict}}$ (the bonus-augmented margin) but σ is computed from M_{physics} inputs (per-band σ_g plus the situational widenings of Section 7.3.1). When the bonuses fire, the bonus terms themselves carry uncertainty that is not folded into σ: $B_{\text{TEP}} = 15\,\text{dB} \pm 5\,\text{dB}$ at the published-experience plateau, and B_{scatter} is calibrated through w_{sc} but the $\pm$ on the underlying spread ($\sigma_{f_{\text{oF2}}}$ measurement noise, f/MUF position) is non-trivial. A strictly correct treatment would add the bonus-term variances in quadrature to σ when bonuses are active; under a conservative $\sim$5 dB bonus-term uncertainty the correction is $\sqrt{\sigma_g^2 + 25} - \sigma_g$, which is $+1.0\,\text{dB}$ on a $\sigma_g = 12\,\text{dB}$ upper-band ($\sqrt{169} - 12 = 1.0$) and $+1.4\,\text{dB}$ on a $\sigma_g = 8\,\text{dB}$ low-band ($\sqrt{89} - 8 \approx 1.43$). The current model omits this widening: bonus-active verdicts on both upper HF and the low/mid bands read slightly more confident than they should, with the under-counting at $\sim$1–1.4 dB of σ depending on band. At the operationally most relevant case — 15 m near-MUF with a moderate-cycle TEP firing at $B_{\text{TEP}} \approx 13\,\text{dB}$, $\sigma_g(15\,\text{m}) = 10\,\text{dB}$ — the same $\sqrt{\sigma_g^2 + 25} - \sigma_g$ correction gives $\sqrt{125} - 10 \approx 1.18\,\text{dB}$ of under-counted σ, shifting the z-score by $\sim$10 % at typical near-MUF margins (e.g. $z = 1.0$ nominal becomes $z = 1.0 \cdot 10/11.18 \approx 0.89$ once corrected, dropping the predicted reliability from $\Phi(1.0) \approx 84\,\%$ to $\Phi(0.89) \approx 81\,\%$). The omission is tracked in Section 10; folding it in requires calibrating per-bonus uncertainty estimates against the per-pair WSPR truth (Section 7.2), which is gated on that mode's wiring.

The TEP plateau saturates at 15 dB only at solar maximum ($F_{10.7\mathrm{A}} \to \infty$); at moderate cycle ($F_{10.7\mathrm{A}} = 120$) the plateau sits at $\sim$11.2 dB and at quiet sun ($F_{10.7\mathrm{A}} = 70$) at $\sim$9.0 dB (Eq. 66: $8 + 7/(1 + e^{55/30}) = 8 + 0.97 \approx 8.97$). The scatter upper bound (15 dB at $w_{\mathrm{sc}} = 1.5$ when both $\sigma_{f_{\mathrm{oF2}}} = 1.5$ MHz and $f/\mathrm{MUF} \geq 3$ are simultaneously hit, see Section 6.5) shares the asymptotic numerical ceiling with the solar-max TEP plateau, but in practice the scatter term rarely saturates: a typical TEP-eligible 15 m cell at 17 LT with $f/\mathrm{MUF} = 1.3$ and $\sigma_{f_{\mathrm{oF2}}} = 0.5$ MHz produces $B_{\mathrm{scatter}} = w_{\mathrm{sc}} \cdot \min(1, 0.5/1.5) \cdot \min(2, 1.3-1) \cdot 5 = 1.5 \cdot 0.333 \cdot 0.3 \cdot 5 \approx 0.75$ dB, more than an order of magnitude below the moderate-cycle TEP plateau ($\sim$11 dB at $F_{10.7\mathrm{A}} = 120$). In a TEP-active cell, $\max(B_{\mathrm{TEP}}, B_{\mathrm{scatter}})$ therefore predominantly selects B_{TEP}; the value of the max versus a sum is to prevent the small-but-non-zero B_{scatter} from stacking on top of the calibrated TEP plateau. The $\max(\cdot)$ choice is a policy made on physics grounds rather than an empirical retune: the calibration harness operates on the climatology code path and does not compute either bonus at replay time, so it cannot directly distinguish $B_{\mathrm{TEP}} + B_{\mathrm{scatter}}$ from $\max(B_{\mathrm{TEP}}, B_{\mathrm{scatter}})$. Per-cell attribution of which mechanism dominates on TEP-eligible 15 m paths (EU–Brazil, NA–Brazil, NA–Argentina, EU–Argentina, EU–South Africa) is deferred to a future harness extension that incorporates the alternative-mode bonuses into `replayMarginFromCell`, with VOACAP cross-checks against the 7-path fixture set in `scripts/tests/voacap.mjs` (regenerated by the heavy `voacap-fixtures.mjs` suite). Pure physics drives the verdict; the N0NBH-style `heuristicTier` function in `src/physics/physics.js` is retained as an independent reference predictor (and is what `tune-blend.mjs` exercises as a regression canary) but does not enter the verdict path. Tier verdicts are defined by ITU-R P.842 reliability buckets in σ-units rather than by a fixed dB-width tier scheme, so the per-band absolute margin bias has no constant width to fit against.

7.2 Offline Constant Calibration

Two layers, one set of constants. The ionocast pipeline runs in two distinct contexts. The *online* layer is the browser at runtime: it fetches live data feeds (SWPC, kc2g, GIRO, GFZ, Kyoto, DONKI, wspr.live), evaluates the physics in `src/physics/`, and emits the band-table verdict. The *offline* layer is `scripts/harness.mjs` on a developer workstation: it imports the same `src/physics/` code (no inline copy that can drift), replays the budget over the 35-path basket against 30 d of WSPR aggregates, and writes the calibration baselines (`harness.baseline.json`, `spot-baselines.mjs`) that the online layer ships with. The two layers exchange a single committed-artefact bundle: the calibrated constants in `src/constants.js` and the WSPR spot baselines in `src/data/spot-baselines.mjs`, both written by the offline harness and shipped with the online build. Everything else (downstream live data, the ClickHouse query, the per-(path, band) baselines used for regression detection) lives entirely on one side or the other. A formal data-flow diagram is candidate documentation work tracked in Section 10; until then this paragraph is the explicit description.

Several of the budget's empirical constants — the lumped ionospheric loss L_{iono}, the per-extra-hop defocusing constant D (Section 3.8), and the F2-scatter weight w_{sc} (Section 6.5) — were originally set from published ITU-R decompositions and ham-operator experience rather than measured in place. The per-band σ table (Table 9) is *not* fitted by the harness: σ is anchored to the ITU-R P.533 day-to-day spread (literature range 6–12 dB; ionocast picks 8–12 dB across the per-band table, see Table 9) and held fixed. The harness scores binary accuracy and Brier against WSPR but does not adjust σ, since under the P.842 reliability framework the tier boundary scales with whatever σ is set, and a machine-fitted σ would chase the WSPR-truth artefact rather than the underlying physics spread.

The current calibration harness is `scripts/harness.mjs`. It imports the production physics directly from `src/physics/` (no inline copy that can drift), replays the SNR budget over a 35-path basket (Appendix C) spanning 1–5 hops and every major continent pair plus polar / transequatorial / NVIS-tail geometries against 30 d of WSPR hourly spot counts (`wspr.live`, ClickHouse), aligned hour-by-hour with the SWPC K_p history, the $F_{10.7}$ 81-day mean, and pre-fetched 30 d GIRO foF2 / foEs histories from 27 digisonde stations (Appendix D) spanning Europe, North America, the South Atlantic, Australia, the Pacific, and an expanded equatorial-belt set (Ascension and Jicamarca for the Atlantic / Americas equatorial-trough anchor pair — Jicamarca on the dip equator, Ascension at dip lat $\sim -3°$ also deep in the trough region; Townsville and Niue for the Pacific southern crest; BVJ03 Boa Vista at $+12°$ for the lone northern-crest sample). The Pacific / Australia block (Townsville, Niue, Perth, Canberra, Hobart) is included to cover the South Pacific, and the multi-snapshot subcommand `scripts/harness.mjs t1` (emitting `scripts/data/.cache/t1-snapshots.jsonl`) is the per-station bias diagnostic used when adding stations. The fetcher chunks queries 3-at-a-time with retry-on-503 because DIDB rate-limits parallel fetches; coordinates are verified against the kc2g station registry by `scripts/harness.mjs verify`. Per-(path, timestamp) cells precompute band-independent state once (climatology, geometry, solar zenith) and the ∼25,000 unique cells are reused across all 10 HF bands per scoring call, keeping a full sweep cycle well under 1 s.

For each candidate config the harness computes the Brier loss of the predicted *open* probability against a binary "alive" floor on the WSPR aggregate. This is a sanity check on the model's binary verdict, not a calibration target: WSPR-spot activity and link viability for the configured station are correlated but not identical, so binary accuracy is reported with that caveat (see Section 8). An earlier per-band tier-midpoint residual emitter targeted a WSPR-percentile tier truth; that target was retired in favour of ITU-R P.842 reliability buckets defined by the model's own σ, so per-band absolute-margin biases are no longer fitted. In its place the harness writes a per-(path, band) margin/σ baseline and flags any cell that drifts more than 2 dB in mean margin or 5 pp in P(open) across runs — a self-consistency regression detector that needs no external truth.

The "alive" floor itself comes in two forms, selectable via `-ground-truth={global,per-path}`:

- **Global** (default, baseline file `harness.baseline.json`): band-level "alive somewhere in the world" floor of ≥ 50 spots/hour, computed from a single bbox-unrestricted WSPR aggregate query. Cheap (one query per harness run, ∼7 k rows for a 30-day window), but every reference path sees the same open-rate truth, so the per-(path, band) drift signal is dominated by the path-independent open-rate bias.
- **Per-path** (baseline file `harness.baseline.perpath.json`): TX bbox + RX bbox aggregate per reference path (bidirectional, default half-width 5°), with a much sparser floor of ≥ 1 spot/hour reflecting the per-pair WSPR density. Adds 35 queries per harness run, but the open-rate truth is path-specific, so the per-(path, band) drift signal directly reflects per-pair behaviour. Used to flag a per-mechanism regression that the global mode averages out — for example, the §4.3 asymmetric-MUF retest, where the bias was concentrated on the minority of cells where kc2g sat above climatology.

Both modes write to separate baseline files so an operator can maintain both anchors simultaneously and compare drift in either register.

A separate driver script, `scripts/tests/tune-r7.mjs`, runs coordinate-descent joint optimisation over the seven-parameter set $\{L_{\text{iono}}, D, w_{\text{sc}}, w_{\text{NVIS-tail}}, w_{\text{Es-prim}}, \sigma\text{-scale}, \text{fusion-flag}\}$ from three deterministic starting configurations (Appendix E; "baseline", "fusion-up", "modes-on"). Production values: $L_{\text{iono}} = 1\,\text{dB}$, $D = 0.25\,\text{dB}$, $w_{\text{sc}} = 1.5$, $w_{\text{NVIS-tail}} = 1.0$ (no scaling vs the bare NVIS-tail formula of Eq. 52), $w_{\text{Es-prim}} = 1.0$ (no scaling vs the bare Es-mode formula in §6.1), σ-scale $= 1.0$ (per-band σ_g from Table 9 held), and fusion-flag false (`FUSION_PRIMARY_MUF = false` in `src/constants.js`; the R3 per-hop foF2 fusion path is implemented but kept off in production until the per-pair WSPR ground-truth wiring of §7.2 can calibrate it without exploiting the global-truth structural ceiling). At the 2026-04-28 calibration freeze, all three seeds converged to the same basin in ≤ 2 iterations: the final $\{L_{\text{iono}}, D, w_{\text{sc}}\}$ landed within ±10% across seeds; σ-scale and the two $w_{\text{NVIS-tail}}/w_{\text{Es-prim}}$ weights settled at 1.0 from every seed at that freeze (their natural-units fixed point at the time); fusion-flag converged to *false* on every seed. Final Brier landed within ±0.001 of the median seed. *Drift caveat:* solar activity has moved the harness target since the freeze (see the "Solar-driven drift on re-run" note in Table 8). Re-running the same coordinate descent against later 30-day windows pulls σ-scale and $w_{\text{Es-prim}}$ off the 1.0 fixed point: a 2026-05-06 re-run lands σ-scale ≈ 2.0 and $w_{\text{Es-prim}} \approx 1.5$ as the local optima against that window. Production constants stay at their 2026-04-28-freeze values per the explicit no-auto-commit policy below; the re-run divergence is a calibration-frame drift signal, not a regression. Lift across the freeze cycle was concentrated on 12 m / 10 m, driven entirely by the $w_{\text{sc}} = 1.5$ scatter recovery; the other six parameters showed no measurable Brier change at the freeze when swept individually.

The fixed reference values cited throughout the paper are consolidated in Table 8 so each percentage can be tied to a specific basket / config / metric.

Table 8: Canonical harness numbers cited in the paper, each tied to its (basket, ground-truth mode, configuration) so a reader can disambiguate. Binary "open" threshold is ≥ 50 spots/h on the global aggregate or ≥ 1 spot/h on the per-path aggregate.

Run	**Basket**	**Global truth**		**Per-path truth**	
		accBin	**Brier**	**accBin**	**Brier**
Pre-scatter baseline (2026-04 freeze)	35 paths	91.08 %	0.0625	—	—
Post-scatter sweep (2026-04 freeze)	35 paths	92.35 %	0.0535	—	—
Pre-B6 production (2026-04-28 freeze)	35 paths	94.25 %	0.0386	30.97 % §	0.6353 §
Post-B6 production (2026-05-08, current)	35 paths	84.08 %	0.1081	40.17 % §	0.5575 §

§ Per-path-truth metric uses TX/RX-bbox-restricted aggregates with the ≥ 1 spot/h floor; the resulting accBin / Brier numbers are *not directly comparable* to the global-truth column because the per-path floor is far stricter (most cells read closed) and because operator-activity sparsity dominates the binary outcome on minority-traffic bands. The pre-B6 freeze headline 94.25 %/0.0386 and the post-B6 current headline 84.08 %/0.1081 are both the global-truth metric and both inherit the structural "open somewhere" ceiling that the per-path mode was specifically built to expose (Section 10 #9); the post-B6 numbers are *lower* on global truth because the new D-region formula predicts 160 m closed more often (correctly, per the per-path metric: 160 m accBin rose from 19.9 % to 37.6 % across the B6 cut). The per-path column is the more rigorous metric and is the one that should drive constant retunes; B6 improved it materially on 160 m and left the other bands unchanged. Treat the global number as a regression-detection signal, not a quality score; the per-path column is for cell-level drift detection against the saved baseline.

Solar-driven drift on re-run. The headline numbers are the values produced by the 30-day window centred on the 2026-04-28 calibration freeze. Re-running the same harness against a later 30-day window will give different absolute numbers as solar activity shifts the band-by-band open-rate distribution: a 2026-05-06 fresh run ($F_{10.7}$ rose 105 → 156 over the intervening week) produced 86.06 % / 0.0937 global and 38.29 % / 0.5740 per-path. This is calibration-frame drift, not regression: the sweep-derived constants are unchanged but the truth signal itself has moved with the sun. The committed `scripts/data/harness.baseline.json` is the regression anchor and the ~2 dB / 5 pp cell thresholds are the actual drift gates.

The harness does not auto-commit: a human reviews the suggested constants and pastes the winners into `src/constants.js`.

7.3 Circuit Reliability

Given the margin M and spread σ, the circuit reliability is the probability that the achieved SNR meets or exceeds the mode's required SNR — in other words, the probability the link is usable on a given attempt:

$$R(M, \sigma) = \Phi\left(\frac{M}{\sigma}\right) \tag{77}$$

where Φ is the standard normal CDF. JavaScript has no built-in Φ, so the runtime evaluates it via the Abramowitz & Stegun 26.2.17 rational approximation (`src/physics/tier.js`, `normCdf`); maximum absolute error is $\sim 7.5 \times 10^{-8}$ in the tails, well below the verdict-relevant resolution at the ionocast tier boundaries ($z = \pm 1.2816, +0.2533, -0.3853$, where $\partial R/\partial z = \varphi(z) \approx 0.18, 0.39, 0.37$

respectively). The approximation choice is therefore not a budget-shaping parameter; it is fixed for reproducibility. Eq. 77 is the Gaussian-spread basic-circuit- reliability model that underlies ITU-R P.842 [8]: the recommendation itself presents the calculation in piecewise / tabular form rather than the literal $\Phi(M/\sigma)$ identity used here, but both reduce to the same Gaussian assumption on day-to-day SNR fluctuations. ionocast evaluates this model with live nowcast inputs — M is the current-hour budget margin (Eq. 76), and σ is the per-band base σ_g (Table 9, set within the ITU-R P.533 day-to-day spread range 6–12 dB; ionocast picks 8–12 dB by band) augmented in quadrature with the situational widenings of §7.3.1. R drives the tier verdict (Section 8) but the band-table UI surfaces a different percentile alongside the tier label, the *Stability* metric defined in Section 7.3.1.

Strict P.842 vs. instantaneous: an operator-relevant divergence. P.842 as written is a *climatological* predictor: the formula is evaluated with the monthly-median margin at a given hour and the day-to-day spread, so R answers "on what fraction of days does the median link at this hour exceed the requirement?" ionocast evaluates the same algebra with live data — the current-hour observed kc2g MUF, the live $F_{10.7A}$, K_p^{eff} from the past 7 d — and so R answers "what is the probability the link is viable *right now*, given the model's residual uncertainty?" Both questions use the P.842 formula and a P.842-grade σ, but they are not the same question. An operator deciding whether to call CQ at this minute wants the latter; a frequency-coordination engineer planning circuit allocations across a season wants the former. ionocast surfaces the nowcast interpretation, and the σ widenings of §7.3.1 (near-MUF, storm, forecast storm, cross-terminator, storm-recovery TID) plus the Es-active contribution defined in §6.1 capture the within-hour spread that P.842's static day-to-day σ does not. The verdict bucket boundaries (Table 11) use the same $R = \Phi(M/\sigma)$ formula as P.842 but apply ionocast-specific reliability targets (0.90/0.60/0.35/0.10) chosen for operator failure-rate intuition rather than the classical 0.95/0.80/0.50/0.20 schedule; see Table 11.

Tier thresholds (Section 8) are themselves expressed as constant levels of R, so the boundary in dB scales with the band's σ rather than being pinned to a hand-set number. This makes the verdict self-consistent with the model's own uncertainty: a band whose σ is small (low-band, climatology dominated) reaches "excellent" at a smaller dB margin than a band whose σ is large (10 m near MUF, where prediction spread is intrinsic).

What R does and does not measure. R is the probability that the *physics* of the link permits a QSO at the operator's configured rig, conditional on an operator on the other end transmitting and listening at that hour. It does not include operator-availability factors (band-time-of-day social bias, WSPR-spot population density on the destination side, holiday patterns, contests, etc.), which are independent of ionospheric state and not represented in the SNR budget. An empirical calibration against 30 d of WSPR per-path aggregates (TX/RX bbox restricted, floor 1 spot/h; the per-path scoring path inside `scripts/harness.mjs` when run with `-per-path`) finds that predicted tier ranks correctly — excellent > good > fair > poor > closed in observed open-rate ordering — but the absolute per-path completion rate runs ∼60–65 pp below the predicted reliability across the upper four tiers (the predicted "excellent" bucket sees a ∼30 % per-path open rate against the WSPR cache; "good" ∼14 %; "fair" ∼3 %; "poor" ∼1 %; "closed" ∼0.1 %; these per-tier rates are from an ad-hoc post-processing of the per-cell `harness.report.json`, not from a byTier field that the standard report currently emits). The held-out window (2026-04-26 onwards, post the 2026-04-25 loss-constant calibration) matches the in-sample window within 1 pp on every tier, ruling out a calibration-drift explanation. The gap is a physics-vs-operator-availability semantics issue: R captures "the path's physics permit a QSO with the typical operator's rig" (physics-permitted reliability), while WSPR per-path open-rate captures "a WSPR-active operator pair completed a transmit at this hour" (operator-attempt completion rate), which is uniformly lower because operators do not attempt every available (band, hour, path) cell.

7.3.1 Verdict Stability

The band-table surfaces a per-band *Stability* percentage, defined as

$$S(M, \sigma) = \Phi(\min_{b \in \mathcal{B}} |z - b|), \quad z = M/\sigma, \tag{78}$$

where $\mathcal{B} = \{-1.2816, -0.3853, +0.2533, +1.2816\}$ is the set of tier boundaries in z-space (Table 11). Operator reading: *how likely is the verdict to NOT cross its nearest tier boundary if the true margin moves to its expected value?* The metric is one-sided, it asks about the closest boundary only, not both flanks of the bucket, and bucket-width-independent: a Fair verdict Δz from its nearest boundary reads the same percentage as an Excellent verdict Δz from its boundary.

Reachable values. What S can show depends on the bucket, because the maximum nearest-boundary distance depends on bucket width:

- **Closed** ($z < -1.28$) and **Excellent** ($z \geq 1.28$): open-ended on one side. S rises monotonically the further z moves into the tail: $\sim 84\%$ at 1σ past the boundary, $\sim 97\%$ at 2σ, $\to 1.0$ deep in the tail.
- **Poor** ($\sim 0.90\sigma$ wide), **Fair** ($\sim 0.64\sigma$ wide), **Good** ($\sim 1.03\sigma$ wide): max nearest-boundary distance is half each bucket's width, so S caps at $\Phi(0.45) \approx 67\%$ for Poor, $\Phi(0.32) \approx 63\%$ for Fair, and $\Phi(0.51) \approx 70\%$ for Good at each bucket centre. $S = 50\%$ on a boundary; the centred-bucket maxima are the most stable a middle-tier verdict in that bucket can read.

Read S on the open buckets against an absolute scale (deep Excellent / Closed approach 100%); read S on the middle buckets relative to their per-bucket centred maxima.

One-sided semantics. S measures the probability that pure-σ noise does not cross the *nearest* boundary; it ignores the far boundary on a finite-width bucket. For a centred Fair verdict ($z = -0.066$, halfway between -0.385 and $+0.253$) the far boundary is also 0.32σ away and is not counted, so $S = 63\%$ rather than the two-sided in-bucket probability $2\Phi(0.32) - 1 \approx 25\%$. The one-sided form is the correct "how locked in is this verdict against the most likely flip" indicator and is what operators typically want; the two-sided form is the literal $P(\text{predicted} = \text{true})$ retained as `tierConfidence` in the source. The earlier band-table column showed the two-sided form, which capped at $\sim$ 25–35% for centred middle-tier verdicts and required a " (peak)" annotation; the new S replaces it.

Operator-attempt completion rate (deferred). The band-table currently surfaces only the physics-permitted tier label. A planned *Expected Completion* (*Comp.*) column would fold the operator-availability gap discussed above into a single percentage; the formulation below documents that deferred display so the activity-prior factor it depends on (already computed via `spotBaselineMean` in `src/derive/spots.js`) has a published target shape:

$$\text{Comp.}(b, h, M, \sigma) = R(M, \sigma) \cdot a(b, h), \tag{79}$$

where $a(b, h) = \text{baseline}(b, h) / \max_h \text{baseline}(b, h)$ is the per-band operator-availability prior, derived from the 30-day per-(band, UTC-hour) WSPR baselines that already drive the spot-override (§8.1). $a(b, h) = 1$ at the band's peak activity hour; $a(b, h) \to 0$ deep in the band's dead-of-night for that band's typical user population. Comp. would read as "what fraction of attempts at this rig in this hour actually complete a QSO," which is the question an operator picking a band to call CQ on is asking. Tier and Comp. together would disambiguate the physics-vs-availability gap: an Excellent tier on 12 m at 0300 UTC reads ~95 % physics-permitted (R) and ~5–10 % Comp. — the band is open, but nobody is listening. Once the column ships, the intended operator guidance is: use Tier when planning around the physics, use Comp. when picking a band to actually call CQ on right now.

The factorisation $R \cdot a(b, h)$ assumes physics-permitted reliability and operator-availability are independent. This holds strictly within a band in steady state – $a(b, h)$ is normalised against the band's own 24-hour maximum, so it encodes *intra-band* hourly activity variation and not inter-band reliability. Across bands the absolute Comp. values are not directly comparable (a band with chronically thin WSPR-pair density can never reach Comp. $\to 1$ even when fully open), which is why the planned Comp. column would surface the percentage per row rather than as a cross-band ranking. In the limit of a future per-pair availability prior with band-specific ceilings, the factorisation would need refining to account for the fact that bands with high WSPR activity have those operators *because* the band is reliably open — a feedback that the current intra-band normalisation sidesteps but does not eliminate.

Storm-time correlation breaks the independence assumption. During a major geomagnetic storm both R drops (physics) and $a(b, h)$ should drop too (operators on the affected bands stop calling because they observe the band is dead) — i.e. R and a would be positively correlated under exactly the disturbance conditions where Comp. would be most operationally relevant. The factorisation $R \cdot a(b, h)$ treats them as independent, which would bias Comp. *high* during storms (the product of the depressed R with the steady-state a overstates expected completion versus the true $R \cdot a_{\text{actual}}$ where $a_{\text{actual}} < a$). The intra-band normalisation softens but does not remove this: storm-time activity dropouts span all bands roughly together, so the per-band 24-hour maximum baseline still encodes a quiet-day shape that storm hours genuinely depart from. When the column ships, Comp. during $K_p^{\text{eff}} \geq 5$ conditions should be read as an upper bound on expected completion, not a point estimate; the storm-aware refinement is tracked in Section 10 and is gated on the per-pair availability data wiring of Section 7.2.

The spread σ is *per-band and condition-dependent*, computed as the RSS of a per-band base σ_g (Table 9; values anchored within the ITU-R P.533 day-to-day spread literature range 6–12 dB, with ionocast picking the upper-half subset 8–12 dB across the table) and seven situational penalties (six enumerated below plus the Es-active mode contribution defined in §6.1).[3]

- **Near-MUF:** when $f/\text{MUF} > 0.85$, fading and multipath intensify. Adds up to +4 dB at $f = \text{MUF}$.
- **Storm:** $K_p^{\text{eff}} \geq 5$ (the storm-lagged effective K_p from Section 5.3.1, after Bz-forward and Dst bumps) produces ionospheric irregularities. Linear in K_p^{eff} above the activity threshold: $\sigma_{\text{storm}} = 3 + 0.75\,(K_p^{\text{eff}} - 5)$ dB for $K_p^{\text{eff}} \geq 5$, zero otherwise; that gives +3 dB at $K_p^{\text{eff}} = 5$ and +6 dB at $K_p^{\text{eff}} = 9$. Using the effective rather than live K_p means the storm σ persists into the recovery phase (when the live index has already dropped but the F-region thermosphere is still recovering). σ_{recovery} below fires only when the phase classifier identifies recovery, so the two penalties stack during recovery (covering both the residual lagged-Kp irregularities and the LSTID ripple) and the storm penalty alone applies during main phase. The same K_p^{eff} also drives the auroral amplification described later in this section.[4]

[3]Notation: σ is overloaded across the paper for distinct quantities — the prediction-spread family in this section (σ, σ_g, σ_{MUF}, σ_{storm}, σ_{forecast}, σ_{term}, σ_{recovery}, σ_{night}, σ_{Es}); ground conductivity in Eq. 22 (in S/m); the Gaussian kernel widths σ_{cr}, σ_{tr} in the EIA term (Eq. 45, in degrees); the per-hop foF2 standard deviation σ_f in the F2-scatter recovery (Section 6.5); the standard-normal CDF argument in Eq. 77's $\Phi(M/\sigma)$; and the dimensionless σ-scale tuning parameter on the coordinate-descent grid in Appendix E (Table 17, which uniformly multiplies the per-band σ_g table during calibration sweeps; held at 1.0 in production). The unit-ramp clamp formerly written as $\sigma(x)$ has been renamed $R(x)$ in §5.4 to avoid collision. The remaining distinct meanings are recoverable from local context (units, subscripts) but the overloading is real.

[4]In the source code, `snrMarginHf` (and the auroral loss terms it consumes) take a parameter named `opts.kp`. This is *always* K_p^{eff} (i.e. the storm-lagged value with Bz-forward and Dst bumps applied) rather than the live 3-hour K_p sample; the bump-and-lag composition happens upstream in `src/derive/conditions.js` before the budget is invoked. A reader auditing the code without the data-flow context could mistake the unqualified `kp` name for the live index.

- **Forecast storm:** the SWPC 3-day Kp forecast shows a peak $K_p^{\text{peak}} \geq 5$ within the next 6 h. Sharp electron-density gradients build hours before the index reading, so σ widens ahead of the disturbance. Same magnitude as σ_{storm} above evaluated at K_p^{peak} (+3 dB at $K_p^{\text{peak}} = 5$, +6 dB at $K_p^{\text{peak}} = 9$), attenuated by 0.7× for forecast confidence, then ramped down linearly as the current K_p catches up to the forecast peak:

$$\sigma_{\text{forecast}} = 0.7\,\sigma_{\text{storm}}\left(K_p^{\text{peak}}\right) \cdot \min(1,\, \max(0,\, K_p^{\text{peak}} - K_p^{\text{eff}})) \tag{80}$$

  The catch-up ramp is gated on K_p^{eff} rather than the live 3-hour K_p, matching the σ_{storm} branch above and preserving the smooth-ramp philosophy applied at every other gate; using the live index here would step the forecast bump down on each 3-hour tick rather than as the thermosphere actually catches up. Without this gap-based gate, summing in quadrature with the current-Kp σ_{storm} branch would have produced $\sqrt{\sigma^2_{\text{storm}} + 0.49\,\sigma^2_{\text{storm}}} \approx 1.22\,\sigma_{\text{storm}}$ at the moment of catch-up (a 22% inflation), instead of the intended hand-off to σ_{storm} alone. The 1 Kp-unit saturation width of the catch-up ramp is empirical: K_p^{eff} typically climbs by ∼1 unit per hour during onset, so a 1-unit gap maps to roughly an hour of forecast prominence ahead of the live index, which matches the operationally-useful early-warning horizon of the SWPC 3-day forecast. A wider saturation window would extend the forecast bump deeper into peak hours and double-count with σ_{storm}; a narrower window would cut the forecast bump off before the live index has caught up at slow-onset events.
- **Cross-terminator:** when the path midpoint has $|\cos\chi| \leq 0.15$ (sunrise/sunset line), sharp electron-density gradients increase variance by +3 dB (the variance contribution is 9 dB^2, added in quadrature to the per-band σ_g^2). Above $|\cos\chi| \geq 0.20$ the contribution is zero; in the $|\cos\chi| \in [0.15, 0.20]$ band the variance contribution ramps linearly from 9 dB^2 down to 0, matching the smooth-ramp philosophy applied at every other gate (auroral $c(\phi)$, Bz bump, twilight $r(\cos\chi_k)$, near-MUF, Es threshold) and avoiding a tier-confidence cliff at terminator approach.
- **Storm-recovery TID:** during the recovery phase of a geomagnetic storm (Dst depressed but K_p^{eff} trailing the live K_p, see Section 5.3.1), large-scale travelling ionospheric disturbances ripple through the F-region for hours after the surface-field index settles. The mean margin is no longer suppressed but the path variance is visibly elevated; ionocast adds +4 dB to σ across all paths until the storm phase classifier exits "recovery".
- **Night-time low/mid HF inflation:** ITU-R P.533 within-month variability for night-time multi-Mm paths on 160 m through 30 m runs ∼8–10 dB; the per-band σ_g table sits at the 8 dB floor on those bands, anchored to a quiet-day reference. Without an extra bump, an "excellent" verdict at +11 dB margin on a quiet night-time 30 m / 20 m path over-claims reliability by about one tier. ionocast adds +3 dB in quadrature when the path midpoint is in darkness ($\cos\chi_{\text{path}} < 0$) and $f \leq 16$ MHz (i.e. 17 m and below, where the tabulated σ_g is not already inflated by near-MUF variability). Above 17 m the table value already reflects the higher upper-band spread, so further inflation would double-count.

All penalties are added in quadrature:

$$\sigma = \sqrt{\sigma_g^2 + \sigma^2_{\text{MUF}} + \sigma^2_{\text{storm}} + \sigma^2_{\text{forecast}} + \sigma^2_{\text{term}} + \sigma^2_{\text{recovery}} + \sigma^2_{\text{night}} + \sigma^2_{\text{Es}}}$$

with the night-time inflation $\sigma^2_{\text{night}} = (3\,\text{dB})^2$ entering only when $\cos\chi_{\text{path}} < 0$ and $f \leq 16$ MHz, and the Es-active contribution $\sigma^2_{\text{Es}} = (2\,\text{dB})^2$ entering only when Es is selected as the propagation channel (§6.1; the Es-mode budget consumes the same RSS via the shared `_conditionalSigmaDb` helper, so storm-time, terminator, recovery, and night-low/mid penalties all propagate into Es-mode verdicts the same way they do for F2).

Why σ_{storm} and σ_{recovery} are path-blind. σ_{MUF} and σ_{term} are evaluated at the path midpoint (so they vary path-by-path across the 10-path basket), but σ_{storm}, σ_{forecast}, and σ_{recovery}

are global: they fire on the same K_p^{eff} and storm-phase signals regardless of whether any hop on the path actually crosses an auroral CGM latitude. This is intentional rather than a simplification. Storm-time variance arises from at least three mechanisms whose footprints are not confined to high latitude: large-scale travelling ionospheric disturbances (LSTIDs) launched at the auroral oval propagate equatorward at $\sim$300 m/s to $\sim$1000 m/s and reach midlatitudes within hours; thermospheric Joule heating shifts the global [O]/[N_2] ratio, depressing foF2 by 10–20 % on midlat paths during main phase; and plasma-bubble / spread-F activity at low and equatorial latitudes during the recovery phase produces irregularity-driven fading on transequatorial paths that share no auroral hops. The global σ widening captures this collectively. The *path-localized* component of storm response — additional auroral D-region absorption on hops crossing the storm-expanded ϕ_{CGM} threshold — is captured separately in L_{aur} (Section 3.7, with the storm-main amplification described below), so a midlat path during a G3 storm sees the global σ_{storm} widening but no L_{aur} contribution, while an auroral path sees both. The combination correctly under-widens neither: midlat paths get the global-irregularity penalty without the absorption term, and auroral paths get both the absorption (mean) and the widening (spread). The remaining bias is a slight over-widening of *deep equatorial* midlat paths during main phase (where LSTID arrival is delayed by an hour or two relative to the global signal), worth at most $\sim$1 dB of σ inflation on those paths — below the per-band σ_g resolution and accepted as the cost of the global formulation.

Storm-phase auroral amplification. Independent of the σ widening above, the auroral absorption term L_{aur} (Section 3.7) is incremented by +4 dB per hop during the storm *main* phase, then re-clamped against the 30 dB per-hop cap (`AUR_PER_HOP_CAP_DB`). During main, the auroral oval expands equatorward and the particle precipitation that drives auroral D/E absorption is strongest; the steady-state K_p-keyed L_{aur} undercounts the depth. The +4 dB additive bump is calibrated to operator-experience reports of typical extra absorption (3–6 dB) on high-latitude amateur HF paths during peak Dst depression. ITU-R P.533 [1] Annex 1 Table 2 tabulates climatological auroral-and-other losses (L_h) keyed on geomagnetic latitude, season, and local time — with values reaching 12–16 dB on auroral-zone (62.5–72.5° geomagnetic) winter dawn paths and under 3 dB on midlatitude paths — but does not isolate a "main-phase extra" increment as a separate column. The 3–6 dB range cited here is therefore an operator-experience figure consistent with the disturbed-vs- quiet difference implicit in P.533's seasonal climatology rather than a literal P.533 entry. An earlier draft used a 1.4× multiplier, which would have produced +10 dB on a 25 dB nominal L_{aur} (well past the 3–6 dB range cited as the target); the additive form lands at the cited midpoint regardless of the steady-state value, with the per-hop cap as the hard ceiling for already-saturated terms. Recovery and quiet phases keep the steady-state value. Storm phase is classified from Dst trajectory plus the storm-lag kernel (initial / main / recovery / quiet / active); main is identified by Dst ≤ -50 nT with K_p^{eff} still loading.

Table 9: Per-band base σ_g, anchored to the ITU-R P.533 day-to-day spread range and validated against per-spot wspr-snr residual standard deviation bucketed by (band, distance, hour-of-day, Kp, tx-latitude band) at $\geq$10 samples per bucket; the measured within-bucket σ was stable across three bucket-granularity settings (coarse 1500 km / 6 h vs tight 500 km / 1 h), confirming the spread is within-condition rather than cross-condition contamination. 12 m / 10 m fall back to operator-experience values (fewer than five buckets per band; the WSPR signal is too weak to drive a within-condition fit there).

Band	σ_g (dB)	within-cond data (dB)
160 m	8	8.5
80 m	8	8.0
60 m	8	8.0
40 m	8	8.5
30 m	9	9.0
20 m	9	9.5
17 m	9	9.0
15 m	10	10.0
12 m	12	(n=2 weak)
10 m	12	(n=3 weak)
6 m	8	(no VHF data)
2 m	8	(no VHF data)

σ_g sensitivity. σ_g is held at the P.533 anchor rather than fitted, on the grounds that machine-fitted σ would chase the WSPR-truth structural ceiling (Section 10 #9) instead of physical day-to-day spread. The cost of the fixed-σ choice is that any miscalibration in σ_g propagates directly into the tier-boundary placement, which matters most for verdicts that already sit close to a tier boundary. Table 10 reports the verdict shift on the three near-MUF upper bands when σ_g is swept by ± 2 dB around its assigned value, holding the nominal margin at $M = +2.5$ dB, chosen to land 17 m's verdict near the Fair/Good boundary at $z = 0.2533$, where the σ leverage is concentrated. At any value deep inside a bucket the table would show all-zero flips on every band; at any value near a boundary the band whose $\sigma_g \cdot 0.2533$ Fair/Good gap is narrowest in dB will flip first.

Table 10: Verdict-bucket shift on 17 m / 15 m / 12 m at $M = +2.5$ dB when σ_g moves ± 2 dB around its assigned value (per Table 9: σ_g=9 on 17 m, σ_g=10 on 15 m, σ_g=12 on 12 m). "Δ tier (full sweep)" counts the tier flips across the three sweep points. 17 m and 15 m both flip; 12 m does not. At 17 m the Fair/Good boundary at the assigned σ_g=9 sits at $z \cdot \sigma_g = 0.2533 \cdot 9 = 2.28$ dB, so $M = +2.5$ dB lands just inside Good ($z = 0.278$, $R = 0.61$); at σ+2=11 the boundary moves to 2.79 dB and the same M falls into Fair ($z = 0.227$, $R = 0.59$). At 15 m the boundary at σ_g=10 sits at 2.53 dB, just above $M = +2.5$ dB ($z = 0.25$, $R = 0.60$, Fair); at $\sigma-2$=8 the boundary tightens to 2.03 dB and $M = +2.5$ dB lands above it ($z = 0.313$, $R = 0.62$, Good). 12 m has a $0.2533 \cdot 12 = 3.04$ dB Fair/Good gap which $M = +2.5$ dB sits below at all three sweep points, so it reads Fair throughout.

Band	$\sigma_g - 2$	assigned σ_g	$\sigma_g + 2$	Δ tier (full sweep)
17 m	$\sigma = 7$, $R = 0.64$, Good	$\sigma = 9$, $R = 0.61$, Good	$\sigma = 11$, $R = 0.59$, Fair	1 tier (Good → Fair)
15 m	$\sigma = 8$, $R = 0.62$, Good	$\sigma = 10$, $R = 0.60$, Fair	$\sigma = 12$, $R = 0.58$, Fair	1 tier (Good → Fair)
12 m	$\sigma = 10$, $R = 0.60$, Fair	$\sigma = 12$, $R = 0.58$, Fair	$\sigma = 14$, $R = 0.57$, Fair	0 tiers

The takeaway is that σ_g leverage is concentrated where M is close to a tier boundary; deep inside a bucket the verdict is robust to $\sigma_g \pm 2$ dB on every band. 17 m has the narrowest Fair/Good gap in dB ($0.2533 \cdot \sigma_g = 2.28$ dB at σ_g=9) of the three near-MUF bands, so it crosses a boundary at lower M values than 15 m or 12 m and is the most sensitive to σ_g miscalibration in practice. This makes 17 m the prime candidate for re-anchoring once per-pair WSPR ground truth is wired (§10 #9) and a fitted σ_g becomes defensible without exploiting the structural ceiling.

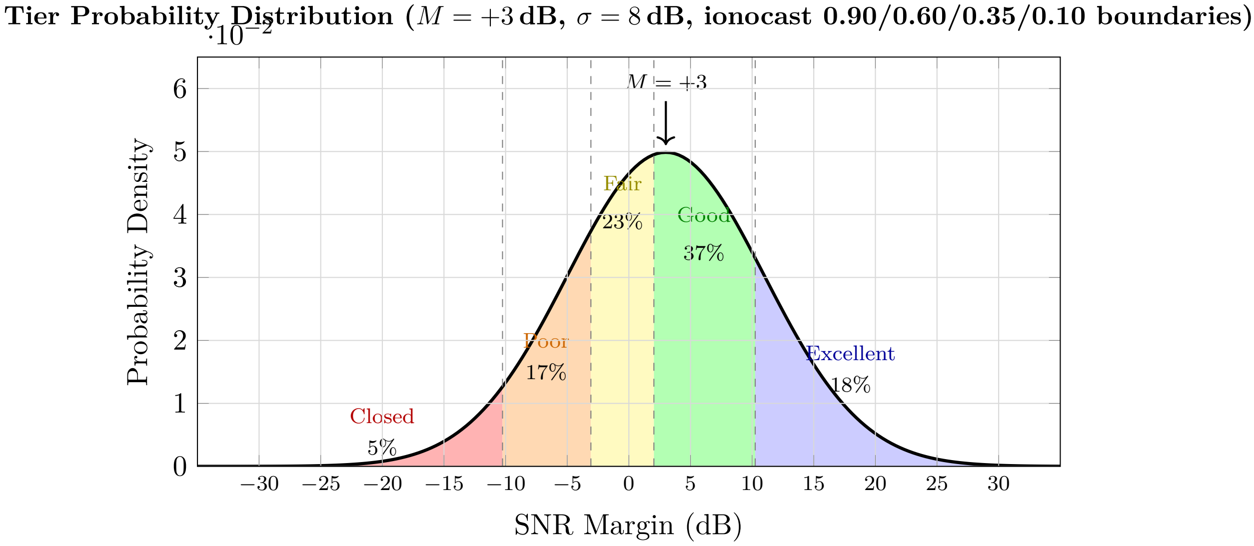

Figure 4: Tier probability distribution for a margin of +3 dB with $\sigma = 8$ dB under the tier boundaries in Table 11. The Fair band sits at $z \in [-0.385, +0.253]$ (-3.08 dB to $+2.03$ dB at $\sigma = 8$, asymmetric); the Excellent / Closed cutoffs are symmetric at $\pm 1.282\sigma$ (± 10.25 dB). Most likely tier is Good (37%), with 18% Excellent and 23% Fair — the distribution is asymmetric because $M = +3$ sits squarely inside Good. The $\sigma = 8$ dB shown here is the 6 m / 2 m placeholder value (Table 9); the HF bands use $\sigma_g \in \{8, 9, 10, 12\}$ dB depending on band. Read the figure as a tier-distribution illustration at an in-between σ rather than as a representative HF prediction; tier widths scale with σ and the corresponding dB-space boundaries shift accordingly.

8 Tier Mapping

Verdicts are computed *per band* (160 m through 10 m individually, plus 6 m and 2 m for VHF), not in groups. For each band the SNR margin is evaluated on every reference path (short and long, across all five destinations); the **best-margin** path wins and its margin drives the verdict, while the reason string and `openDirs` list record which destination that was. If a single path is viable, the band is not "closed" — it is open to that destination, which is material information for the operator. A median-margin aggregation would dominate the verdict with the failing long paths because most reference paths are long transcontinental hops that are routinely over-MUF or night-crossing, burying genuinely open short paths (e.g. a 944 km EM79–NYC path reporting +27 dB margin would have its verdict overwritten by the median of eight failing long paths).

Best-path summary, full-detail companion. The band-table row shows the best-margin destination only (the "Best Path" column), which is a deliberate compression: displaying all ten path columns per row would be visually unworkable on small screens. An operator who wants the underlying per-destination breakdown consults the Reference Paths panel in the Ionosphere section, which lists the kc2g MUF on each of the 10 paths with a colour-coded f/MUF column for the band currently selected. The two surfaces share the same underlying budget; the band-table row is the headline, the Reference Paths panel is the source-of-truth. The Best-Path column itself is therefore only a hint — if it reads "Tokyo" on 20 m and the operator wants Argentina, the Reference Paths panel is what to consult next.

The continuous margin is discretised into five tiers using fixed-dB cuts (Table 11). The cuts are absolute and do not scale with the per-band prediction uncertainty σ_g. The tier label answers "does the model expect a usable signal on this band right now, with the station the operator has configured?" and is therefore a property of the user's actual rig. The defaults (100 W SSB, 5 dBi horizontal dipole, suburban noise) match the canonical first-visit operator profile; any change to power, mode, antenna, or noise environment recomputes the SNR budget and reshapes the verdict accordingly.

Tier verdicts are driven by the SNR budget plus the additive mode bonuses (Eq. 76: $B_{\text{grayline}} + \max(B_{\text{TEP}}, B_{\text{scatter}})$); the WSPR spot-override of §8.1 can additionally promote a band-group summary tier from below-Good up to Good when the activity ratchet fires, but does not enter the per-band row's tier or stability columns. The harness retains a binary "≥ 50 spots/h" WSPR [34] activity check as a sanity floor only (it does not feed the runtime verdict) since spot activity measures who is transmitting rather than whether a QSO would complete on the configured station; the per-path mode of the harness uses a sparser ≥ 1 spot/h floor over TX/RX bbox-restricted aggregates, documented in §7.2 (Table 8).

Table 11: Tier thresholds. Margin is the dB difference between predicted SNR and the mode's required SNR (Table 4) including all additive mode bonuses; the cuts are absolute dB values, identical across all bands. The corresponding circuit reliability $R = \Phi(M/\sigma)$ varies with the per-band σ_g but is no longer the boundary criterion.

Tier	**Margin**	**Meaning**
Excellent	$\geq +18$ dB	Loud and easy, almost any antenna
Good	$\geq +6$ dB	Clear copy, normal QSO conditions
Fair	≥ -5 dB	Workable with effort, careful timing
Poor	≥ -14 dB	Long shot, weak signals at the floor
Closed	< -14 dB	Band effectively closed

Why fixed dB rather than R-percentile cuts. An earlier revision of the verdict mapped tiers to fixed reliability buckets ($R = 0.90/0.60/0.35/0.10$, equivalent to margin $\geq 1.2816\,\sigma/0.2533\,\sigma/-0.3853\,\sigma/-1.2816\,\sigma$). It read well as operator-facing prose ("Good = QSO completes 3 in 5 attempts") but a 2026-05-10 field-day audit by the author showed it labelled bands as Good or Fair that were unworkable on-air; only 20 m, 17 m, and 15 m carried real traffic at the time the model was reading five HF bands as Good or better. The mismatch came from two sources. First, the percentile cut at $+0.2533\,\sigma$ puts Good at margin $\approx +2$ dB on lower HF ($\sigma_g = 8$ dB); a 2 dB model margin disappears under the 3 to 6 dB of QRM that a contest environment adds on top of the ITU-R P.372 [2] baseline noise the model assumes. Second, the formal operator-readable framing turned out to over-promise: "3 in 5 QSOs complete" encoded a 60% floor that does not survive contact with on-air reality where adjacent-channel splatter, fading, and operator-side antenna or noise mis-estimation all push outcomes downward. The fixed-dB cuts take operator intuition as primary: +6 dB as Good means the model expects to clear required SNR by 6 dB *before* real-world noise adds anything, which lines up with what an operator calls a usable band on-air. The boundaries are now σ-independent; band-specific σ_g continues to drive the stability and confidence columns.

8.1 WSPR Activity Override

WSPR spot counts give an independent observational signal that the physics budget does not consume directly. When observed activity for a band exceeds 1.3× the band's 30-day mean for the current UTC hour *and* the physics-derived tier is currently below "Good" (i.e., Fair, Poor, or Closed), the verdict is promoted to Good. This catches openings outside the budget — weak sporadic-E, off-axis scatter, propagation modes that the climatology underweights.

The override is a strict one-way ratchet *capped at Good as a promotion ceiling, not a global ceiling*: a physics-Excellent verdict is preserved (the tier-rank condition is false when the prior tier is already at or above Good, so the override never fires); a physics-Fair verdict can be promoted to Good but no higher; physics-Poor and physics-Closed can also only be promoted to Good. Low or absent activity does not invalidate the physics; it might just mean nobody is on the air, so the override never demotes.

Where the override appears in the UI. Two surfaces display per-band conditions and they handle the override differently — by design, to avoid pathologies where the row tier and the row Stability percent appear to disagree:

- **Per-band table rows** (HF Bands panel): the Tier column and the Stability column both come from the physics-derived M, σ via `bestPerBand[name].tier` and `bestPerBand[name].stability`. The override does *not* enter individual rows. Tier and Stability are therefore internally consistent within each row: a centred physics-Fair row reads "Fair" near the 66 % middle-tier ceiling; a tail-deep physics-Excellent row reads "Excellent" near 100 %.
- **Band-group summary line** (e.g., "20 m: Good · margin 5 dB · unusually active: 100 spots/h vs avg 30"): the Tier comes from the override-promoted value when the override fires. In that case, the Stability percent is *suppressed* from the line and replaced by the activity note — the operator sees *why* the group was promoted rather than a numerically inconsistent Stability computed against the wrong tier bucket.

The architecture deliberately splits per-band detail (physics truth) from group-level summary (override-aware) so neither display ever forces the operator to read "Good · 27 %".

Data-driven baselines. The reference is the 30-day *mean* hourly spot count per (band, UTC-hour), regenerated daily by the `wspr-baselines.yml` GitHub Actions workflow (a 06:00 UTC cron job that runs `node scripts/harness.mjs wspr-baselines` and commits the diff to `src/data/spot-baselines.mjs`, providing a true sliding 30-day window that tracks solar-cycle progression and seasonal shifts without manual intervention):

```
SELECT band, hour_utc, sum(hourly_count) / 30.0 AS avg_spots
FROM (
 SELECT band, toStartOfHour(time) AS h,
        toHour(time) AS hour_utc,
        count() AS hourly_count
 FROM wspr.rx
 WHERE time >= now() - INTERVAL 30 DAY
 GROUP BY band, h, hour_utc
) GROUP BY band, hour_utc
```

The outer aggregator divides total spots in the (band, hour) cell by the window length in days rather than averaging only over the hours that had at least one spot, so quiet cells (e.g. a band with rare openings at a given UTC hour) get a baseline that includes the zero-spot days rather than only the days that opened. The inner `GROUP BY band, h, hour_utc` groups by both `toStartOfHour(time)` (full date+hour timestamp) and `toHour(time)` (hour-of-day in [0, 23]); the redundancy is intentional, since `h` drives the per-day-per-hour `hourly_count` and `hour_utc` is what the outer group averages over to collapse the 30-day window into a 24-hour cycle.

This produces a 10×24 matrix of typical hourly spot rates across all days of the week. The override fires when observed activity exceeds 1.3× the matrix entry for the current (band, UTC-hour) cell — "the band is *meaningfully* busier than usual for this time of day." Weekends are included in the rolling baseline so that the live observation (which includes weekends) is compared against a like-for-like mean; major contest weekends are $< 10\,\%$ of weeks in a 30-day window so their inflation of the mean is bounded, and they affect the override ratio in both directions (a current contest day reads proportionally less anomalous against a slightly inflated mean). WSPR hourly spot counts are over-dispersed Poisson with strong diurnal structure: spot density depends on operator population (heavily clustered in NA + EU, see "Reception-population bias" below) and on whether the band is open at all, so the variance scales faster than $\sqrt{\text{mean}}$ and a per-cell coefficient of variation in the ∼0.3–1.5 range is typical (lower on heavily-trafficked daytime cells, higher on quiet upper-band predawn cells). At that CV range the 1.3× threshold corresponds to roughly $\text{mean} + 1\sigma$ on the busy cells and $\text{mean} + 0.2\sigma$ on the quiet ones; the per-cell CV distribution itself is not separately tabulated in the paper (see Section 10's deferred-empirical-claims backlog item for the harness extension that would compute it). Tightening to a bare spots > mean comparison would fire on every above-median hour and over-promote borderline cases; the multiplier was empirically chosen as the smallest value that filters out routine above-average hours without missing genuinely anomalous activity. The choice was made by sweeping candidate thresholds in $\{1.0, 1.1, \ldots, 2.0\}$ against the same 30-day WSPR baseline data used to construct the matrix (`wspr.rx` table, 2026-04 window): 1.3× was the smallest ratio at which the override fired on roughly 5–10% of (band, hour) cells per day — low enough to read as "unusual" rather than "daily routine", high enough to keep the override actionable on the cells where it matters (off-axis sporadic-E days, evening 15 m enhancement during HSS recovery, upper-band cells whose 30-day-mean is structurally low for the current solar phase but where live counts climb sharply on a genuine opening). Combined with the Good promotion ceiling, the threshold keeps the override an "unusually active" signal across all (band, hour) cells.

The baseline matrix is emitted as a plain ES module at `src/data/spot-baselines.mjs` and imported at runtime by the derive pipeline. It is regenerated daily at 06:00 UTC by the `wspr-baselines.yml` GitHub Actions workflow (running `node scripts/harness.mjs wspr-baselines` and committing the diff), giving a true sliding 30-day window that tracks solar-cycle progression and seasonal shifts without manual intervention. The committed-and-shipped form means fresh checkouts get valid baselines out of the box without needing to hit ClickHouse on first run.

Reception-population bias in the baseline. The `wspr.rx` aggregate counts spots received by listening stations. WSPR receivers are heavily concentrated in North America and Europe (operator-population statistics from `wsprnet.org`'s active-stations roster; ∼60% of worldwide WSPR receivers sit in those two regions, with another ∼15% in the AU/NZ/JA cluster), so a 30-day baseline of "∼100 spots/h on 20 m at 0900 UTC" reflects predominantly North American daytime listening rather than a globally-uniform "activity at this hour". The override fires uniformly across the operator population, but the activation threshold reflects NA/EU/Pacific listening windows. Operators outside those regions (AF, low-income tropical-belt countries, Central Asia) see a baseline whose 24-hour shape is set by other regions' UTC clocks rather than by their local-time activity rhythm; the override correctly fires when the global aggregate runs hot, but the "unusually active for this hour" interpretation holds most crisply in the receiver-dense regions. A per-region renormalisation is candidate work, gated on enough per-pair receiver coverage to fit regional baselines without over-fitting noise.

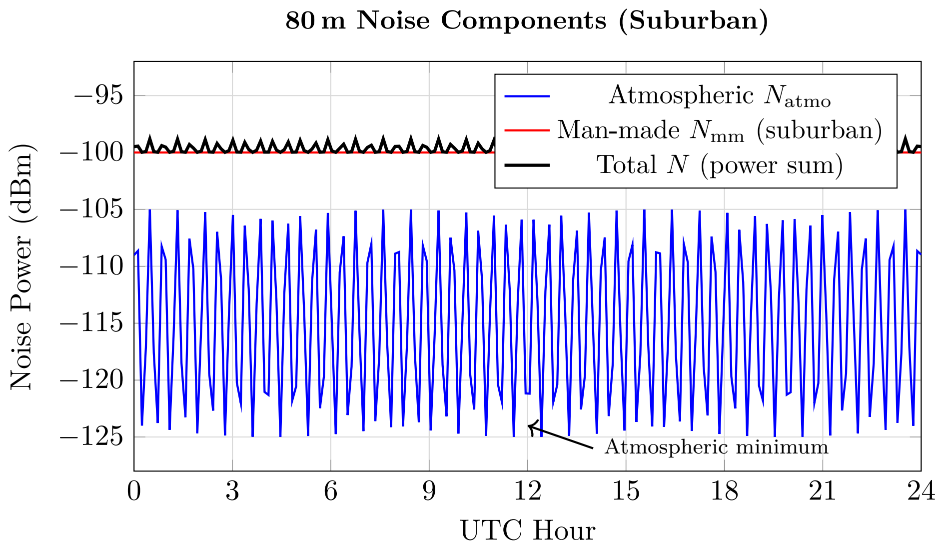

Figure 5: Noise power components on 80 m in a suburban environment. The man-made noise (-100 dBm) dominates at all hours; the atmospheric diurnal swing barely affects the total. Power-summing the two components in linear units (the model used here) gives the black curve, which sits essentially at the man-made level since $10^{-115/10} + 10^{-100/10} \approx 10^{-100/10}$ when the components differ by ~15 dB. An earlier formulation that combined the two by an incorrect dB-domain operation produced a noise estimate several dB above the man-made-dominated truth on suburban / urban sites, inflating the margin and biasing verdicts open; the power-sum form is therefore a correctness constraint, not a stylistic choice.

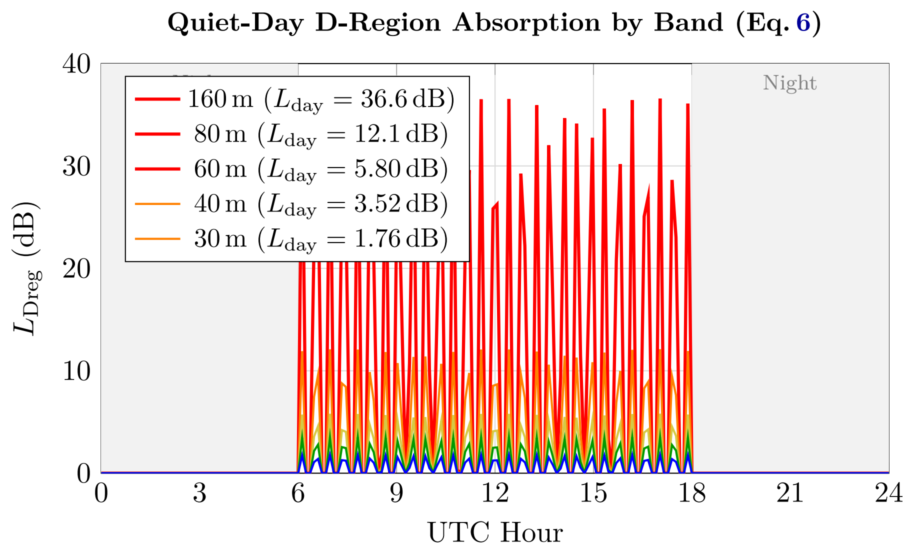

Figure 6: Quiet-day D-region absorption by band under the $L_{\text{day}}(f) \cdot \cos^{0.7}(\chi)$ form of Eq. 6. 160 m reaches 36.6 dB at local noon; 30 m and above are essentially unaffected. Night absorption is zero. The illustrative curves show a smooth descent toward zero at low $\cos\chi$, but the actual implementation hard-gates $L_{\text{Dreg}} = 0$ at $\cos\chi < 0.05$ (Eq. 6); the figure smooths that gate for readability rather than showing the cliff. The grayline bonus of §5.5 adds the $L_{\text{day}}(f) - L_{\text{Dreg}}(f, \chi)$ refund on top of the day-side charge plotted here, capped at 25 dB.

9 Output Formatting and Alerts

The tiers and indices produced by the foregoing pipeline are surfaced across an eight-panel web UI:

- **Active Alerts** (§9.2): SWPC-bulletin feed plus rule-derived "soft" alerts in one severity-sorted list at the top of the page.
- **HF Bands**: the ten HF amateur bands as a per-band table. Each row shows the band frequency, the predicted verdict tier (Excellent / Good / Fair / Poor / Closed per Section 8), the SNR margin in dB, the Stability percent (Eq. 78: Φ of σ-distance to the nearest tier boundary; bucket-width-independent and so directly readable on an absolute scale — 50 % at a boundary, ∼84 % one σ inside, → 100 % deep inside), the dominant propagation mode (F2, NVIS, Es, TEP, F2-scatter, MS, or VHF Auroral-E [Section 6.4], displayed as "Aurora"), and the best-margin destination from the path basket. Distinct from circuit reliability R of Eq. 77: R is the probability the link is viable, Stability is the probability the verdict *label* is robust to within-σ noise. The "Best Path" cell is a compression of the underlying 10-path evaluation; for the per-destination breakdown consult the Reference Paths sub-panel in the Ionosphere section (cross-referenced by the cell's info popover).
- **VHF Bands**: 6 m and 2 m mirrored from the HF layout, with Es-MUF (§6.1), meteor scatter (§6.3), and an "aurora" regime label (path contributes a non-zero L_{aur} from §3.7 and the E-region scatter that accompanies auroral activity is what the path is reading on, rather than F2 refraction; this is a UI mode-label rather than a separate physics treatment). A second panel below the band table renders per-station tropospheric ducting status from the 24-station UWyo radiosonde basket (Section 6.6): each row shows the lowest-1 km $\mathrm{d}N/\mathrm{d}h$ and an ITU-R P.453 classification (standard / super-refractive / ducting), sorted ducting first so an active event surfaces at the top.
- **Upcoming Disruptions** (outlook): SWPC's 3-day R/S/G probability table, the SWPC 3-h K_p forecast, Earth-directed CMEs from NASA DONKI, IMO-catalogue meteor showers, top active solar regions, and the 27-day ISES SFI / A_p outlook.
- **Ionosphere**: nearest-GIRO-station digisonde readings (foF2 / foEs / hmF2), GNSS TEC, 30 MHz cosmic noise, the Reference Paths panel — the per-QTH path table listing kc2g MUF on the five short-path destinations and their long-path counterparts, ten paths total — plus the kc2g global foF2 map and the GloTEC TEC map. The Reference Paths panel is the source-of-truth that Section 8's band-table Best-Path column compresses; an operator who wants per-destination detail rather than the headline winner consults this panel.
- **Geomagnetic**: K_p / A_p / D_{st} / H_{p30} / Sym-H / AE / aurora HP indices, a 7-day K_p trend chart, and the OVATION aurora forecast and D-RAP map.
- **Solar**: SFI / SN / X-ray class / proton flux indices, real-time DSCOVR / ACE solar wind (B_z / V_{sw} / n), and SDO imagery (HMI visible-light sunspots, EUV 193 Å corona / coronal holes, EUV 304 Å chromosphere; per `src/ui/sections.js`).
- **Links**: data source attribution, license credits, and links to the upstream services.

This section documents the assembly logic for the two surfaces that depend on derived numeric thresholds (the alerts panel soft-rule grader, and the per-band tier verdict feeding the HF / VHF tables); the layout choices for the other panels are data-source documentation rather than algorithm.

9.1 Algorithmic surfaces and data-source surfaces

Of the eight panels above, three have non-trivial assembly logic that this section documents:

- *Active Alerts* (§9.2, below) combines SWPC bulletins with rule-derived "soft" alerts whose thresholds are model-side numeric constants rather than upstream fields.
- the *HF Bands* and *VHF Bands* tables surface per-band $(M, \sigma) \to$ tier verdicts already documented in detail in Sections 3, 7.3, and 8; their layout is just a transposition of those

into table rows.

- the *Reference Paths* sub-panel (inside the Ionosphere panel) computes per-QTH great-circle midpoints for five canonical destinations and their long-path counterparts (10 paths total), looks up the nearest kc2g ionosonde for each midpoint subject to a 30 min freshness gate, and renders the per-path MUF with a colour-coded f/MUF column for the band currently selected by the band table. This is the full-detail companion to the band table's compressed Best Path column (Section 8).

The remaining five panels (Upcoming Disruptions, the rest of Ionosphere outside Reference Paths, Geomagnetic, Solar, Links) are data-source documentation: each block is a labelled rendering of a fetched upstream (SWPC / DONKI / IMO / kc2g / GIRO / SDO) without ionocast-side processing. Their assembly logic lives in the data fetchers and the `src/ui/sections.js` layout description rather than in this section; an operator-facing description of each block sits inline in the panel's `interp` attribute and is displayed when the panel's info badge is clicked.

Reading the Stability column at a glance. The band-table column displays $S(M, \sigma)$ from Eq. 78: Φ of the σ-distance to the nearest tier boundary. Bucket-width-independent, so a centred Fair verdict and a centred Excellent verdict at the same σ-offset from their respective nearest boundaries read the same percentage. Operator reading: 50 % at the boundary itself, ∼84 % one σ inside the bucket, asymptotic to 100 % deep inside; finite-width middle buckets (Poor / Fair / Good) cap at distinct per-bucket centred ceilings, $\Phi(0.45) \approx 67\,\%$ for Poor, $\Phi(0.32) \approx 63\,\%$ for Fair, $\Phi(0.51) \approx 70\,\%$ for Good (average ∼66 %, often referenced as a single round figure in plain-prose contexts), so read mid-tier values relative to their own bucket-specific ceiling and open-ended Closed / Excellent values against the absolute scale. The two-sided literal $P(\text{predicted} = \text{true})$ form $C(M, \sigma)$ retained in source as `tierConfidence` caps at $\sim 32\,\%$ on centred middle tiers; the production UI (`src/ui/builders/tables.js`) surfaces S instead.

9.2 Active Alerts Panel

The alerts panel combines two rendering pipelines into one severity-sorted list.

Official SWPC feed. A polling fetch of `products/alerts.json` emits up to four recent items. Severity is derived from the canonical 6-letter Space Weather Message Code embedded in the bulletin body (e.g. `ALTTP2`, `WARK04`); the 3-letter prefix maps as `ALT*` / `WAR*` → *alert*, `WAT*` → *watch*, and `SUM*` / `FOR*` / `ADV*` → *info*. When the message code is missing or unrecognised, the parser falls back to the leading severity word in the body ("ALERT:", "WARNING:", "WATCH:", "SUMMARY:", "FORECAST:", "ADVISORY:") with the same mapping. Anything else lands in *info*. The mustard *warn* tier was retired during the pill-revamp pass and is no longer emitted by either the SWPC pipeline or the soft-alert rules.

Rule-derived (soft) alerts. Eleven rules run on every refresh against the same data bundle the per-band budget consumes (K_p, K_p^{eff}, B_z, D_{st}, storm phase, aurora HP, DRAP, GOES X-ray, GOES $\geq 10\,\text{MeV}$ protons, solar-wind speed, and forecast σ). Each rule is *graded* and short-circuits at the highest matching tier, so $K_p = 9$ does not simultaneously fire G5, G4, G3, G2, and G1. Thresholds reuse the constants documented in Sections 3–5; no new magic numbers are introduced.

Intra-tier sort order. Alerts are sorted primarily by severity tier (*extreme* > *alert* > *watch* > *info*); within a single tier the rule iteration order is the deterministic tiebreak: FLARE first (most time-urgent — X-ray ionisation peaks within seconds of the flare event, operator response window is minutes), then geomagnetic indices (K_p, B_z, D_{st} in that order), then aurora, DRAP, PCA, storm recovery, HSS arrival, forecast σ inflation, and finally the geomagnetic-storm composite chip (storm-phase summary). The geomagnetic block is sorted by operator familiarity rather than causal chain: K_p first because it is the most-cited storm-severity index in amateur literature (the NOAA G-scale headlines map directly to it), B_z next as the L1 forward-warning input that leads geomagnetic activity at Earth by ∼30–60 min via solar-wind transit ($1.5e6\,\mathrm{km}/v_{\mathrm{sw}}$), and D_{st} last as the ring-current depth signal. The underlying causal chain runs in the opposite direction: $B_z \to D_{\mathrm{st}} \to K_p$ (§5.3.2 records that D_{st} responds within minutes of storm onset, 1–3 h before the 3-hour K_p index publishes). The fixed display order means repeated refreshes show the same chip arrangement, so an operator scanning the panel between refreshes can pattern-match position rather than re-read each line.

Table 12: Soft-alert rule set. Severity maps to the same *info* / *watch* / *alert* / *extreme* levels used by the SWPC renderer; *extreme* ranks highest. Rule order matches the deterministic intra-tier tiebreak in the prose above. Subscripts on labels are the NOAA G/R/S scale codes the rule emits as a clickable inline term in the chip body.

Trigger	**Threshold**	**Level**	**Label**
X-ray flare	class ≥ X20	extreme	FLARE R5
	class ∈ X10–X19.9	extreme	FLARE R4
	class ∈ X1–X9.9	alert	FLARE R3
	class ∈ M5–M9.9	alert	FLARE R2
	class ∈ M1–M4.9	watch	FLARE R1
Geomagnetic K_p	$K_p \geq 9$	extreme	G5
	$K_p \geq 8$	extreme	G4
	$K_p \geq 7$	alert	G3
	$K_p \geq 6$	alert	G2
	$K_p \geq 5$	watch	G1
IMF B_z southward	$B_z \leq -10\,\mathrm{nT}$	alert	BZ SOUTH
	$B_z \leq -5\,\mathrm{nT}$	watch	BZ SOUTH
Ring current D_{st}	$D_{\mathrm{st}} \leq -250\,\mathrm{nT}$	extreme	DST EXTREME
	$D_{\mathrm{st}} \leq -100\,\mathrm{nT}$	alert	DST STRONG
	$D_{\mathrm{st}} \leq -50\,\mathrm{nT}$	watch	DST MODERATE
Aurora HP	HP ≥ 100 GW	alert	AURORA
	HP ≥ 50 GW	watch	AURORA
D-region absorption at QTH	DRAP ≥ 10 MHz	alert	ABSORPTION
	DRAP ≥ 5 MHz	watch	ABSORPTION
Polar cap absorption	flux ≥ 100 000 pfu	extreme	PCA S5
	flux ≥ 10 000 pfu	extreme	PCA S4
	flux ≥ 1 000 pfu	alert	PCA S3
	flux ≥ 100 pfu	alert	PCA S2
	flux ≥ 10 pfu	watch	PCA S1
Storm recovery tail	$K_p^{\mathrm{eff}} - K_p \geq 1$ & $K_p^{\mathrm{eff}} \geq 4$	info	STORM TAIL
HSS arrival	$v_{\mathrm{sw}} \geq 800\,\mathrm{km/s}$	alert	HSS
	$v_{\mathrm{sw}} \geq 600\,\mathrm{km/s}$	watch	HSS
Forecast σ inflation	$\sigma_{\mathrm{forecast}} \geq 4\,\mathrm{dB}$	watch	FCAST
	$\sigma_{\mathrm{forecast}} \geq 2\,\mathrm{dB}$	info	FCAST
Geomagnetic-storm composite	`stormPhase = "main"` (per §5.3.2)	alert	GSTM

Soft alerts render alongside SWPC notices rather than only when the official feed is empty: the derived context — southward B_z, falling D_{st}, DRAP blanketing QTH, an HSS front arriving — is material information SWPC does not publish until formal watch or warning bulletins are issued, which may lag the underlying event by several hours.

10 Known Limitations and Future Work

The items below are the model's currently-known shortcomings and the cleanups they imply. They mix three classes that a reader prioritising follow-up work may want to disentangle: *physics gaps* where the model omits a propagation mechanism or approximates one (#2 TX pattern, #3 NVIS height, #4 RX-side noise rejection, #5 EIA crests, #6 flare SID decay, #7 storm-lag kernel, #10–#11 equatorial / polar residuals, #13 Es coverage, #15 generic chordal-hop propagation outside TEP and F2-scatter); *validation gaps* where the metric or the calibration target has a known ceiling (#9 binary-accuracy ceiling, #12 w_{sc} tied to a metric with a structural cap); *process / calibration cadence* items where the infrastructure rather than the physics is the bottleneck (#1 in-app feedback loop not yet built, #8 upper-band scatter-recovery dependence as an implementation status note); and *closed items retained for calibration history* (#14 N_{base} re-anchor, kept as the most recent ground-truth alignment but not gating any current verdict). The numbering is chronological by when the item entered the backlog; readers wanting priority order should weight #1–#7 (active gaps) above #9, #12, #14 (calibration / metric-ceiling items).

1. **No per-user bias correction loop yet.** An on-device validation feedback loop (operator scores predictions, per-(band, forecast-horizon) bias and σ tighten as samples accumulate) is the intended target shape but is not yet implemented in the live runtime; every operator currently sees the same global tune. Until such a loop ships, the per-band σ is the static base from Table 9 (8 dB on 160 m–40 m, 9 dB on 30 m / 20 m / 17 m, 10 dB on 15 m, 12 dB on 12 m / 10 m, 8 dB on the VHF placeholders; the unconditional 8 dB fallback is reached only when the band-frequency lookup misses entirely, which the BANDS table makes impossible in production), and the offline harness (Section 7.2) is the only feedback mechanism — it validates the global tune against 30 d of WSPR aggregates but does not specialise to any one operator.

2. **TX antenna pattern is an elevation-only cut.** The type + height + λ model (Section 3.14) captures the dominant TX-side physics – ground-reflection lobe position, free-space directivity, and polarisation – but misses second-order effects on the transmit budget: azimuth is not modelled (we don't know where a beam is pointed, so beams get their elevation-cut gain only); ground conductivity is fixed at "average soil" for all verticals (salt-water / sea-shore verticals are $\sim$3 dB better); the real-earth pseudo-Brewster null on verticals ($\sim$5–8 dB dip near the pseudo-Brewster angle, $\sim$13–15° elevation over average soil per the ARRL Antenna Book [13] pseudo-Brewster-angle tabulation; the sea-water / very-good-ground angles drop to 3–8°) is absent; and beams are treated as having the same elevation-lobe shape as a horizontal wire at the same height, which is correct for the main lobe but misses the additional off-broadside suppression a multi-element array provides.

3. **NVIS secant model is single-layer.** For paths < 500 km the model substitutes GIRO f_{oF2} with a secant correction (Eq. 51) anchored at the observed h_{mF2} when available, defaulting to 300 km. A more accurate NVIS model would account for E-/F-layer mode coexistence at low elevations and the full angular dependence up to $\sim 70^\circ$ off-vertical, where the simple secant overestimates the oblique gain.

4. **No receive-side directional noise rejection.** A physically distinct effect from #2 above: the transmit-side pattern affects the radiated signal toward the destination; the receive-side

pattern affects the *noise* captured. A Yagi pointed at the target rejects off-axis noise by its front-to-back ratio (5 dB–15 dB), which improves SNR even though signal-side gain on a one-way link is unchanged by the orientation choice. This effective SNR improvement is not yet modelled. Splitting transmit and receive antenna directivity would also let the budget account for asymmetric station geometry (e.g. Yagi at one end, dipole at the other).

5. **Equatorial anomaly crests are approximated, not measured.** The Gaussian correction in Eq. 45 captures the daytime crest enhancement at dip latitude $\pm 15\,°$ (+50% of the base foF2 at $F_{10.7\mathrm{A}} = 70$, saturating to +85% at $F_{10.7\mathrm{A}} \geq 120$) plus the equatorial-trough depression on the dip equator (-30% at $F_{10.7\mathrm{A}} = 70$, saturating to -50% at $F_{10.7\mathrm{A}} \geq 170$), but uses a single Gaussian width and a single crest position for all seasons and cycle phases. The true EIA migrates in latitude and varies with the equatorial $E \times B$ drift; a data-driven crest locator would be more faithful.

6. **Flare SID decay not tracked.** The flare absorption term uses the current X-ray class as an instantaneous proxy and therefore misses the 10–60 minute exponential decay of the SID after an X-ray peak. The 10-minute model refresh picks up the decayed class, so the error is bounded, but minute-scale transitions are smeared.

7. **Storm-lag kernel uses two coarse parameters.** The Laplace-kernel-weighted effective K_p (Eq. 60) has τ_{peak} fixed at 2 h and τ_{decay} switched between 8 h (CME) and 24 h (CIR / HSS) per the storm-type classifier of §5.3.1. Real F-region dynamics depend on storm intensity, latitude, and season; a per-storm-intensity τ_{decay} schedule and a latitude-dependent τ_{peak} are good targets for the offline calibration harness once a longer archive is available.

8. **Upper-band tracking is now scatter-recovery-bound.** After the climatology rebuild (Section 4.2), the H-pol Fresnel multi-hop model (Section 3.8), the per-band σ refit (Table 9), and the F2-scatter recovery weight calibrated by joint coordinate descent (Section 6.5, Section 7.2), the upper-band binary-accuracy lift is concentrated on 12 m and 10 m, where the F2-scatter recovery channel is the dominant remaining lever. Lower bands (60 m and below) saturate at $> 98\,\%$ accuracy under the global-truth metric and are not budget-limited. Headline binary-accuracy values for each calibration run are in Table 8; *read those numbers as regression-detection signals, not as operator-facing per-path accuracy* (see #9 below for the structural ceiling that the per-path column exposes).

9. **The harness's binary-accuracy metric has a structural ceiling that permissive predictors clear without effort.** The "open ≥ 50 spots/h" target is an OR over every path on the band: most hours of most upper bands are "open somewhere" globally, so a predictor that errs toward "open" scores well even when its per-path verdicts are wrong. The sharpest evidence is the gap between the two ground-truth modes *on the same harness run*: at the pre-B6 2026-04-28 freeze the global aggregate read 94.25 % accBin while the per-path mode (TX/RX-bbox restriction with ≥ 1 spot/h floor) read 30.97 % on the identical predictions; the post-B6 2026-05-08 baseline reads 84.08 % global vs 40.17 % per-path, narrower but still substantial. A 40–60 pp gap across two encodings of the same underlying truth is not a model property; it is a metric property. The headline number is sitting on the "somewhere is open" ceiling, not on a per-path-correct floor. Replacing the WSPR aggregate with TX-RX-pair queries (per-path ground truth) and replacing the climatology fallback with a real-time IRI assimilation (IRTAM) are the next two highest-leverage improvements; both require a server-side fetcher with periodic grid retrieval, which the current Cloudflare Pages-only architecture does not host.

10. **Equatorial-belt foF2 validation has a residual gradient.** The equatorial set is Boa Vista (BVJ03, northern crest, dip lat $\sim +12°$ N, BR), Ascension (Atlantic, dip lat $\sim -3°$, deep in the trough region), Townsville and Niue (southern crest, Pacific), and Jicamarca

(essentially on the dip equator, a second equatorial-trough anchor rather than a crest one). The 2026-04-29 BVJ03 addition closed the one-sided crest fit; the EIA grid sweep (Section 4.2) now bottoms out near max|bias| $\approx$ 1.18 MHz rather than the pre-BVJ03 $\sim$2.2 MHz floor. Further reduction is bounded by station density across the longitude-varying dip equator: Indian-sector and SE Asian dip-latitude stations (Trivandrum, Hyderabad, Madimbo, etc. are candidates — none currently return DIDB data) or a real-time IRI assimilation are the next leverage points. This residual is coupled to the polar over-prediction in #11 below: both are residuals on the same $\mathcal{P}(\phi) \cdot d(\cos\chi)$ dayBump and $A_{\text{EIA}}(F_{10.7\text{A}})$ term, so a clean fix needs to retune them jointly.

11. **Polar over-prediction and equatorial residual are coupled in the same model term.** High-latitude stations Tromsø, Gakona, and Eielson AFB over-predict daytime foF2 by $\sim$1–1.7 MHz, and the equatorial bias (#10 above) floors at $\sim$1.18 MHz post-BVJ03. These are listed as separate items because their candidate fixes differ (polar wants a steeper sigmoid above 60°, partly addressed by Eq. 41; equatorial wants additional Indian / SE-Asian dip-latitude stations), but structurally they are residuals on the same $\mathcal{P}(\phi) \cdot d(\cos\chi)$ dayBump and $A_{\text{EIA}}(F_{10.7\text{A}})$ term in Eq. 39: fitting the polar fall-off and the EIA Gaussian amplitude / width separately — as a sequence of single-region tunes — can each regress the other. A clean closure needs a joint refit holding both residuals visible in the cost function; the BVJ03-extended basket makes this tractable now and is tracked in `docs/BACKLOG.md` under the latitude-gradient joint refit item.

12. **Scatter-recovery weight $w_{\text{sc}} = 1.5$ is defensible-but-not-final.** The scatter weight was set by a coordinate-descent sweep against the harness's binary "$\geq$ 50 spots/h" WSPR target (Section 6.5 / Section 7.2). An unrestricted sweep keeps improving as $w_{\text{sc}} \to 4$, but that range is known to be exploiting the global-vs-path-specific structural mismatch in the harness rather than a real propagation effect, so the shipped weight is held at the 15 dB-cap physical edge. The paper acknowledges this in Section 6.5, but the limitation is worth surfacing as its own line in this list: the calibrated w_{sc} has been validated against a metric whose calibrators do not fully trust beyond a certain point. Closing the loop properly requires re-running the sweep against the per-pair WSPR ground truth (Section 7.2) once that mode is wired into `replayMarginFromCell`'s alternative-mode bonuses; the new optimum is then reported against a metric that the calibrators *do* trust at all weights in its range.

13. **Sporadic-E budget under-fits rare 30 m / 40 m openings by design.** The HF Es budget (`snrMarginHfEs`) returns null below 14 MHz regardless of f_{oEs} (Section 6.1), even though the algebraic Es MUF gate $f \leq 5 f_{\text{oEs}}$ would otherwise admit lower bands on strong-Es days ($f_{\text{oEs}} \geq 4$ MHz pushes $5 f_{\text{oEs}} \geq 20$ MHz). The hard 14 MHz floor matches amateur-radio literature: Es is essentially unobserved below 20 m at the canonical 2000 km single-hop distance the budget assumes. But rare Es openings on 30 m / 40 m *do* appear in the WSPR record during peak-cycle summer evenings with $f_{\text{oEs}} \geq 8$ MHz, typically on shorter-than-2000 km paths where the geometry makes lower-band Es viable. These events are not captured by the budget. An operator who sees 30 m WSPR activity that the model calls F2-only should treat the verdict as "F2 explains most of it; rare Es is possible on top and the budget will not tell you." The Es budget is also *single-hop only* ($d_{\text{Es}} = 2000$ km hard-coded regardless of actual path length, per §6.1); multi-hop Es chains and Es-plus-F2 mode coexistence on long paths are not modelled. Removing the lower-band floor and adding multi-hop geometry would require a distance-aware Es model and a recalibrated $L_{\text{iono,Es}}$ for the shorter-hop regime; both are deferred to a future VHF-Es-aware harness.

14. **N_{base} table re-anchored to P.372 quiet-rural (closed 2026-04-30).** Pre-2026-04-30, N_{base} was being compared against eyeball P.372 references that sat below the cosmic galactic floor. Once the `tests.mjs` noise-floor suite was corrected to evaluate $N_{\text{base}} + \Delta N_{\text{diurnal}}$

at midnight against properly-anchored P.372 references, the model came out quieter than P.372 quiet-rural midnight by 0–11 dB on the upper bands. The 2026-04-30 retune ports the corrected anchors directly into N_{base} (net shift: 17–10 m all +11 dB, 30 m +5, 20 m +8, 80–40 m +1–2, 160 m +10, 60 m unchanged). WSPR per-spot residual mean improved from −23.97 to −18.73 dB; per-pair Brier improved 0.64→0.57. The item is retained in this list as the most recent calibration ground-truth alignment; it is closed for forward work and does not gate any current verdict.

15. **Generic chordal-hop propagation not modelled outside TEP and F2-scatter.** ionocast explicitly models one chordal mode — trans-equatorial propagation (§6.2, `tepBonusDb`) — which captures the chordal afternoon / evening F-region mode across the magnetic equator that skips multiple D-region traversals. The F2-scatter recovery (§6.5, `scatterBonusDb`) covers adjacent physics: above-MUF off-axis recovery via F-region irregularities and short-chord modes near MUF. Other chordal regimes are *not* separately modelled: generic mid-latitude chordal hops (signals reflecting between F-region tilts without intermediate ground bounces in non-equatorial regions), polar chordal modes along auroral-oval gradients, and long-haul N–S evening chordal modes outside the TEP-eligibility window (cross-equator + 17–23 LT + $|\phi_{\text{dip}}| \geq 10°$ on both endpoints). The link budget assumes ground reflection between F2 hops (Fresnel L_{gr} + defocus D per intermediate bounce, plus $n_{\text{hops}} \cdot L_{\text{iono}}$), so any chordal trajectory that skips a ground bounce is over-charged on the loss side compared to reality. TEP and scatter bonuses partially compensate where their eligibility gates fire; on a long-haul N–S midlatitude path outside those gates (e.g. a ZL–JA evening 20 m path that anecdotally reads stronger than the standard ground-reflection budget would suggest), the verdict reads slightly pessimistic. Adding a generic chordal-mode bonus would require a calibration anchor distinct from the equatorial TEP record — candidate paths are sparse in the WSPR basket and the bonus would risk over-rewarding paths the standard budget already calls open — so the gap is acknowledged here rather than papered over. Closing it cleanly is gated on an extension of the per-pair WSPR ground-truth wiring (#9) plus a separate calibration sweep specifically for non-equatorial chordal candidates.

A Default Settings

Table 13: Default operator settings.

Parameter	Default Value	Notes
Transmit power P_{tx}	50 dBm (100 W)	Typical HF transceiver
Antenna gain G_{ant} (peak)	+5 dBi	Installed dipole w/ ground reflection; effective ga
Mode	SSB	$S_{\text{req}} = +10$ dB
Noise environment	Suburban	$F_a = +15$ dB
Reference bandwidth	2.5 kHz	SSB passband
Spread σ_g (base)	8–12 dB per band	ITU-R P.533 anchor, not fitted (Table 9)
kc2g freshness gate	30 min	Stale data discarded
Hop distance ceiling	4000 km (at $h_F = 300$ km)	The runtime ceiling is h_F-dependent per Eq. 19;
$L_{\text{iono,HF}}$	1 dB per F2 reflection	Lumped median-conditions residual (§3.11; `L_IO`
$L_{\text{iono,Es}}$	15 dB per Es hop	Patchy / aspect / polarisation aggregate (§6.1; `L`
$L_{\text{iono,aur}}$	25 dB per VHF auroral-E hop	Empirical lump (§6.4; `L_IONO_AUR_DB`)
D (defocus per extra hop)	0.25 dB	Calibrated against 30-day WSPR (§3.8; `DEFOCUS`
D-region prefactor K_D	$200\,\text{dB} \cdot \text{MHz}^2$	$L_{\text{day}}(f) = K_D/(f + 0.5)^2$ (Eq. 6; `D_REGION_PREF`

B Noise Floor Table

The reference noise data feeding the noise model (Section 3.12) is in Table 3 of the main text. The operator's noise environment selection determines how the man-made channel of Eq. 25 fires: rural sites gate the channel off entirely (the indicator $\mathbb{1}_{\text{env}\neq\text{rural}}$ is zero); suburban sites add the man-made channel at $N_{\text{base}} + 15\,\text{dB}$; urban at $N_{\text{base}} + 25\,\text{dB}$. F_a is a single additive term, so a per-environment expansion of Table 3 is unnecessary; the reader can apply the offset mentally.

C Calibration Path Basket

The 35-path basket used by the offline calibration harness (`scripts/harness.mjs`, Section 7.2). Each row is a great-circle TX→RX pair specified by endpoint coordinates; the basket is intentionally diverse across hop count (1 to 5 hops, NVIS through long-path), midpoint geometry (near-station vs ocean), and latitude regime (midlatitude / polar / equatorial / transequatorial). The same basket is the input to the sweep that produced the binary-accuracy figure quoted in Section 7.3.1.

Table 14: The 35 calibration reference paths. Columns: path name, TX endpoint (ϕ, λ) in degrees, RX endpoint (ϕ, λ) in degrees, and the geometry / use-case note from `scripts/paths.json`.

Name	**TX** ϕ	**TX** λ	**RX** ϕ	**RX** λ	**Geometry / use case**
EU–EU short	38.72	-9.14	50.45	30.52	3000 km midlat 1-hop
EU–EU west	55.95	-3.20	37.40	-5.99	2400 km midlat 1-hop
EU–EU east	60.17	24.94	37.97	23.73	2500 km midlat 1-hop
EU–NA mid	38.72	-9.14	40.71	-74.01	5400 km transatlantic 2-hop
EU–NA north	55.95	-3.20	45.51	-73.57	5100 km transatlantic 2-hop
EU–NA south	37.40	-5.99	25.76	-80.19	7100 km transatlantic 2-hop
NA–EU east	40.71	-74.01	50.45	30.52	7600 km transatlantic 2-hop
NA–NA west-east	37.77	-122.42	40.71	-74.01	4100 km transcontinental 2-hop
EU–Asia	38.72	-9.14	35.68	139.69	11144 km 3-hop, midpoint Siberia
NA–Asia	37.77	-122.42	35.68	139.69	8300 km transpacific 3-hop
Hawaii–VK	21.31	-157.86	-33.87	151.21	8200 km transpacific equator-crossing 3-hop
Hawaii–NA	21.31	-157.86	37.77	-122.42	3900 km midlat 1-hop, midpoint Pacific
Hawaii–JA	21.31	-157.86	35.68	139.69	6200 km midpoint Pacific 2-hop
EU–South Africa	38.72	-9.14	-26.20	28.04	8500 km TEP-eligible 3-hop
NA–Brazil	40.71	-74.01	-23.55	-46.63	7700 km TEP-eligible 2-hop
EU–Brazil	38.72	-9.14	-23.55	-46.63	8700 km TEP-eligible 3-hop
VK–EU	-33.87	151.21	50.45	30.52	14944 km Asia-transit 5-hop, midpoint SE Asia
VK–EU west	-33.87	151.21	38.72	-9.14	18200 km 5-hop, midpoint Indian Ocean
VK–NA	-33.87	151.21	37.77	-122.42	11900 km transpacific 4-hop
JA–VK	35.68	139.69	-33.87	151.21	7800 km 2-hop, equator-crossing
JA–EU	35.68	139.69	50.45	30.52	7800 km 2-hop, midpoint Siberia
NA–Argentina	40.71	-74.01	-34.61	-58.38	8500 km TEP-eligible
EU–Argentina	38.72	-9.14	-34.61	-58.38	10100 km 3-hop
Polar W–UA	40.71	-74.01	55.75	37.62	7500 km via near-pole 2-hop
Polar W6–EU	37.77	-122.42	50.45	30.52	9700 km polar 3-hop
Polar W–NA	40.71	-74.01	37.77	-122.42	4100 km, hop-count consistency check
Equator NA	21.31	-157.86	-23.55	-46.63	13013 km equator-crossing 4-hop
Asia–VK	22.32	114.17	-33.87	151.21	7400 km 2-hop
EU–Asia short	55.75	37.62	35.68	139.69	7500 km 2-hop, midpoint Siberia/China
NVIS DE–NL	52.52	13.41	52.37	4.90	600 km NVIS-tail short
NVIS GB–FR	51.51	-0.13	48.85	2.35	350 km true NVIS short
NVIS US–CA	40.71	-74.01	45.42	-75.70	530 km NVIS-tail
NA–IT	40.71	-74.01	41.90	12.50	6900 km transatlantic midlat
EU–Mid-East	50.45	30.52	25.20	55.27	4200 km 2-hop midlat-equatorial
ZL–W6	-41.29	174.78	37.77	-122.42	11200 km transpacific 4-hop

Endpoint coordinates are read by the harness from `scripts/data/paths.json` (the source of truth); this table is a paper-side mirror. "Polar W–NA" is an intentional duplicate of NA–NA west-east, retained as a hop-count regression canary, so the basket has 35 entries but only 34 unique (TX, RX) pairs.

D GIRO Digisonde Stations

The 27 ionosonde stations queried by the harness for foF2 / foEs ground-truth. The list lives in two files that must stay in sync: `scripts/harness.mjs`'s `GIRO_STATIONS` (URSI code plus coordinates, used by the offline calibration harness) and `functions/_handlers/giro.js`'s `GIRO_STATIONS` (URSI code, display name, coordinates, and provider attribution, used by the live UI). Coordinates are verified against the kc2g station registry by `scripts/harness.mjs verify`.

Table 15: The 27 GIRO digisonde stations used for foF2 / foEs ground-truth and the EIA grid sweep. Coordinates are (ϕ, λ) in degrees (East $>$ 0, West $<$ 0). Dip latitude ϕ_{dip} is computed via the tilted-dipole approximation of Eq. 17 with the IGRF-13 north magnetic pole at (80.7°N, 72.7°W); positive values are in the northern magnetic hemisphere. Region groupings follow the harness's structural blocks; "Provider" is the institution operating the digisonde feed via the GIRO / DIDB network.

URSI	**Name**	ϕ	λ	ϕ_{dip}	**Provider**
Europe / midlatitude (10)					
PQ052	Pruhonice, CZ	50.0	14.6	49.5	Inst. Atmospheric Physics, Czech Acad. Sci.
JR055	Juliusruh, DE	54.6	13.4	54.2	IAP Kühlungsborn
EB040	Roquetes, ES	40.8	0.5	42.9	Observatori de l'Ebre
FF051	Fairford, GB	51.7	-1.5	53.8	Rutherford Appleton Lab / STFC
SO148	Sopron, HU	47.6	16.7	46.9	Geodetic and Geophysical Inst., HAS
AT138	Athens, GR	38.0	23.5	36.4	National Observatory of Athens
RO041	Rome, IT	41.8	12.5	41.9	INGV Rome
GM037	Gibilmanna, IT	37.9	14.0	37.8	INGV Gibilmanna
NI135	Nicosia, CY	35.0	33.2	32.0	GIRO / DIDB
DB049	Dourbes, BE	50.1	4.6	51.2	Royal Meteorological Inst. of Belgium
North America (4)					
MHJ45	Millstone Hill, MA	42.6	-71.5	51.9	MIT Haystack Observatory
BC840	Boulder, CO	40.0	-105.3	47.6	NOAA / NCEI Boulder
WP937	Wallops Island, VA	37.9	-75.5	47.2	NASA Wallops Flight Facility
IF843	Idaho Natl Lab, ID	43.8	-112.7	50.6	Idaho National Laboratory
Polar / high-latitude (3)					
TR169	Tromsø, NO	69.6	19.2	67.4	EISCAT / UiT Tromsø
GA762	Gakona, AK	62.4	-145.0	63.8	HAARP / Gakona
EI764	Eielson AFB, AK	64.7	-147.1	65.6	GIRO / DIDB
Equatorial / low-latitude (3)					
AS00Q	Ascension Island	-7.9	-14.4	-3.0	UK Ionospheric Monitoring Service
BVJ03	Boa Vista, BR	2.8	-60.7	11.9	Instituto Nacional de Pesquisas Espaciais (INPE)
JI91J	Jicamarca, PE	-11.9	-76.9	-2.6	Radio Observatorio de Jicamarca / IGP
Africa / southern midlatitude (1)					
LV12P	Louisvale, ZA	-28.5	21.2	-28.7	SANSA
South America / southern midlatitude (1)					
CN53M	Cachoeira Paulista, BR	-22.7	-45.0	-14.4	INPE
Australasia / Pacific (5)					
TV51R	Townsville, AU	-19.6	146.8	-26.6	GIRO / DIDB
ND61R	Niue	-19.1	-169.9	-20.0	GIRO / DIDB
PE43K	Perth, AU	-32.0	116.1	-41.2	GIRO / DIDB
CB53N	Canberra, AU	-35.3	149.0	-42.0	GIRO / DIDB
HO54K	Hobart, TAS	-42.9	147.3	-49.7	GIRO / DIDB

Coverage gaps. Boa Vista (BVJ03) anchors the northern equatorial-anomaly crest at dip latitude $\sim +12°$ N (Brazilian sector, where the South Atlantic Anomaly shifts the magnetic equator south of its geographic counterpart, lifting the dip latitude of geographic- equatorial stations onto the northern magnetic flank). Indian- sector and SE-Asian dip-latitude stations near $\phi_{\text{dip}} \approx +15°$ N (Trivandrum, Hyderabad, Madimbo, etc.) would close the longitude-coverage gap but do not currently return DIDB data. The southern hemisphere is well-covered (Townsville, Niue, Perth, Canberra, Hobart for the Pacific block), but Africa is sparse outside SANSA's Louisvale, and Asia / Indian Ocean coverage is essentially absent. This shapes the residual EIA-amplitude bias documented in Section 10.

E Reproducibility Manifest

The harness-fitted constants $\{L_{\text{iono}}, D, w_{\text{sc}}, w_{\text{NVIS-tail}}, w_{\text{Es-prim}}, \sigma\text{-scale}, \text{fusion-flag}\}$ were produced by joint coordinate descent (Section 7.2, `scripts/tests/tune-r7.mjs`) starting from three deterministic seed configurations chosen to enter the parameter space from distinct basins. This appendix lists those seeds, the per-parameter sweep grid, and the on-disk artefacts needed to reproduce the harness-fitted numbers end-to-end.

Table 16: Coordinate-descent starting configurations. All three seeds run the same descent over the grid below; the winner is the seed whose final state has the lowest binary Brier loss against the WSPR "alive somewhere" truth. "L_{iono}" here is the per-hop HF lumped ionospheric loss in dB; "defocusDbPerExtraHop" is the D constant of Section 3.8; the four mode weights are the dimensionless multipliers controlling whether F2-scatter / NVIS-tail blending / Sporadic-E-as-mode contributions fire; "fusionEnabled" is the per-hop foF2 fusion flag (deferred path of Section 6.5); "sigmaScale" uniformly scales the per-band σ_g table (Table 9) to expose whether the harness wants a wider or tighter prediction spread.

Seed	L_{iono}	D	**fusion**	w_{sc}	w_{NVIS}	w_{Es}	**σ-scale**
baseline	1	0.25	off	0	0	0	1.0
fusion-up	4	0.25	on	0	0	0	1.0
modes-on	1	0.25	on	1.5	1.0	1.0	1.3

Three deterministic starting configurations. The three names encode the experimental hypothesis each seed tests: *baseline* freezes alternative-mode bonuses to zero and asks whether the F2-only budget alone is sufficient; *fusion-up* asks whether enabling per-hop foF2 fusion plus a deeper L_{iono} shifts the Brier optimum; *modes-on* asks whether all alternative-propagation bonuses turned on at moderate weights, plus a wider σ, finds a different basin. All three converged to the same basin within ≤ 2 iterations (Section 7.2), final $\{L_{\text{iono}}, D, w_{\text{sc}}\}$ within $\pm 10\,\%$ across seeds.

Per-parameter sweep grid. Each coordinate-descent step sweeps one parameter through its grid below while holding the others fixed at their current best values. Grids were chosen wide enough to cover both physical-edge values and obvious-overshoot values; the descent converges inside the physical region.

Table 17: Coordinate-descent sweep grid (`scripts/tests/tune-r7.mjs`'s `GRID`).

Parameter	**Sweep values**
L_{iono} (`lIonoHfDb`)	0, 0.5, 1, 2, 4, 6, 8
D (`defocusDbPerExtraHop`)	0, 0.25, 0.5, 1.0
`fusionEnabled`	false, true
w_{sc} (`scatterWeight`)	0, 0.5, 1.0, 1.5
$w_{\text{NVIS-tail}}$ (`nvisTailWeight`)	0, 0.5, 1.0, 1.5
$w_{\text{Es-prim}}$ (`esWeight`)	0, 0.5, 1.0, 1.5
σ-scale (`sigmaScale`)	0.7, 1.0, 1.3, 1.6, 2.0

On-disk calibration artefacts. The harness writes two baseline files and one report file; together with the source configurations above and the calibrated-constants file `src/constants.js` they specify the calibration completely.

`src/constants.js`. The single source of truth for every harness-fitted dB constant (L_{iono}, D, w_{sc}, $w_{\text{NVIS-tail}}$, $w_{\text{Es-prim}}$, σ-scale, fusion-flag above plus EIA amplitudes / TEP plateau coefficients / Es persistence kernel constants / Bz bump table / storm-floor schedule). Hand-edited after the harness reports a winner; every value in the budget traces to this file.

`scripts/data/harness.baseline.json`. Global "open somewhere" baseline computed from a single bbox-unrestricted WSPR aggregate query (cheap, $\sim 7\,k$ rows for a 30-day window). Used as the default truth for the Brier / accBin numbers in the headline metrics (Table 8).

`scripts/data/harness.baseline.perpath.json`. Per-path baseline using TX/RX bounding-box restricted aggregates (35 queries per harness run, ≥ 1 spot/h floor). Adds a path-specific drift signal that the global mode averages out.

`scripts/outputs/harness.report.json`. Latest harness run output. Regenerated each `node scripts/harness.mjs` invocation; `scripts/outputs/` is gitignored so the file is not committed to source control.

`src/data/spot-baselines.mjs`. Per-(band, UTC-hour) 30-day WSPR spot-count baseline used by the spot-override surface (Section 8.1) and the operator Comp. availability prior of Eq. 79. Regenerated daily at 06:00 UTC by the `wspr-baselines.yml` GitHub Actions workflow, which runs `node scripts/harness.mjs wspr-baselines` and commits the diff; this gives a true sliding 30-day window that tracks solar-cycle progression and seasonal shifts without manual intervention.

Auxiliary test, tune, and diagnostic suites. Three script families live alongside the harness in `scripts/tests/`:

Tune suites. Coordinate-descent or grid-sweep scripts that the production calibration pulls from. `tune-r7.mjs` runs the seven-parameter sweep described above; `tune-eia.mjs` sweeps the EIA crest amplitude / width / latitude triplet against the equatorial GIRO basket (§4.2); `tune-blend.mjs` sweeps the legacy ensemble-blend weights as a regression canary against the retired N0NBH-style heuristicTier (§7). All three write report cards to stdout rather than mutating committed files.

Validation suites. Cross-checks against external sources: `wspr-snr.mjs` compares per-spot WSPR SNR to the model's predicted budget on the same path / hour; `psk.mjs` pulls PSKReporter spot density as an independent activity signal; `rbn.mjs` and `rbn-beacon.mjs` use Reverse Beacon Network spot data against CW skimmer references. These cover decode-mode truth that WSPR-only validation misses.

Diagnostic suites. Pre-fix investigation tooling retained for regression detection: `storm-split.mjs` bins per-path margin error by storm phase (CME-onset / main / recovery / quiet); `day-night.mjs` bins by sunlit fraction of the path; `hops.mjs` bins by integer hop count. These produce per-bin Brier / accBin breakdowns and are the source of the band-distribution numbers cited in §10 #8.

What this paper documents and what it does not. Every constant in the SNR budget (Section 3) and MUF estimation (Section 4) is given inline. The σ_g table is in Table 9; the noise-floor table in Table 3; the full path basket in Appendix C; the GIRO station list in Appendix D; the Φ approximation in Section 7.3; the IGRF pole and EIA / winter / equatorial / polar amplitudes inline at their respective equations. What this paper does *not* reproduce is (a) the WSPR ground-truth aggregates themselves — those are queried live from `wspr.live` at harness-run time and their values shift week-to-week as activity rebalances; and (b) the SWPC, GIRO, GFZ, and Kyoto data feeds, which are also live. Reproducing the headline accBin / Brier numbers in Table 8 therefore requires running the harness against contemporaneous data from those feeds, which means the absolute numbers will differ slightly run-to-run; the regression detection signal (drift $> 2\,$dB mean margin or $5\,$pp in P(open) across runs) is the reproducibility guarantee, not bit-exact metric values.

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